ENERGY SHOCKS AND THE MACROECONOMY: EVIDENCE FROM NATURAL GAS MARKETS*

Daniele COLOMBO† Francesco TONI‡

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Abstract

What determines the macroeconomic effects of energy-price fluctuations? We study this question in U.S. and Euro Area natural gas markets by constructing new instruments for gas-demand and supply disturbances. Exploiting exogenous temperature variation and high-frequency price movements around supply-related events, the design allows us to estimate demand and supply elasticities and distinguish realized supply disruptions from news about future scarcity. These shocks have aggregate consequences beyond gas markets. Adverse news about future gas availability generates larger, more persistent consumer-price effects than realized shortages, despite little aggregate industrial-production response. Inventory accumulation and financial-market volatility point to precautionary stockbuilding and heightened uncertainty as transmission mechanisms. Real effects depend on the margin of adjustment in gas use. In a more gas-exposed setting, consumption and investment fall alongside persistent production losses, consistent with aggregate-demand amplification; in less exposed settings, aggregate effects are limited and losses concentrated in energy-intensive industries.

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†Corresponding author. Department of Economics, London Business School, Regent's Park, London NW1 4SA, United Kingdom. Email: dcolombo@london.edu. Phone: +44 7798 637841.

‡Université Côte d'Azur, CNRS, GREDEG, France. Institute of Economics, Sant'Anna School of Advanced Studies, Italy. Email: francesco.toni@unice.fr

I Introduction

Recent disruptions to energy supply routes have renewed interest in the long-standing question of how energy prices affect the macroeconomy, and have made concerns about energy security and future availability especially pressing (IEA, 2026b). Answering this question is inherently challenging because prices respond to economic conditions, yet it is central to policy as higher energy costs can raise inflation while depressing economic activity. The magnitude and persistence of these effects depend on the forces driving prices (Kilian 2009) and on the margins through which economies can adjust, including supply, inventories, substitution, and spending. We study these mechanisms in the natural gas markets of the United States and the Euro Area.

Natural gas is an essential input, central to Europe's 2022 energy crisis and again exposed to supply risks amid ongoing disruptions in the Strait of Hormuz. It is characterized by infrastructure constraints that leave regional markets relatively segmented, providing a natural setting to examine how macroeconomic transmission varies with differences in the available margins of adjustment. The United States relies largely on domestic production, whereas the Euro Area depends heavily on imports and is therefore more exposed to risks affecting external supplies. Gas is also storable, so expectations about future availability can influence inventory decisions and current prices (Kilian and Murphy 2014).

This paper develops new instruments to disentangle demand and supply forces in natural gas markets. For demand, we exploit unusually large temperature deviations from seasonal norms, which generate shifts in heating-related gas demand that are not driven by economic conditions. Heating accounts for a substantial share of total natural gas consumption: 29% in the Euro Area and 18% in the United States.[1] Consistent with a heating-demand channel, the demand instruments are most strongly correlated with gas prices when heating needs are high. They remain informative after accounting for other weather-related conditions.

For supply, we use high-frequency movements in gas prices around supply-related events. Both disruptions to current gas flows and news about future availability are closely monitored by financial markets and can result in significant price fluctuations. We manually review thousands of gas-related news reports to identify plausibly exogenous news about gas availability, screening out events accompanied by contemporaneous demand-side or other confounding developments. We then measure gas-price changes within narrow windows around each event to isolate the price response to new supply information. Broader economic conditions are typically already reflected in prices and are unlikely to change discretely

[1] Authors' calculations for space heating using IEA data (2026a), averaged over 2004–2022 for the Euro Area and 2010–2022 for the United States.

within the event window. In both regions, we retain unexpected changes in current gas flows that are not driven by economic conditions. For the Euro Area, we also retain a separate set of events revealing new information about the availability and reliability of future supplies, reflecting its exposure to transit disputes and geopolitical developments affecting external supplies. Using the resulting series in regional structural Bayesian VARs, we identify gas *demand* and *flow-supply* shocks and, for the Euro Area, *supply-security news* shocks, and trace their macroeconomic effects. A broad set of sensitivity checks shows that our main findings are robust to observable confounding information and to changes in instrument construction and estimation method. The instruments are also not systematically correlated with standard measures of macroeconomic and financial shocks.

We first use the instruments to estimate short-run demand and supply elasticities, which characterize the scope for adjustment to gas-market shocks. Demand is substantially less price-elastic than supply in both regions: -0.08 versus 0.31 in the Euro Area and -0.04 versus 0.23 in the United States. This limits the extent to which lower consumption can absorb supply disruptions without substantial price increases, as also reflected in our impulse responses. Consistent with this, our variance decompositions show that supply shocks account for a larger share of gas-price fluctuations than demand shocks.

Second, we show that these shocks have economically significant effects beyond gas markets. Their transmission reflects both what drives the price increase and how economies adjust. Gas-demand shocks highlight the role of supply structure in transmission: they raise domestic gas production and industrial activity in the United States, whereas in the Euro Area supply expands through imports and consumer-price increases are larger and more persistent. Flow-supply shocks generate modest consumer-price effects, with little aggregate industrial-production response in the United States and only a temporary decline in the Euro Area.

Adverse news about future gas availability generates persistent gas-price increases, induces inventory accumulation, and increases financial-market volatility. Flow-supply shocks, by contrast, draw down inventories and leave volatility broadly unchanged. These contrasting responses matter for inflation: for the same initial gas-price increase, supply-security news generates larger and more persistent consumer-price effects than realized flow disruptions, despite little aggregate industrial-production response. A 10% gas-price increase raises the consumer-price level by around 0.5% over the medium run. Together, these findings point to a strong expectations and uncertainty channel: precautionary stockbuilding sustains gas-price pressures rather than cushioning shortages, while increased financial-market volatility signals heightened uncertainty about future supply. This contrasts with Känzig's (2021) news about planned OPEC oil supply, which does not raise measured financial uncertainty.

Third, we study the mechanisms shaping the real effects of gas-supply shocks in the Euro Area, where pronounced inflationary effects coexist with limited aggregate production responses. We find that production losses remain concentrated in energy-intensive industries, while higher input costs spread more broadly through prices. Gas-supply shocks transmit from electricity to producer and then consumer prices, including core inflation, with supply-security news accounting for most of this pass-through.

This broad transmission to prices raises a further question: can higher energy costs trigger a broader contraction in spending and production? We examine these effects in the Euro Area and Germany, whose pre-crisis dependence on Russian gas placed it at the center of the debate over the costs of a supply cutoff (Deutsche Bundesbank 2022; Bachmann et al. 2024; Di Bella et al. 2024; Albrizio et al. 2025). Gas use, consumption, and investment decline in Germany, alongside a broader and more persistent contraction in output.

The real costs of supply disruptions depend on how economies adjust. With small gas expenditure shares and scope for alternative sourcing or substitution, reductions in gas use can entail limited aggregate costs (Moll, Schularick, and Zachmann 2023). When higher energy costs erode real income and depress spending, however, losses can propagate beyond directly exposed industries through aggregate demand (Guerrieri et al. 2022; Auclert et al. 2023). The German responses are consistent with this amplification mechanism. Model-based benchmarks provide quantitative support for this interpretation. Calibrating the supply-side approximation of Moll, Schularick, and Zachmann (2023) with gas-import expenditure shares and our estimated elasticities yields smaller losses than those implied by our empirical responses. Because our elasticities are close to those used in the original crisis assessments, this gap is unlikely to reflect lower assumed substitutability. Instead, the German losses are comparable to those predicted by models incorporating aggregate-demand amplification (Auclert et al. 2023; Pieroni 2023).

Together, these findings help reconcile the Euro Area's aggregate resilience to the Russian gas cutoff with persistent inflation, concentrated industrial losses, and prolonged economic weakness in Germany.[2] Gas prices began rising as concerns about future supply intensified, before physical flows were substantially curtailed, and increased further as deliveries fell. The Euro Area replaced part of the lost gas and sharply reduced consumption without experiencing the aggregate output collapse projected in the most adverse early scenarios (Moll, Schularick, and Zachmann 2023). Yet inflation rose sharply and remained elevated even as the acute physical supply crisis eased (see Online Appendix Figure B.11). Our results

[2]Draghi (2024) documents a 10–15% decline in European energy-intensive industrial production since 2021. German Federal Statistical Office (2026) reports contractions in German GDP in 2023 and 2024, followed by only slight growth in 2025.

show how concerns about future supply can sustain inflationary pressures even when current shortages are contained. Our historical decompositions further show that gas-supply shocks were a key driver of the Euro Area's 2021–2023 inflation surge, while oil shocks contributed more in the United States. The German evidence, in turn, is consistent with gas-supply shocks contributing to the country's post-2022 stagnation.

Finally, we examine the environmental consequences of gas shocks, highlighting the role of price persistence. Demand and realized flow-supply shocks raise emissions in the short run. Following flow disruptions, electricity generation shifts toward oil in the United States, where some gas-fired plants can switch fuels, and toward coal in the Euro Area, where utilities can shift generation across gas- and coal-fired plants (EIA 2022; Reuters 2024). Supply-security news, by contrast, raises gas prices more persistently, and emissions fall below baseline at longer horizons. This pattern is consistent with a shift from immediate fuel switching toward longer-run adjustment to cleaner energy.

Related literature and contribution. We contribute to the literature estimating energy-market elasticities and distinguishing demand from supply forces. Most of this literature focuses on crude oil and uses a range of identification strategies (Kilian 2009; Kilian and Murphy 2012; Baumeister and Hamilton 2019; Caldara, Cavallo, and Iacoviello 2019; Kilian 2022; Baumeister and Hamilton 2024b). Within the smaller VAR literature on natural gas, elasticities are estimated using recursive identification (Nguyen and Okimoto 2019; Rubaszek and Uddin 2020) or frameworks combining zero, sign, and magnitude restrictions with prior beliefs (Wiggins and Etienne 2017; Rubaszek, Szafranek, and Uddin 2021; Casoli, Manera, and Valenti 2024), building on methods developed for oil markets (Baumeister and Hamilton 2015; Baumeister and Hamilton 2019).[3] We provide new estimates of aggregate gas demand and supply elasticities for both the United States and the Euro Area using instrumental-variable identification (Lunsford 2015; Stock and Watson 2018). Our instruments provide identifying information from outside the VAR, allowing estimation without imposing theory-based restrictions. Our estimates lie toward the less elastic end of previous short-run demand estimates and the more elastic end of supply estimates, implying less immediate adjustment in gas use and greater supply responsiveness. These short-run market elasticities provide empirical targets for models of energy use (Atkeson and Kehoe 1999; Casey 2024) and carbon pricing (Fried 2018), addressing the need for empirical elasticity estimates highlighted by Moll, Schularick, and Zachmann (2023). The region-specific estimates also account for differences in adjustment across fragmented natural gas markets (Szafranek and Rubaszek

[3]Other approaches include simultaneous-equation estimates of world gas demand and supply elasticities (Krichene 2002) and cross-country estimates of gas-demand elasticities using gas reserves as instruments for prices (Burke and Yang 2016).

2024), which a single global elasticity may fail to capture.

Our identification strategy connects research on weather and supply-related events in gas markets with the analysis of natural gas shocks and their macroeconomic effects. For the demand instruments, we build on studies linking temperature variation to gas prices (Mu 2007; Nick and Thoenes 2014; Wang, Zhang, and Broadstock 2019; Chen, Hartley, and Lan 2023). While this variation has been used to estimate sectoral demand elasticities (Davis and Muehlegger 2010; Hausman and Kellogg 2015), to the best of our knowledge we are the first to use it to identify aggregate gas demand shocks and trace their macroeconomic effects. For the supply instruments, we build on event studies examining how gas prices respond to market-relevant announcements and disruptions (Gay, Simkins, and Turac 2009; Halova, Kurov, and Kucher 2014; Bartelet and Mulder 2020; Goodell et al. 2023; Goodell et al. 2024). We integrate this event-study literature with VAR analysis by constructing instruments from price movements around supply events. Our approach adapts high-frequency identification methods developed in monetary-policy research (Kuttner 2001; Gertler and Karadi 2015; Altavilla et al. 2019) and subsequently applied to other domains (Wu and Cavallo 2012; Känzig 2021; Känzig 2023). We also construct informationally robust versions of the instruments to account for observable confounding information (Romer and Romer 2004; Nakamura and Steinsson 2018; Miranda-Agrippino and Ricco 2021).

Existing work on gas-shock transmission highlights the roles of precautionary inventory demand (Adolfsen et al. 2024) and inflation expectations (Boeck and Zörner 2025) using restriction-based identification. Closest to our approach, Alessandri and Gazzani (2025) construct a high-frequency instrument for Euro Area gas-supply shocks and find sizable stagflationary effects. Their instrument combines realized flow changes with news about future supply availability and security. Our separate instruments for realized flow disruptions and supply-security news clarify when precautionary inventory demand arises and how it shapes macroeconomic transmission. For the same initial gas-price increase, realized disruptions raise consumer prices and temporarily reduce aggregate production, while drawing down inventories. News about future supply instead induces inventory accumulation and generates larger and more persistent inflationary effects, with little aggregate production response.

Our cross-region comparison further shows how inflationary transmission differs across natural gas markets, extending existing evidence on European gas-price pass-through (Adolfsen et al. 2024; Casoli, Manera, and Valenti 2024; López et al. 2026). Following gas-demand shocks, headline consumer-price pass-through reaches around 4% after two years in the Euro Area, compared with a peak of 1.5–2% in the United States. Flow-supply pass-through is around 1% in both regions, while supply-security news generates pass-through rising from

around 2% on impact to 5% over the medium run. Our European evidence also shows that gas-supply shocks propagate to core inflation. By contrast, U.S. studies find little pass-through to core inflation from oil and broader energy shocks, with natural-gas-specific estimates also small and imprecise (Gao, Kim, and Saba 2014; Kilian and Zhou 2022; Kilian and Zhou 2023). Our historical decompositions further contribute to the debate on the drivers of the post-pandemic inflation surge (see, among others, Bańbura, Bobeica, and Martínez Hernández 2023; Baumeister 2023; Giannone and Primiceri 2024; Bernanke and Blanchard 2025; Barro and Bianchi 2026; Coibion and Gorodnichenko 2026) by quantifying the contributions of gas and oil disturbances in the United States and the Euro Area.

Finally, we contribute to the literature on the environmental effects of natural gas prices. Previous studies find that, in the short run, lower gas prices reduce emissions through coal-to-gas substitution (Knittel, Metaxoglou, and Trindade 2015; Cullen and Mansur 2017; Brehm 2019), while over longer horizons higher gas prices can promote cleaner investment and reduce emissions (Stock and Zaragoza-Watkins 2024). This tension underlies the "gas trap" discussed by Harstad and Holtsmark (2024): while gas can reduce emissions in the near term, its continued use may hinder the transition to cleaner energy sources (Acemoglu et al. 2023). Our findings speak directly to this trade-off by showing how different shocks generate price increases of different persistence: temporary increases encourage switching toward more carbon-intensive fuels, whereas persistent increases are associated with lower emissions over time, consistent with adjustment toward cleaner energy. These findings underscore that gas-specific price increases are not equivalent to a broad carbon price.

Layout. The remainder of the paper is structured as follows. Section II provides background on natural gas markets. Section III details the identification strategy for gas demand and supply shocks. Section IV outlines the econometric framework and estimation approach. Section V presents the empirical findings, and Section VI concludes. The Online Appendix provides additional details on the data, instrument construction, and Bayesian estimation, along with supplementary results and sensitivity analyses.

II Natural gas markets

This section outlines key characteristics of the natural gas markets in the Euro Area and the United States, emphasizing important regional differences. These distinctions motivate the use of a region-specific approach to analyzing natural gas shocks and provide the foundation for our identification strategy. Additional descriptive evidence is reported in Online Appendix B.

Natural gas is a critical energy source, accounting for about one quarter of total energy supply in both the Euro Area and the United States. It plays a crucial role as a primary fuel in residential and commercial heating, and it also serves as an important input for industrial activities and electricity generation. In industry, natural gas is used not only as a fuel for heat and steam generation, but also as a direct production input in gas-intensive sectors such as chemicals, ceramics, paper, and basic metals. Residential consumption represents a substantial share of total natural gas demand, 35% in the Euro Area and 22% in the United States, primarily for heating purposes. Gas demand is highly sensitive to temperature fluctuations, particularly in winter (Chen, Hartley, and Lan 2023). Consequently, anomalous weather-driven variability can be exploited as a source of exogenous variation to study the effects of gas demand shocks. Despite accounting for a sizable share of total energy use, natural gas represents a relatively small expenditure item at the aggregate level: final gas consumption accounts for approximately 0.5% of nominal GDP in the Euro Area and 0.4% in the United States.[4]

While the demand side of the natural gas market is broadly comparable across the two regions, the supply side exhibits substantial differences. European gas supply relies overwhelmingly on imports. Domestic production has fallen steadily and now stands below 100 petajoules per month. As a result, import dependence has risen sharply, increasing from approximately 50% of total consumption in the early 1990s to over 90% in 2019, with imports concentrated among a limited set of major suppliers. Before the sharp reorientation of import flows after 2022, these included Russia, Norway, Algeria, and the United States (Council of the European Union 2023). This dependence on external suppliers means that both disruptions to current gas flows and news about future availability or supply security are closely monitored by financial markets and can result in significant price fluctuations. The price fluctuations that follow market-relevant events can be leveraged to study the effects of gas price shocks using high-frequency identification techniques.

The United States presents a very different supply structure: it is the world's largest natural gas producer. Domestic production has grown substantially, doubling from approximately 2,000 PJ per month in the early 2000s to 4,000 PJ in 2023. This remarkable expansion has been primarily driven by advancements in shale gas extraction (Hausman and Kellogg 2015; Acemoglu et al. 2023; Stock and Zaragoza-Watkins 2024), with the United States gradually emerging as a net exporter, particularly of LNG to European and Asian markets.

[4]Consumption shares refer to average residential natural-gas consumption over 2004–2022 and are based on own calculations using IEA data (2024a, 2024b); see Online Appendix Figure B.3 for a detailed breakdown by end use. Expenditure shares are averages over 2004–2022 and are computed by valuing total final natural-gas consumption at TTF prices for the Euro Area and Henry Hub prices for the United States, divided by nominal GDP from the IMF and FRED.

While natural gas prices in the United States exhibit lower volatility than in the Euro Area, they remain sensitive to domestic production disruptions.

The global natural gas market remains regionally segmented because transportation constraints and limited liquefaction and regasification capacity limit the fungibility of natural gas across regions. As a result, prices can diverge substantially across regions, despite the expansion of global LNG trade. For instance, the United States is less exposed to gas price shocks originating abroad, owing to its relative self-sufficiency and constraints on LNG export capacity, which limit the ability of domestic production to reach international markets (IMF Blog 2023). This degree of market segmentation highlights the importance of a region-specific analysis of gas price shocks, beginning with the identification of the relevant regional price benchmarks. The United States pioneered natural gas market liberalization in the 1970s, adopting a gas-on-gas pricing model, with the Henry Hub (HH) serving as the benchmark price since 1990. In Europe, although liberalization occurred more recently, the Dutch Title Transfer Facility (TTF) has emerged as the benchmark for gas prices and is recognized as the most liquid trading hub. Online Appendix B.3 presents supporting evidence of price integration within the European gas market. Because futures contracts on these benchmarks are liquid and actively traded, daily movements in HH and TTF futures prices provide the event-window variation used in our identification strategy.

III Identification strategy

Building on these market features, this section describes how we construct monthly instruments for structural gas demand and supply shocks. Demand shocks are isolated using exogenous temperature deviations from seasonal norms, which generate variation in heating demand. Supply shocks are identified using market-relevant news combined with high-frequency movements in natural gas spot and futures prices, distinguishing between events that affect current gas flows and, for the Euro Area, news about future availability and supply security.

III.A Demand shocks

We construct the demand instruments using deviations of temperature from seasonal norms as a proxy for heating demand. Unusually warm weather lowers heating demand and tends to reduce natural gas prices, whereas unusually cold weather raises heating demand and prices (see, e.g., Colombo and Ferrara 2025, for evidence on temperature effects on energy production and prices).

The monthly instruments are constructed as follows. First, we compute a gas-consumption-weighted average of local temperatures, using predetermined weights based on past gas consumption.[5] Second, we remove regular seasonal patterns by subtracting, for each calendar month, the average temperature for that month computed over the full sample. Equivalently, the monthly series can be interpreted as the average of daily temperature deviations from the relevant seasonal benchmark within each month. Third, we retain only unusually large deviations from these seasonal norms by setting small fluctuations to zero. This reduces noise and isolates months in which temperature movements are most likely to generate meaningful shifts in heating demand. Moreover, unlike typical seasonal temperature fluctuations, large deviations from average temperatures are less likely to be anticipated by economic agents. As a result, these deviations are less likely to be incorporated into trading decisions and affect gas prices primarily through the heating-demand channel, supporting the validity of the instrument.[6] Formally, for each region, the monthly temperature index $TI_{y,m}$ is defined as

$$TI_{y,m} = T^{SA}_{y,m}\,\mathbf{1}\left\{\left|T^{SA}_{y,m}\right| > \sigma_T\right\}, \qquad T^{SA}_{y,m} = T_{y,m} - \overline{T}_m, \tag{1}$$

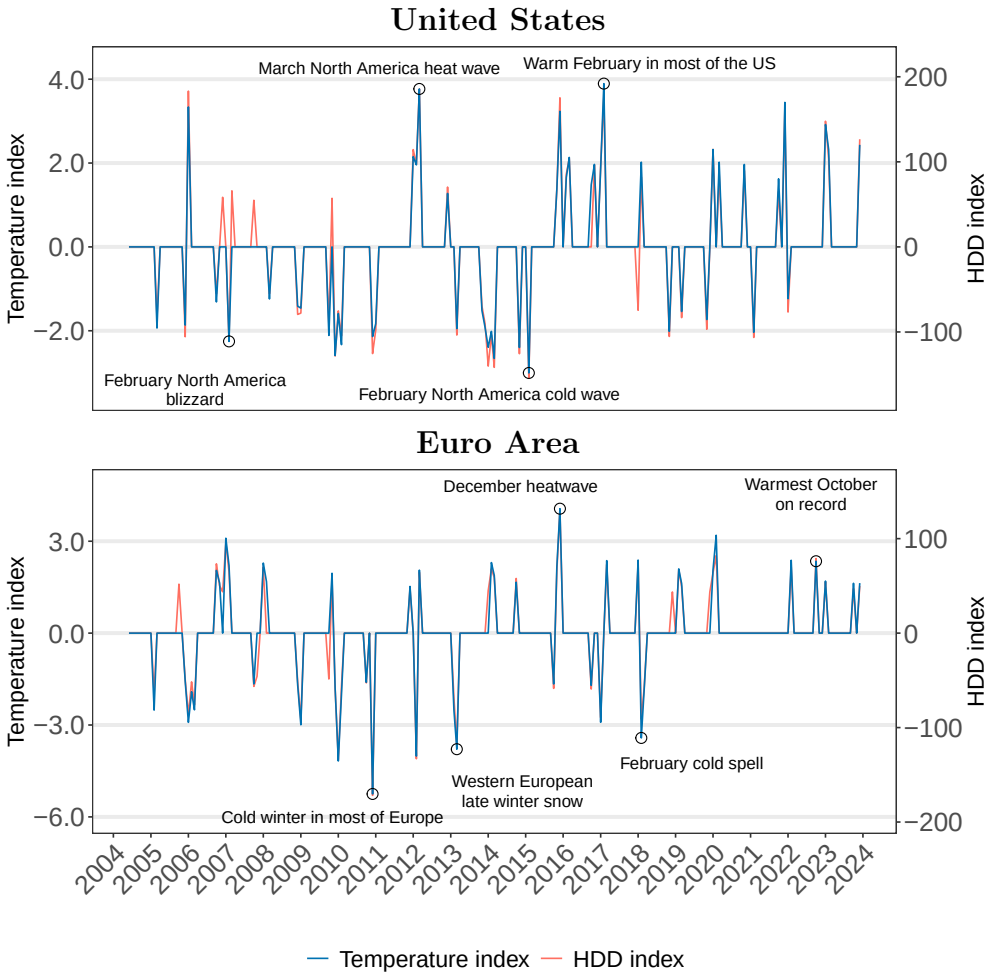
where $T_{y,m}$ denotes the regional monthly temperature in year y and month m, and $\overline{T}_m$ is the sample average for calendar month m. The term σ_T denotes the standard deviation of the seasonally adjusted temperature series over the full sample. Additionally, we set the index to zero in warmer months (April to September), when the relationship between temperatures and gas prices is ambiguous: higher temperatures may either increase electricity demand for air conditioning or reduce heating demand.[7] Finally, we exclude months affected by the COVID-19 pandemic. In practice, this adjustment only removes a single spike in the U.S. series, in March 2020, and has no material effect on the results. Online Appendix C.1 provides additional details on the data sources and alternative constructions of the series. Figure I presents the resulting index for each region.

[5] Weights are based on a three-year moving average of lagged gas consumption, from $y-3$ to $y-1$, so the aggregation is not driven by contemporaneous demand conditions. The Euro Area measure averages temperatures in Belgium, Germany, France, Luxembourg, and the Netherlands, the countries most directly linked to TTF pricing; the U.S. measure aggregates temperatures across the contiguous U.S. states using the same weighting scheme.

[6] Temperature forecasts deteriorate significantly as the forecast horizon extends, remaining relatively unreliable even with advanced forecasting methods; see, for example, Lopez-Gomez et al. (2023). Online Appendix Figure C.2 provides evidence that anticipation effects are limited in both magnitude and timing, supporting the use of monthly temperature deviations as instruments.

[7] This choice may be especially relevant for the United States, where cooling demand during the warmer months constitutes an important driver of electricity consumption (Sailor and Pavlova 2003; Bartos et al. 2016). However, as shown in Section V.C, including warmer months in the demand instrument does not materially affect the results.

Figure I: *Temperature-based demand instruments*



Notes: The figure reports the demand instruments for the United States (top panel) and the Euro Area (bottom panel). The blue line denotes the baseline temperature index, while the red line denotes the analogous index constructed from heating degree days. Circles highlight selected temperature-related gas-market episodes. For the United States, these are: 2007M2, a winter storm affecting much of eastern North America; 2012M3, record-breaking heat across many U.S. states; 2015M2, a severe cold wave particularly impacting the eastern United States; and 2017M2, a late-winter heat wave across the central and eastern United States. For the Euro Area, these are: 2010M12, the coldest December in 100 years and the coldest winter month since February 1986; 2013M3, a late-season snow event that impacted Western Europe; 2015M12, the warmest December on record for several countries; 2018M2, the cold spell commonly referred to as "The Beast from the East," characterized by cold winds and low temperatures; and 2022M10, the warmest October on record in Europe.

Validation of the demand channel. Positive spikes in the index tend to coincide with declines in gas prices, and conversely. This relationship remains strong after controlling for standard macroeconomic variables. The negative association is consistent with the proposed heating-demand channel and supports the relevance of the instrument. If temperature affects

gas prices primarily through heating demand, the relationship should be strongest when absolute temperatures are low enough for heating needs to matter. By contrast, unexpectedly cold temperatures during warmer periods should have a limited effect, since temperatures may still remain too high to trigger widespread heating use. To assess this mechanism, we compare the correlation between the temperature index and gas prices across months with different levels of heating and cooling needs, measured using *heating degree days* (HDD) and *cooling degree days* (CDD).[8] The correlation is substantially weaker in months with low HDD than in months with high HDD. This indicates that temperature fluctuations affect gas prices mainly when they translate into changes in heating demand. Symmetrically, the negative correlation weakens in months with high CDD, indicating that cooling-related demand does not generate the same gas-price relationship. The same pattern appears when comparing winter and warmer months, reinforcing the choice to focus on winter temperature deviations for instrument construction, which reduces noise and improves instrument relevance. Online Appendix C.1 reports the corresponding correlations by heating and cooling needs. As an additional check, we construct an alternative index using heating degree days rather than temperature deviations. Once rescaled, this series closely tracks the baseline measure, as shown in Figure I, reflecting that HDD is itself constructed from temperature and captures similar heating-demand variation. Consistent with this, Online Appendix E.1 shows that replacing the temperature-based demand instruments with HDD-based instruments leaves the impulse responses nearly unchanged.

III.B Supply shocks

To identify gas supply shocks, we employ a high-frequency identification strategy, drawing on methodologies originally developed in the monetary policy literature (Cochrane and Piazzesi 2002; Nakamura and Steinsson 2018; Altavilla et al. 2019) and more recently adapted to oil and energy-market applications (Wu and Cavallo 2012; Känzig 2021; Känzig 2023). Specifically, we analyze changes in natural gas prices around market-relevant supply events. We first construct flow-supply instruments based on events that reveal realized or near-term changes in the physical availability of natural gas. These events provide a natural source of identifying variation: they include, for example, unplanned outages, pipeline disruptions, force majeure declarations, production shut-ins, and unexpected restorations. For these events, identification requires that the underlying change in gas availability is not driven

[8] HDD and CDD proxy the energy required to heat and cool buildings. For definitions, see Eurostat (2023); the data are from European Commission (2025). We aggregate them using the same gas-consumption weights as in the baseline temperature index, at the country level for Europe and at the state level for the United States.

by contemporaneous macroeconomic conditions, and that the event-window price response primarily reflects gas-supply information rather than other confounding news.

For the Euro Area, we also construct an instrument based on future-supply news. This reflects the import-dependent structure of European gas markets, where prices respond not only to current disruptions but also to news about transit routes, geopolitical developments, and expected future availability. This news need not affect current gas flows immediately. Instead, it reveals new information about expected gas availability and supply security, so the corresponding instrument captures gas supply-security news shocks rather than pure contemporaneous physical supply shocks. As with flow-supply events, identification requires that markets react to the relevant information within a narrow event window. This allows the resulting price surprises to capture information not yet incorporated into gas prices. The narrow window also limits reverse causality from broader economic conditions, which are typically already priced and unlikely to change discretely within the event window. For this instrument, the relevant exclusion restriction is that the price reaction is primarily driven by gas-supply information rather than broader confounding news.

Unlike monetary policy announcements or OPEC meetings, gas-supply news does not originate from a single institutional source, but is dispersed across producers, pipeline and LNG infrastructure operators, regulators, governments, and geopolitical events. We therefore construct the event set from financial newswires rather than from a single announcement calendar. Specifically, we collect gas supply-related news via Factiva and manually assess whether each candidate event is gas-supply-specific, newly revealed, and plausibly unconfounded.[9]

Event selection and classification. The sample covers the 20-year period from January 1, 2004 to December 31, 2023, starting with the first full year after the establishment of the TTF and the first year for which daily European gas-price series are consistently available. We use Henry Hub and TTF spot and front-month prices to construct daily percentage price changes. These maturities are used to screen for candidate event days because they are liquid and highly responsive to news about current or near-term gas-market conditions. We exclude days on which neither the spot nor the front-month price change can be computed because of missing observations.[10] To construct a clean set of candidate event days, we first apply mechanical ex ante filters that remove dates likely to be confounded by forces

[9]The Factiva search uses the keywords "Natural Gas" and "LNG" in Reuters Newswires, Dow Jones Newswires, and S&P Global Insight Daily Analysis. Searches are geographically restricted to Europe and North America, respectively. To capture region-specific institutional news, we additionally include "Gazprom" for the Euro Area and "FERC" for the United States.

[10]If a closing price is missing for only one maturity, we retain the date and set that maturity's daily price change to zero; this affects 45 U.S. dates and 436 Euro Area dates.

unrelated to gas supply. We exclude months in which the temperature instrument records demand-driven movements, OPEC and monetary-policy announcement days, which can generate immediate gas-price revisions through oil-market or financial channels unrelated to gas supply, and March–May 2020, the onset of the COVID-19 pandemic.[11] For the United States, we also exclude days with large same-direction movements in oil and gas prices, since such comovements may reflect shocks common to domestic oil and gas production.[12]

We then retain only dates on which the absolute gas-price surprise exceeds 2% in either the spot or front-month contract.[13] This conservative threshold limits background price noise while avoiding the premature exclusion of potentially relevant supply news. For the remaining dates, we use the Factiva search described above to identify those with natural-gas-related news. Dates without any such news are discarded, leaving 2,590 U.S. dates and 2,434 Euro Area dates for manual review. We then carefully read the corresponding articles one by one to identify clean gas-supply events and classify the associated news.

In the manual review, we retain flow-supply events such as explosions, unplanned outages, force majeure declarations, production shut-ins, and disruptions to fields, pipelines, storage, or LNG infrastructure.[14] We exclude anticipated maintenance and voluntary production adjustments, which are more likely to be already priced or endogenous to contemporaneous market conditions. For Euro Area future-supply news, we retain events with a clear bearing on expected gas availability, such as gas-specific sanctions, transit disputes, and import-infrastructure decisions, while excluding broader geopolitical or macroeconomic events whose gas-price effects cannot be attributed primarily to gas supply. We also assess same-day confounders beyond the mechanical filters: minor unrelated news does not disqualify a date if the gas-supply event is clearly the dominant driver of the price movement, but otherwise valid supply events are excluded when demand-side developments or other energy-market shocks appear to dominate, or materially contribute to, the gas-price movement.

[11] Scheduled releases of inflation, activity, and labor-market conditions are a potential concern because they may reveal aggregate economic information that affects both gas prices and macroeconomic outcomes through non-gas channels. Excluding release dates would substantially narrow the candidate pool, so we retain them and assess their influence. Online Appendix E.1 shows that absolute gas-price changes on release days are not significantly different from those on control days, and that excluding the four Euro Area and six U.S. overlapping events yields impulse responses similar to the baseline.

[12] We define these as days on which the WTI spot or front-month price change exceeds one standard deviation and oil and gas prices move in the same direction.

[13] Section V.C shows that the results are robust to stricter 3% and 4% thresholds, with the latter being close to a one-standard-deviation daily movement in the post-filtering sample and therefore providing a considerably more conservative event screen.

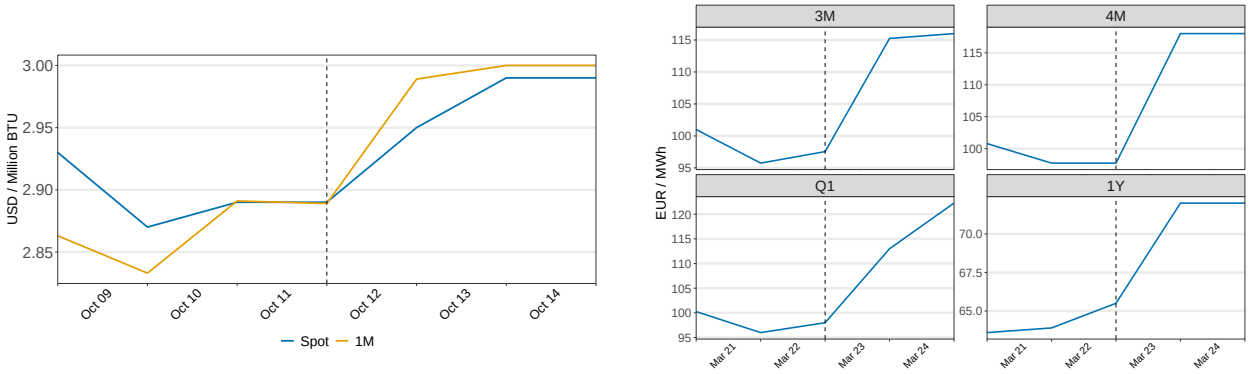
[14] Weather events are generally excluded because they may affect gas prices through gas demand as well as through direct macroeconomic disruption. We retain only U.S. offshore storm events that do not make landfall, where the relevant news concerns offshore production shut-ins rather than direct onshore economic disruption.

To make event selection transparent and assess its sensitivity to researcher judgment, we document each candidate event and the reason for its inclusion or exclusion in an event ledger. To probe that sensitivity, we apply a large language model to the same Factiva material, using instructions based on the selection criteria above. We then reconstruct the instruments using three event sets: baseline events also selected by the classifier, baseline events supplemented with classifier-selected events, and classifier-selected events alone. The resulting impulse responses remain similar to the baseline. Online Appendix C.2 details the event ledger and the LLM-based sensitivity checks.

Event characterization. The above procedure identifies 25 clean events for the United States and 63 for the Euro Area. The Euro Area set comprises 28 realized or near-term flow disruptions and 35 broader supply-security news events about expected gas availability. The full list of retained events is reported in Table I. In the United States, retained events concern domestic production and infrastructure, consistent with the market's reliance on domestic supply. These include pipeline and transmission disruptions, facility outages, storage events, Gulf of Mexico production shutdowns, and regulatory or restoration events. Events affecting LNG export facilities are classified by their effect on domestic gas availability, since export disruptions leave more gas in the U.S. market. For the Euro Area, flow events are instead centered on import-relevant supply disruptions, including outages at gas fields, processing plants, pipelines, and import terminals, as well as unexpected changes in Russian and Norwegian flows. Future-supply news includes sanctions, transit and payment disputes, LNG contracts and cargo competition, infrastructure and storage decisions, reverse-flow capacity, and supply-diversification agreements relevant for future gas availability. We do not define a separate future-supply-news category for the United States: unlike the Euro Area, the U.S. market is largely self-sufficient, and retained U.S. events are naturally classified by their effect on domestic gas availability rather than by supply-security expectations.

Figure II provides examples of the event-day price movements used to construct the supply instruments. The left panel shows a U.S. flow-supply event: the October 12, 2017 Texas Eastern Transmission force majeure, which sharply restricted Appalachian gas flows and was followed by a clear increase in Henry Hub spot and front-month prices over the daily event window. The right panel shows a Euro Area supply-security news event: the March 23, 2022 announcement that Russia would require payments for gas in rubles. This announcement did not immediately curtail physical flows, but it sharply repriced longer-dated TTF futures, consistent with a revision in expected gas availability and supply security. A Euro Area flow-supply event is shown in Online Appendix Figure C.4.

Figure II: *Illustrative price movements around gas supply events*



Notes: The left panel shows Henry Hub spot and front-month prices around the October 12, 2017 Texas Eastern Transmission force majeure, a U.S. flow-supply event. The right panel shows longer-dated TTF futures prices around the March 23, 2022 ruble-payment announcement, a Euro Area supply-security news event. The dashed vertical line marks the beginning of the price-relevant event day. Event-window surprises are measured from the previous trading-day close to the close of that day.

Construction of gas supply instruments. Using the retained gas-supply events, we construct event-day price surprises as the log change in natural gas prices from the previous trading-day close to the event-day close. For each event date d and contract maturity h, the surprise is

$$\Delta Gas_d^h = P_d^h - P_{d-1}^h, \tag{2}$$

where P_d^h denotes the log price of the gas contract with maturity h on date d, with $h = 0$ corresponding to the spot price. Following Känzig (2021), we use a daily event window. Because gas-supply news is dispersed across multiple sources and is not tied to a single announcement calendar, an intraday window is difficult to define consistently and may fail to capture the full market reaction. A multi-day window, by contrast, would increase exposure to unrelated market developments. The daily window therefore balances the need to capture the price response to the event with the need to limit background noise.

We construct the flow-supply instruments from spot and front-month prices, because these maturities are most sensitive to disruptions in current or near-term physical availability and align with the monthly timing of the empirical analysis. For the Euro Area supply-security news instrument, we instead use longer-dated TTF futures, up to one year ahead, since these events shift expectations about future gas availability rather than contemporaneous flows. For each instrument, we extract the common component across maturities using the first principal component of the corresponding event-day surprises. The first component explains 77% of the variation in the price changes used to construct the U.S.

flow-supply instrument, 89% for the Euro Area flow-supply instrument, and 58% for the Euro Area supply-security news instrument. These shares indicate that the retained events generate meaningful common repricing across the relevant maturities, rather than only isolated movements in individual contracts. We then aggregate the resulting daily surprises to the monthly frequency by summing them within each calendar month. In months with no relevant events, the corresponding entry is set to zero. Figure III displays the resulting monthly series.[15]

III.C Diagnostics of the instrument series

We next report diagnostic checks that assess the strength and cleanliness of the instrument series. These checks compare supply-event days with control days, examine comovement with other structural shocks, construct informationally robust instruments that account for observable confounders, and test for autocorrelation and forecastability.

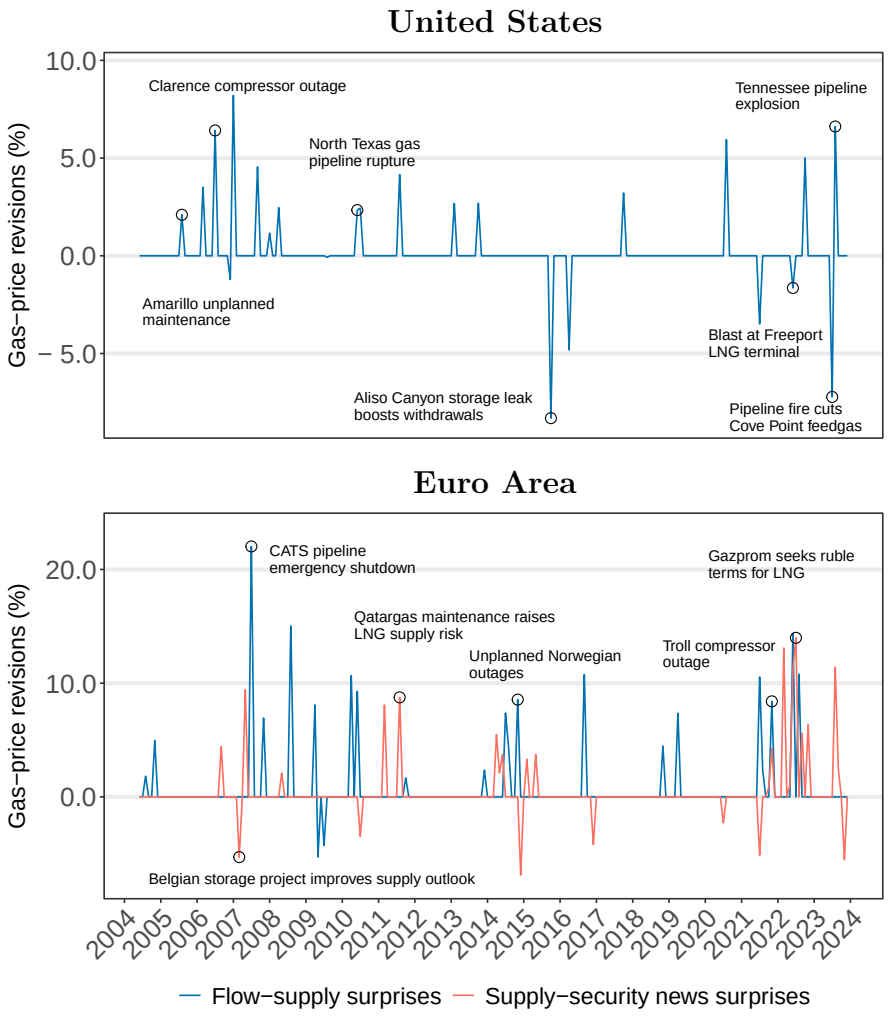
Background noise. Event-day price changes may still contain background noise unrelated to gas supply. We therefore compare daily changes in gas prices on gas-supply event days with those observed on randomly selected control days with no gas-supply events. As Figure IV shows, price changes on event days are markedly larger than in the control sample. The empirical CDFs of absolute price changes are shifted to the right on event days, indicating that gas-supply events are more likely to generate large price movements. Taken together, these results suggest that background noise is limited.

Informationally robust instruments. The control-day comparison above suggests that the retained event days contain a distinct gas-market signal. The remaining concern is whether the associated price movements also convey information about other shocks that affect macroeconomic outcomes through channels unrelated to natural gas. This concern is analogous to the information-effects problem emphasized by Nakamura and Steinsson (2018) in the monetary policy context. We therefore first examine whether the instrument series are systematically related to existing measures of other structural shocks. As reported in Table C.5 in the Online Appendix, the instruments display little systematic comovement with a broad set of external shock measures and indicators.[16]

[15] For the sectoral analysis in Section V.D, we also construct a combined Euro Area supply instrument that pools the retained flow-supply and supply-security events and extracts the first principal component from TTF price surprises across all maturities from spot to one year ahead.

[16] The table reports correlations with oil supply, aggregate demand, and oil-specific demand shocks from Kilian (2009); oil supply and oil demand shocks from Baumeister and Hamilton (2019); oil supply expectations from Känzig (2021); the carbon-policy shock from Känzig (2023); monetary policy surprises for the Euro Area and the United States from Altavilla et al. (2019) and Bauer and Swanson (2023), respectively;

Figure III: *Gas supply surprise series*

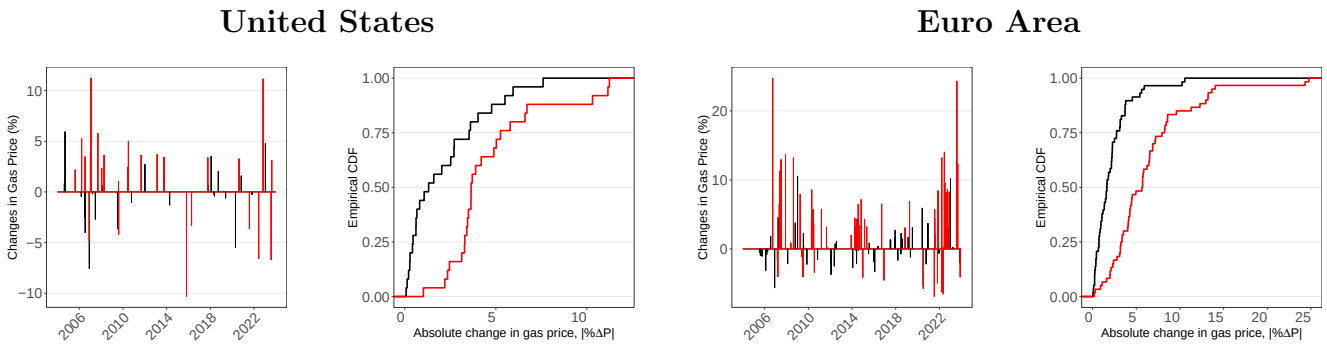


Notes: The top panel reports the U.S. flow-supply surprise series. The bottom panel reports Euro Area flow-supply surprises in blue and supply-security news surprises in red. Each series is constructed as the first principal component of event-day gas-price changes and scaled to match the average volatility of the underlying price surprises, so that the y-axis can be interpreted as percentage gas-price revisions. Black circles highlight selected gas-supply events.

To go beyond simple correlations, we construct robust versions of the instruments by residualizing them with respect to observable confounders. For the supply-side instruments, event-window gas-price surprises may also reflect contemporaneous movements in related commodity markets, especially food and crude oil prices. Food prices rose alongside natural gas prices during the Russia–Ukraine conflict (FAO 2022; World Bank 2022), while crude oil prices are closely linked to natural gas and have broader macroeconomic effects. More

financial uncertainty and credit-spread indicators from Bloom (2009) and Gilchrist and Zakrajšek (2012); and geopolitical risk from Caldara and Iacoviello (2022).

Figure IV: *Price changes on supply-event days (red) and control days (black)*



Notes: Left panels within each region: daily changes in the one-month gas futures price on supply-event and control days. Right panels within each region: empirical cumulative distribution functions of absolute percentage daily price changes. A one-sided Kolmogorov–Smirnov test rejects equality of the distributions in favor of larger event-day absolute price changes in both the Euro Area ($D = 0.69$, $p < 0.001$) and the United States ($D = 0.68$, $p < 0.001$). A permutation test based on mean absolute price changes with 10,000 draws also rejects the null that event-day price movements are no larger than control-day movements in both regions ($p < 0.001$).

generally, the residualization ensures that the gas-market events underlying the surprise series do not primarily capture broader shocks operating through channels other than current or expected natural gas availability.

To this end, we adapt the standard approach from the monetary policy literature that isolates asset-price surprises from their informational component by controlling directly for observable confounders (e.g., Romer and Romer 2004; Nakamura and Steinsson 2018; Miranda-Agrippino and Ricco 2021). We proceed in two steps. First, at the daily frequency, we construct food and oil surprise series on the same gas-event dates. For each commodity, we compute event-day futures-price surprises over the corresponding maturities and extract their first principal component, mirroring the construction of the gas-price surprise.[17] We then residualize the daily gas instrument with respect to contemporaneous food and oil surprises:

$$Gas_d = \alpha_0 + \phi\, Food_d + \theta\, Oil_d + GasClean_d$$

where Gas_d denotes the daily gas surprise instrument, and $Food_d$ and Oil_d are the corresponding daily food and oil surprise series constructed around the same gas-related events.

[17] To construct food surprises, we use wheat prices as a proxy for broader food-price movements, since wheat is central to the Russia–Ukraine commodity-price channel (OECD 2022) and is among the most actively traded food commodities (CME Group 2024). We use Matif wheat futures for the Euro Area and Hard Red Winter wheat futures for the United States.

Second, at the monthly frequency, we residualize each instrument series with respect to its own lags and contemporaneous and lagged observable confounders. This step extends the static correlation checks above by allowing for dynamic confounding. Let $\tilde{z}_t$ denote the monthly instrument being residualized. The informationally robust instrument, IRS_t, is then defined as the residual from the following monthly regression:

$$\tilde{z}_t = \alpha_0 + \sum_{j=1}^{l} \phi_j \tilde{z}_{t-j} + \sum_{j=0}^{l} \mathbf{x}_{t-j}\Gamma_j + IRS_t$$

where $\mathbf{x}_t$ collects monthly energy-market, macroeconomic, financial, monetary-policy, and power-market controls that may confound the interpretation of the instrument, together with the other gas-market instruments. The monthly residualization is applied to all instruments, including the demand instruments. The control set includes the shock measures used in the correlation checks that are available over the same sample, augmented with weather controls for renewable electricity generation.[18] The lag order is set to $l = 3$. As shown in Online Appendix Figure E.2, the impulse responses obtained using the informationally robust instruments are very similar to those based on the baseline instruments, suggesting that the main results are not materially driven by observable confounding factors.

Other diagnostic checks. Online Appendix C.3 reports additional diagnostic tests on the instrument series, including tests for autocorrelation and Granger causality. The series display little evidence of serial dependence. We also find no evidence that gas-specific, macroeconomic, or financial variables help forecast the supply series, as none of the variables considered Granger-causes the instruments at conventional significance levels. Taken together, these results provide further support for the validity of the instruments.

IV Econometric framework

Consider the following structural VAR(p) model:

$$\mathbf{B}_0\mathbf{y}_t = \mathbf{B}_1\mathbf{y}_{t-1} + \cdots + \mathbf{B}_p\mathbf{y}_{t-p} + \mathbf{w}_t, \tag{3}$$

where $\mathbf{y}_t$ is a $K \times 1$ vector of endogenous variables, assumed to have zero mean without loss of generality. The vector $\mathbf{w}_t$ is a $K \times 1$ vector of structural shocks. Following the standard definition of structural shocks as primitive exogenous disturbances, we assume that they are mutually uncorrelated, so that $\mathbb{E}(\mathbf{w}_t\mathbf{w}_t') = \mathbf{\Sigma}_w$ is diagonal (Ramey 2016, Section 2.1). Since

[18] The additional controls are maximum wind gust, precipitation, solar radiation, and drought conditions, as described in Online Appendix Table B.1.

the matrices $\mathbf{B}_0$ and $\mathbf{w}_t$ are generally unobserved, we rely on the reduced-form representation to estimate the model:

$$\begin{aligned}\mathbf{y}_t &= \mathbf{B}_0^{-1}\mathbf{B}_1\mathbf{y}_{t-1} + \cdots + \mathbf{B}_0^{-1}\mathbf{B}_p\mathbf{y}_{t-p} + \mathbf{B}_0^{-1}\mathbf{w}_t \\ &= \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{u}_t,\end{aligned}$$

where $\mathbf{A}_i = \mathbf{B}_0^{-1}\mathbf{B}_i$ for $i = 1, \ldots, p$, and $\mathbf{u}_t$ denotes the reduced-form innovation, with $\mathbf{\Sigma}_u = \mathbb{E}(\mathbf{u}_t\mathbf{u}_t')$. We estimate the reduced-form VAR parameters and residuals using Bayesian methods, implementing standard Minnesota, sum-of-coefficients, and dummy-initial-observations priors, with their hyperparameters selected using the hierarchical procedure of Giannone, Lenza, and Primiceri (2015). Details are provided in Online Appendix A.5. The key equation linking the reduced-form innovations to the structural shocks is:

$$\mathbf{u}_t = \mathbf{B}_0^{-1}\mathbf{w}_t. \tag{4}$$

Our objective is to identify the dynamic effects of the gas-market structural shocks associated with demand- and supply-side disturbances, which corresponds to recovering the relevant columns of the impact matrix $\mathbf{B}_0^{-1}$. To achieve this, we use the instruments constructed as described in Section III.

Identification via external instruments. We are interested in identifying $r < K$ structural shocks. Without loss of generality, order these shocks as the first r entries of $\mathbf{w}_t$, and write $\mathbf{w}_t = \left(\mathbf{w}_{1:r,t}', \mathbf{w}_{r+1:K,t}'\right)'$, where $\mathbf{w}_{1:r,t}$ collects the target shocks and $\mathbf{w}_{r+1:K,t}$ collects the remaining shocks. Let $\mathbf{z}_t = (z_{1,t}, \ldots, z_{r,t})'$ denote a vector of instruments. In our setting, each instrument is intended to identify one labeled target shock. Thus, for $j = 1, \ldots, r$, the relevance and exogeneity conditions are

$$\mathbb{E}[z_{j,t}w_{j,t}] = \alpha_j \neq 0, \tag{5}$$

$$\mathbb{E}[z_{j,t}w_{\ell,t}] = 0 \qquad \text{for all } \ell \neq j. \tag{6}$$

These conditions state that $z_{j,t}$ is relevant for its own target shock and orthogonal to all other structural shocks, thereby preserving the economic label supplied by the instrument. Let $\mathbf{b}_j$ denote the j-th column of $\mathbf{B}_0^{-1}$, i.e., the impact vector associated with $w_{j,t}$. Multiplying (4) by $z_{j,t}$ and taking expectations gives

$$\mathbb{E}[z_{j,t}\mathbf{u}_t] = \alpha_j\mathbf{b}_j, \qquad j = 1, \ldots, r. \tag{7}$$

Each instrument therefore identifies its labeled impact vector up to sign and scale.

Normalization. The scale can be set via any normalization subject to $\boldsymbol{\Sigma}_u = \mathbf{B}_0^{-1}\boldsymbol{\Sigma}_w(\mathbf{B}_0^{-1})'$. One common approach is to rescale the structural shocks to have unit variance, $\boldsymbol{\Sigma}_{\tilde{w}} = \mathbf{I}_K$. Under this normalization, a unit realization of $\tilde{w}_{j,t}$ corresponds to a one-standard-deviation structural shock, and its contemporaneous effects on the variables in $\mathbf{y}_t$ are given by the j-th column of $\tilde{\mathbf{B}}_0^{-1}$. This leads to the unit-variance representation of the model:

$$\tilde{\mathbf{w}}_t = \boldsymbol{\Sigma}_w^{-1/2}\mathbf{w}_t \tag{8}$$

$$\tilde{\mathbf{B}}_0^{-1} = \mathbf{B}_0^{-1}\boldsymbol{\Sigma}_w^{1/2} \tag{9}$$

$$\mathbf{y}_t = \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \tilde{\mathbf{B}}_0^{-1}\tilde{\mathbf{w}}_t \tag{10}$$

Alternatively, one can fix the scale of each identified impact vector by anchoring it to the contemporaneous response of a chosen variable. Let g denote the position of the real price of natural gas in $\mathbf{y}_t$, and partition $\mathbf{b}_j = (b_{g,j}, \mathbf{b}'_{-g,j})'$ and $\mathbf{u}_t = (u_{g,t}, \mathbf{u}'_{-g,t})'$. Then (7) implies

$$\mathbf{b}_{-g,j} = b_{g,j}\frac{\mathbb{E}[z_{j,t}\mathbf{u}_{-g,t}]}{\mathbb{E}[z_{j,t}u_{g,t}]}, \qquad j = 1, \ldots, r.$$

In the impulse responses reported in Section V, we set $b_{g,j} = c$, with c chosen so that each identified shock raises the real price of natural gas by 10% on impact. The empirical impact vectors are obtained by replacing the population moments in the expression above with their sample analogues.

Orthogonalization. The proxy moment conditions (5)–(6) identify the target impact vectors separately. Since these are population restrictions, the corresponding estimated vectors need not form an exactly orthogonal structural system in finite samples. In particular, if $\mathbf{S}$ collects the initially identified impact vectors, the sample analogue of $\mathbf{S}'\boldsymbol{\Sigma}_u^{-1}\mathbf{S}$ need not be diagonal. This matters because recovering structural shocks, and constructing objects such as historical decompositions and forecast error variance decompositions, requires a mutually orthogonal system of shocks. We therefore apply a polar-whitening transformation to the initially identified impact vectors. Specifically, let $\mathbf{G} = \mathbf{S}'\boldsymbol{\Sigma}_u^{-1}\mathbf{S}$ and set $\mathbf{S}^{\perp} = \mathbf{S}\mathbf{G}^{-1/2}$, so that $\mathbf{S}^{\perp\prime}\boldsymbol{\Sigma}_u^{-1}\mathbf{S}^{\perp} = \mathbf{I}$. This transformation imposes exact orthogonality while minimizing the $\boldsymbol{\Sigma}_u^{-1}$-weighted distance from the initially identified system (Colombo and Ragusa 2026).[19] In Section V.C, we show that the baseline results are robust to not imposing orthogonality,

[19] Intuitively, if the instruments identify the correct shock directions and any lack of orthogonality is due to sampling variation, this transformation leaves the population target unchanged while enforcing exact orthogonality in the estimated system. We prefer this orthogonalization to alternatives that impose additional economic structure, such as recursive schemes based on an ordering of the identified shocks as in Mertens and Ravn (2013), because in our setting we do not see a compelling economic justification for such restrictions.

suggesting that the initially identified shocks are already close to orthogonal in sample. This also implies that a researcher interested in a single shock and using only its corresponding instrument would obtain similar impulse responses.

Invertibility. The external-instrument identification strategy described above assumes invertibility, in the sense that the relevant structural shocks are recoverable from the reduced-form VAR innovations. This assumption matters for the structural elasticities discussed in the next section, which are constructed from contemporaneous structural impact vectors, and for historical decompositions, which require recovering the realized sequence of structural shocks. In Section V.C, we formally test this assumption using the procedures in Plagborg-Møller and Wolf (2022) and Hamilton (forthcoming). We fail to reject invertibility in the baseline specification and also show that the main results are robust to an internal-instrument approach that does not rely on this assumption.

IV.A Estimating demand and supply elasticities

The physical accounting identity for the regional natural gas market is

$$DP_t + IM_t - EX_t = C_t + \Delta I_t, \tag{11}$$

where DP_t is domestic production, IM_t imports, EX_t exports, C_t current domestic gas use, and $\Delta I_t = I_t - I_{t-1}$ net injections into inventories.[20] The left-hand side is total gas available to the regional market, while the right-hand side allocates this gas between current flow demand and storage demand. Together with the price of natural gas at the relevant wholesale hub, these balances summarize the aggregate gas market in each region.

We therefore define total available supply as $Q_t^s := DP_t + IM_t - EX_t$. In the Euro Area, this measure is largely driven by imports; in the United States, it mostly reflects domestic production, with net trade flows also contributing. Because the short-run supply elasticity describes the contemporaneous response of Q_t^s, we estimate it using the flow-supply instrument, which shifts current physical availability. Supply-security news shocks instead capture revisions in expected future supply conditions and are therefore used below in the impulse-response analysis. On the demand side, we define current flow demand as $Q_t^d := C_t$, keeping inventory changes separate because they reflect intertemporal storage decisions rather than final contemporaneous gas use. This distinction is consistent with models of storable commodities, which treat inventory accumulation as an intertemporal demand margin distinct from current consumption (Kilian and Murphy 2014; Knittel and

[20] In official energy balances, this identity may also include distribution losses and statistical discrepancies. We abstract from these residual components, which are small relative to the main flow variables.

Pindyck 2016; Baumeister and Hamilton 2019). Our demand-side instrument, interpreted as an unexpected temperature innovation, provides exogenous variation in this flow-demand component.

To estimate the supply and demand elasticities we consider dynamic demand and supply equations of the form:

$$q_t^d = \alpha_{qp}^d p_t + \mathbf{x}_t' \boldsymbol{\beta}_d + w_t^d, \tag{12}$$

$$q_t^s = \alpha_{qp}^s p_t + \mathbf{x}_t' \boldsymbol{\beta}_s + w_t^s, \tag{13}$$

where p_t denotes the logarithm of the real price of natural gas, while q_t^d and q_t^s denote the logarithms of current gas use and total available supply, respectively. The parameters α_{qp}^d and α_{qp}^s denote the price elasticities of demand and supply. The terms w_t^d and w_t^s are the structural shocks to demand and supply, and $\mathbf{x}_t$ is a vector of control variables that includes a constant, contemporaneous values of other endogenous variables, as well as lags of all endogenous variables in the system. We are interested in estimating the parameters α_{qp}^d and α_{qp}^s.

These structural elasticities are recovered from the relevant rows of the contemporaneous structural matrix $\mathbf{B}_0$.[21] Even when $\mathbf{B}_0^{-1}$ is only partially identified, the relevant rows of $\mathbf{B}_0$ can be recovered from the corresponding identified columns of $\mathbf{B}_0^{-1}$. To see this, consider the unit-variance representation in equations (8)–(10). Since $\boldsymbol{\Sigma}_u = \tilde{\mathbf{B}}_0^{-1}(\tilde{\mathbf{B}}_0^{-1})'$, it follows that

$$\tilde{\mathbf{B}}_0 = (\tilde{\mathbf{B}}_0^{-1})' \boldsymbol{\Sigma}_u^{-1}. \tag{14}$$

Thus, if the j-th column of $\tilde{\mathbf{B}}_0^{-1}$ is identified, the corresponding j-th row of $\tilde{\mathbf{B}}_0$ is $\tilde{\mathbf{b}}_{0,j}' = \left[(\tilde{\mathbf{B}}_0^{-1})_{:,j}\right]' \boldsymbol{\Sigma}_u^{-1}$. Since this row is identified only up to sign and scale, we normalize it so that the coefficient on the relevant quantity is equal to unity. Because the structural equations are written with price on the right-hand side, the elasticity α_{qp} is then given by the negative of the normalized coefficient on price, corresponding to the percentage change in quantity associated with a 1% increase in price, consistent with the standard convention.

IV.B Empirical specification

We estimate monthly Bayesian VARs separately for the United States and the Euro Area over the 20-year period from January 2004 to December 2023.[22] Our baseline specification includes seven variables: the real price of natural gas, total available gas supply, gas con-

[21] The impact matrix $\mathbf{B}_0^{-1}$ describes the joint contemporaneous responses used for impulse response functions. Structural elasticities, instead, are *ceteris paribus* objects and correspond to rows of $\mathbf{B}_0$; see, e.g., Baumeister and Hamilton (2024a, Sec. 2.6).

sumption, gas inventories, the headline consumer price index, industrial production, and the real price of crude oil. This specification combines the key price and quantity variables needed to characterize the natural gas market with aggregate macroeconomic indicators used to assess broader effects of gas price shocks. Including the real price of crude oil also helps control for developments in related energy markets that may otherwise confound the effects of natural gas shocks. To study additional transmission channels, we augment the baseline VAR with variables or small blocks of variables tailored to each exercise, following a common practice in the VAR literature (Beaudry and Portier 2014; Gertler and Karadi 2015; Känzig 2021; Miranda-Agrippino, Nenova, and Rey 2026). When an additional series is not available over the full baseline period, we estimate the corresponding augmented specification over the largest common sample. We estimate the VARs in log levels, so impulse responses can be interpreted as percentage changes. A detailed description of the data and sources is provided in Online Appendix B.1. We include six lags and only a constant term among the deterministic components; Section V.C shows that the results are robust to these choices.

V Results

This section presents the empirical results for the Euro Area and the United States. We begin by estimating the short-run price elasticities of gas demand and supply, and then use forecast error variance decompositions (FEVDs) and historical decompositions to assess how demand- and supply-side forces shape gas price fluctuations. We next examine the dynamic effects of gas price shocks, focusing on macroeconomic responses, transmission channels, and robustness of the baseline results. We then extend the analysis for the Euro Area to study sectoral effects on prices and quantities and evaluate the extent to which gas price shocks generate aggregate real effects. Finally, we quantify their contribution to the post-pandemic inflation surge.

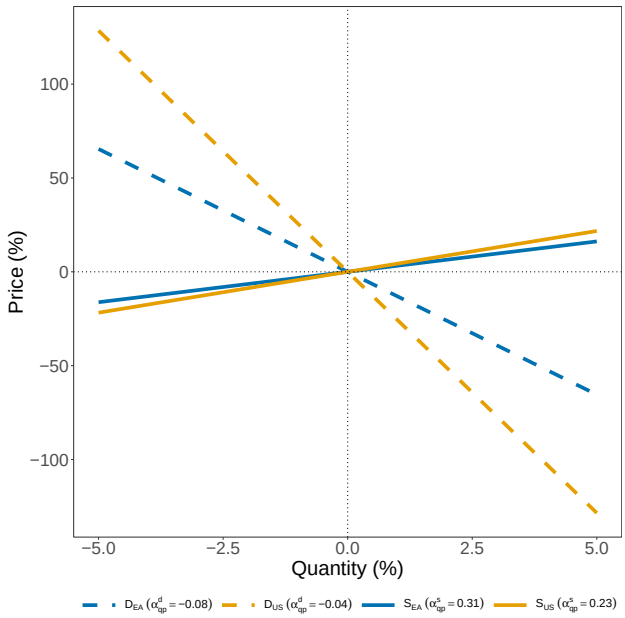
V.A Gas demand and supply elasticities and price volatility

Using the strategy described in Section IV.A, we estimate short-run price elasticities of natural gas demand and supply. For the Euro Area, the demand elasticity is -0.08 and the supply elasticity is 0.31; for the United States, the corresponding estimates are -0.04 and 0.23. Table D.1 in the Online Appendix reports previous estimates based on autoregressive models, several of which adapt the Baumeister–Hamilton framework (Baumeister

[22] Our estimation sample begins in January 2004, the earliest date for which TTF data are available. To ensure comparability, we use the same sample period for the United States specification, although the limiting variable there is gas consumption, which begins in January 2001. As shown in Online Appendix E.3, extending the U.S. sample does not materially affect the results.

and Hamilton 2015; Baumeister and Hamilton 2019). Our estimates rely on instrumented variation and are broadly consistent with this literature, though our demand elasticities are lower in absolute value, while our supply elasticities lie toward the upper end of previous estimates. Figure V illustrates the implied demand and supply curves.

Figure V: *Short-run gas demand and supply curves*



Notes: The figure shows the implied short-run gas demand and supply curves for the Euro Area (blue) and the United States (orange), using the quantity definitions in Section IV.A. Dashed lines denote demand curves; solid lines denote supply curves. Curve slopes reflect the inverse of the estimated elasticities. For illustration, supply and demand quantities are equated; this normalization does not affect the estimated slopes.

Gas demand is less price elastic than gas supply in both regions. This asymmetry implies that, for a given shift in market quantities, supply-side shocks generate larger price movements because adjustment occurs along a relatively inelastic demand curve. The forecast error variance decomposition in Table II is consistent with this implication: gas supply shocks account for a larger share of gas price fluctuations than demand shocks, especially at short horizons. Their contribution declines as the horizon lengthens, but supply shocks remain the larger source of variation. Across regions, supply is more elastic in the Euro Area, consistent with faster adjustment of imports relative to domestic production in the United States. Demand is relatively inelastic in both regions, with the U.S. estimate closer to zero; the FEVDs show that even small elasticity differences can matter for price dynamics when demand is inelastic (Caldara, Cavallo, and Iacoviello 2019). The cross-region difference in demand elasticities is also consistent with sectoral composition: residential gas use accounts

for a larger share of consumption in the Euro Area (see Online Appendix Figure B.3), and existing estimates generally find household gas demand to be more price responsive than industrial gas use (Labandeira, Labeaga, and López-Otero 2017).

Table II: *Forecast error variance decomposition of the real gas price*

Region	Shock	$h = 1$	$h = 6$	$h = 12$	$h = 18$	$h = 24$
United States	Gas demand	11.9	17.2	17.3	17.9	19.0
		[8.3, 15.7]	[11.6, 23.0]	[10.6, 24.0]	[11.4, 24.5]	[11.9, 25.9]
	Gas supply	73.6	41.6	34.5	33.0	31.6
		[63.4, 83.8]	[33.5, 49.5]	[26.0, 42.8]	[24.9, 41.2]	[23.6, 39.4]
Euro Area	Gas demand	15.1	16.2	21.6	23.5	24.0
		[10.9, 19.6]	[11.0, 21.3]	[13.2, 30.0]	[13.2, 33.8]	[13.2, 34.8]
	Gas supply	54.4	35.2	30.7	29.6	28.9
		[42.5, 66.5]	[25.9, 44.6]	[20.0, 41.5]	[18.2, 40.7]	[17.4, 40.2]

Notes: Posterior mean share of the forecast error variance of the real price of natural gas, in percent, explained by gas demand and gas supply shocks at horizons of h months, with 68% credible bands in brackets. The remaining share is due to shocks that are not identified in the model.

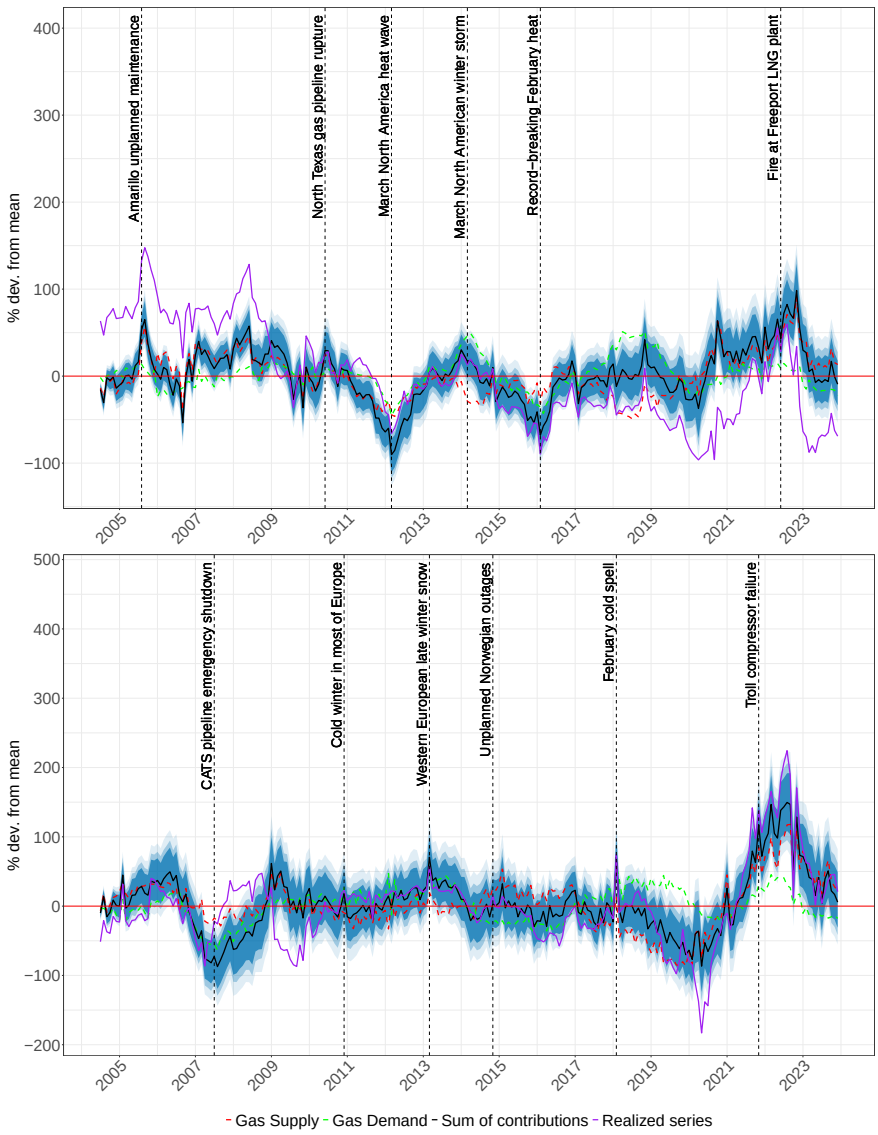
Figure VI presents the cumulative historical contributions of gas flow-supply and demand shocks to the real price of natural gas in the United States and the Euro Area, alongside the observed series. The identified shocks account for a substantial share of the observed price dynamics, with the combined contributions tracking several major episodes, including part of the 2021–2022 surge in European gas prices. Consistent with the FEVD results, gas flow-supply shocks account for a larger share of gas price fluctuations than demand shocks, although demand shocks contribute meaningfully in specific weather-related episodes.

V.B Impulse-response analysis

We now present the dynamic effects of natural gas shocks. Figure VII reports impulse responses from the baseline specification for the United States and the Euro Area. For the Euro Area, it also includes the supply-security news shock introduced in Section III.B. All shocks are normalized to raise the real spot price of natural gas by 10% on impact.[23] Median responses are shown by black lines; blue shaded credible bands refer to the Euro Area and orange shaded credible bands to the United States, at the 68%, 80%, and 90% levels.

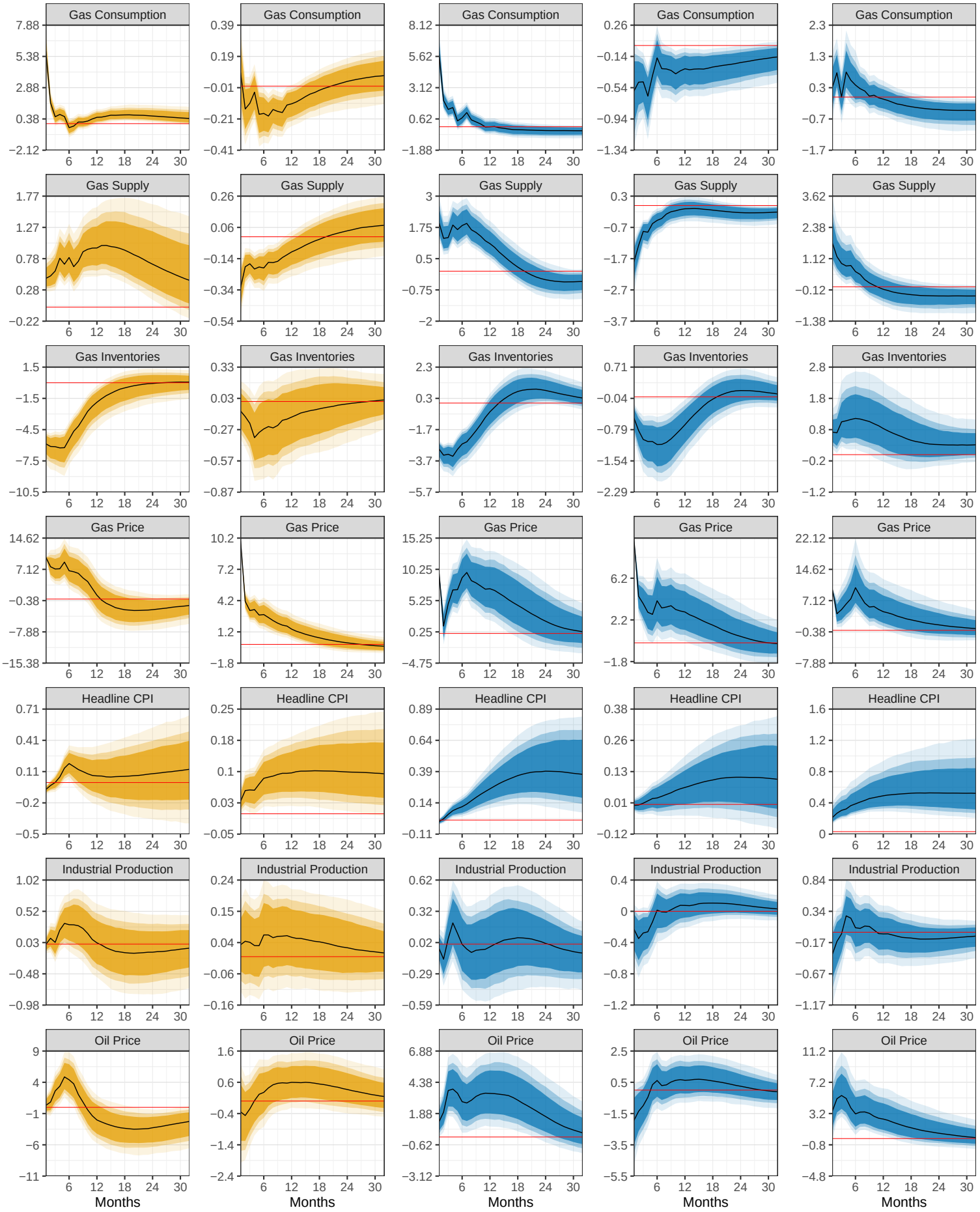
[23]This normalization is close to the size of a typical one-standard-deviation shock. Under a one-standard-deviation shock normalization, the impact responses of the real gas price are 5.8% and 14.7% for U.S. demand and flow-supply shocks, and 8.5%, 15.1%, and 8.0% for Euro Area demand, flow-supply, and supply-security news shocks, respectively.

Figure VI: *Historical decomposition of the real price of natural gas in the United States (top panel) and the Euro Area (bottom panel)*



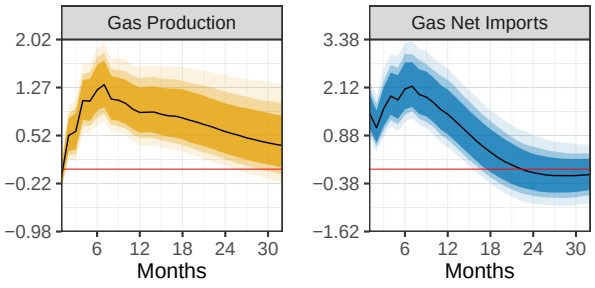
Notes: Estimated contributions of gas flow-supply shocks are shown by the red dashed line, gas demand shocks by the green dashed line, and their combined contribution by the solid black line with blue credible bands. All contributions are expressed as percentage deviations from the mean. The realized real gas price series is shown in purple. In the United States, vertical dashed lines mark unplanned maintenance on the Amarillo Mainline that reduced pipeline capacity in 2005M8; a rupture and explosion on Enterprise's North Texas gas pipeline in 2010M6; a heat wave across North America in 2012M3; a North American winter storm in 2014M3; record-breaking February warmth in 2016M2; and a blast at the Freeport LNG terminal that disrupted operations in 2022M6. In the Euro Area, vertical dashed lines mark the unexpected shutdown of the CATS pipeline due to anchor damage in 2007M7; an abnormally cold winter in Europe in 2010M12; a late-winter snow event in Western Europe in 2013M3; unplanned Norwegian outages in 2014M11; a February cold spell in 2018M2; and a compressor failure at Norway's Troll field in 2021M11.

Figure VII: *Responses to natural gas shocks: United States (orange); Euro Area (blue)*



Responses of gas quantities. We first examine the quantity margins underlying the normalized gas-price responses. Figure VII shows that demand shocks require substantially larger quantity adjustments than supply shocks to generate the same 10% increase in gas prices. In both regions, gas consumption rises by about 6% on impact following a demand shock. By contrast, flow-supply shocks generate the same price increase with much smaller movements in physical quantities, consistent with the short-run inelasticity of gas demand documented above. Following a demand shock, supply adjusts more gradually in the United States, where the response builds over subsequent months, than in the Euro Area, where imports allow for a larger immediate response. Figure VIII reinforces this interpretation: replacing total supply with domestic production for the United States and with net imports for the Euro Area yields similar dynamics. Following a flow-supply shock, gas consumption declines in both regions, but the adjustment remains modest relative to the 10% increase in gas prices and reverses only gradually, consistent with the short-run inelasticity of demand. The decline is larger in the Euro Area, in line with the higher estimated demand elasticity reported above.

Figure VIII: *Supply-component responses to gas demand shocks*



Notes: Impulse responses of U.S. gas production and Euro Area net gas imports following a gas demand shock. The responses are estimated from the baseline specification after replacing total gas supply with the corresponding supply component in each region.

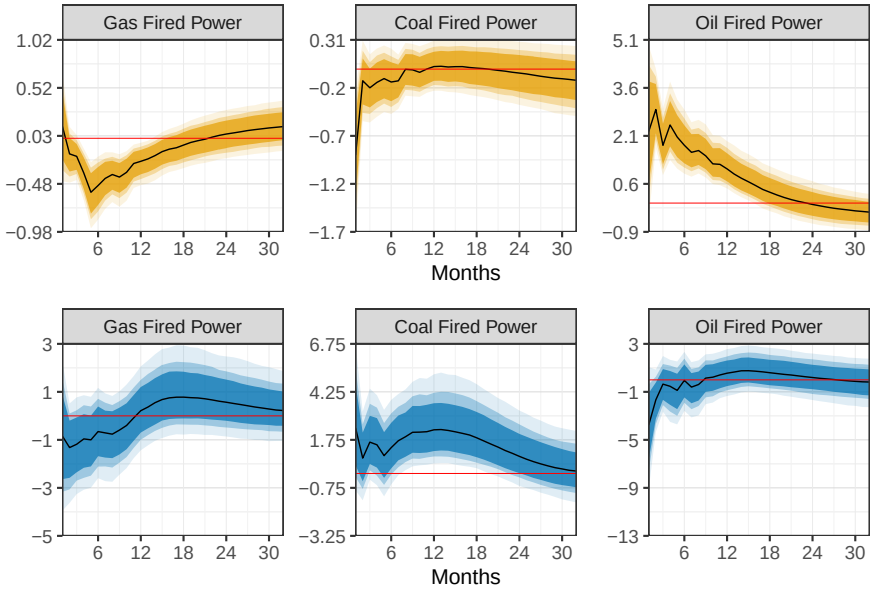
Sectoral composition helps explain the cross-region pattern. As discussed in Section II, gas consumption in the Euro Area has a larger residential component, and existing estimates generally find residential gas demand to be more price responsive than industrial gas use.[24] This is consistent with the larger absolute demand elasticity estimated above.

A second adjustment margin operates through the power sector. Because power generation accounts for a large share of gas use, substitution from gas-fired generation toward oil-

[24]See Asche, Nilsen, and Tveteras (2008), Serletis, Timilsina, and Vasetsky (2010), Andersen, Nilsen, and Tveteras (2011), and Rubin and Auffhammer (2024) for estimates of gas demand elasticities across sectors and regions.

or coal-fired generation can contribute to the short-run decline in aggregate gas consumption after flow-supply shocks. To examine this margin, we estimate alternative specifications that augment the baseline system with fuel-specific power-generation series. The impulse response functions presented in Figure IX show substantial interfuel substitution in the United States following a flow-supply shock, with oil-fired generation showing the clearest response. In contrast, the Euro Area exhibits a more limited degree of interfuel substitution, with coal serving as the main substitute.[25]

Figure IX: *Interfuel substitution in the power sector*



Notes: Impulse responses of gas-fired, oil-fired, and coal-fired power generation to a gas flow-supply shock. Estimates are based on specifications that augment the baseline system with each fuel-specific power-generation series. The Euro Area sample begins in 2007 due to data availability constraints.

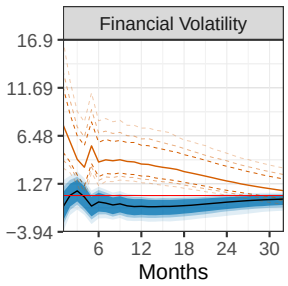
The fifth column of Figure VII provides a final contrast. The Euro Area supply-security news shock differs from flow-supply shocks because it is not tied to an immediate physical shortfall. Gas supply increases on impact, while gas consumption responds only modestly before declining gradually. These dynamics are consistent with adjustment to revised expectations about future availability rather than the direct effect of a contemporaneous disruption to physical flows. The associated inventory response is discussed next.

[25]The power generation sector in the United States is comparatively more flexible, with approximately 20% of natural gas-fired combined-cycle capacity capable of switching to alternative fuels, primarily oil (EIA 2022). In contrast, most gas turbines in Europe are not designed for dual-fuel operation, which significantly constrains short-term interfuel substitution. Coal, however, has remained a key substitute for gas: many European utilities operate mixed portfolios that include both gas- and coal-fired units, enabling them to adjust the fuel mix in response to relative price changes (Reuters 2024).

Inventories, volatility, and uncertainty. The third row of Figure VII shows that gas demand shocks lead to a large decline in inventories in both regions. The decline is larger in percentage terms in the United States; because U.S. inventories are also larger in levels, it corresponds to a larger absolute storage adjustment. Flow-supply shocks also draw down inventories in the short run, but by less than demand shocks. The response is larger in the Euro Area, consistent with storage serving as a buffer against disruptions in external supply. The difference between demand- and flow-supply inventory responses is consistent with the seasonal timing of the shocks. The relevant temperature-driven demand shocks occur in winter months, when heating demand is most important and inventories are more directly used to smooth seasonal fluctuations in consumption. Flow-supply shocks can occur throughout the year, so inventories need not be positioned to absorb them as effectively (Council of the European Union 2022).

The Euro Area supply-security news shock, which captures news about expected gas availability rather than a contemporaneous disruption to physical gas flows, points to a precautionary storage and uncertainty channel. The underlying news events need not reduce current flows immediately; instead, when they raise the perceived risk of future scarcity, firms and storage operators may increase precautionary inventory demand, putting upward pressure on current gas prices. Consistent with this interpretation, the fifth column of Figure VII shows that inventories increase after the shock. This mechanism is closely related to the evidence documented by Kilian and Murphy (2014) for the crude oil market.

Figure X: *Euro Area financial-volatility responses*



Notes: Responses of financial volatility are shown for Euro Area flow-supply shocks (black solid line with blue shaded credible bands) and Euro Area supply-security news shocks (red solid line with red dashed credible bands). Financial volatility is measured by the Euro Stoxx 50 Volatility Index (VSTOXX).

Figure X compares the response of Euro Area financial volatility to flow-supply and supply-security news shocks. Flow-supply shocks generate essentially no volatility response. By contrast, supply-security news shocks lead to a large and persistent increase in volatility. This reinforces the interpretation that these shocks operate not only through expected future

availability, but also through supply-security concerns and uncertainty.[26] By comparison, Känzig (2021) documents an increase in crude oil inventories following a negative oil supply news shock and shows that expectations respond strongly, while financial uncertainty does not. This contrast is consistent with the different nature of the news: OPEC announcements provide information about future supply policy, whereas the Euro Area gas supply-news shocks studied here reflect supply-security concerns that heighten uncertainty about future availability.

Persistence of gas-price responses. The fourth row of Figure VII illustrates the dynamics of the real spot price following shocks normalized to raise it by 10% on impact. The results indicate that price responses to demand shocks are less persistent in the United States than in the Euro Area: the U.S. demand-driven response dissipates within about a year, whereas the Euro Area response declines more gradually. This regional difference is consistent with the adjustment margins discussed above: U.S. inventories respond more strongly to demand shocks, helping absorb the shock more quickly, whereas in the Euro Area consumption and supply adjust more gradually, leaving prices elevated for longer. For flow-supply shocks, by contrast, price dynamics are broadly similar across the two regions. In both cases, the spot price declines after the initial response and gradually converges back toward baseline over the horizon.

Euro Area supply-security news shocks display a distinct pattern. The spot price rises sharply over the first few months and then declines gradually, remaining above baseline for much of the horizon. This response is consistent with the expectation channel discussed above: inventories increase and financial volatility rises, pointing to precautionary inventory accumulation amid heightened uncertainty about future gas availability. Notably, gas prices can remain elevated even as the quantity responses of gas consumption, supply, and inventories move back toward baseline.

Macroeconomic propagation. The fifth and sixth rows of Figure VII report the responses of headline CPI and industrial production. In the United States, gas demand shocks generate a modest increase in headline CPI, peaking at about 0.15–0.2%, and a temporary expansion in industrial production of roughly 0.3%. This implies a price-level pass-through of about 1.5–2% of the initial gas-price increase. U.S. flow-supply shocks also raise headline CPI, with a small but persistent pass-through close to 1%, while industrial production remains essentially unchanged. This CPI response likely reflects the pass-through of domestic

[26] Chiţu, Ferrari Minesso, and Manu (2024) document increased speculation on gas prices, particularly in Europe following the invasion of Ukraine.

gas-supply disruptions into energy costs. The temporary increase in industrial production after demand shocks is consistent with higher activity in energy-related sectors. Online Appendix Figure F.6 supports this interpretation: utilities output rises immediately, while mining, which includes gas extraction, expands more gradually. The latter response is consistent with the rise in domestic gas production documented above.

In the Euro Area, demand shocks generate a persistent consumer-price response, with an implied pass-through of around 4% after roughly two years. Since these are gas-specific demand shocks rather than aggregate demand shocks, aggregate industrial production remains broadly unchanged. Flow-supply shocks have a different profile: their price-level pass-through is similar in magnitude to the U.S. response, around 1%, but is not meaningfully different from zero, whereas industrial production falls by about 0.4% after one month and then recovers within a few months. Thus, contemporaneous supply disruptions generate short-lived real effects rather than a persistent aggregate contraction.

The Euro Area supply-security news shock is instead strongly transmitted to consumer prices, with little evidence of aggregate real effects. The implied pass-through rises from about 2% on impact to roughly 5% over the medium run, while industrial production remains close to baseline. This contrast is central to the interpretation of the shock: news about future gas availability and supply security raises current prices and consumer-price pressures, but it does not require an immediate contraction in contemporaneous gas flows or aggregate production. Consistent with the precautionary storage and uncertainty channel discussed above, the supply-news shock therefore propagates mainly through prices rather than quantities, in line with evidence that natural gas price shocks can generate second-round pass-through via inflation expectations (Boeck and Zörner 2025).

Taken together, the macro responses point to a clear regional asymmetry. Inflationary effects are larger and more persistent in the Euro Area, especially for demand and supply-security news shocks. Aggregate real effects remain limited in both regions, but they are not identical: U.S. flow-supply shocks leave industrial production essentially unchanged, whereas Euro Area flow-supply shocks generate a short-lived decline. This pattern is consistent with the structural differences discussed above: the U.S. market is anchored in domestic production and larger storage buffers, while the Euro Area is more exposed to import dependence and, for supply-news shocks, to precautionary and uncertainty channels related to future gas availability.

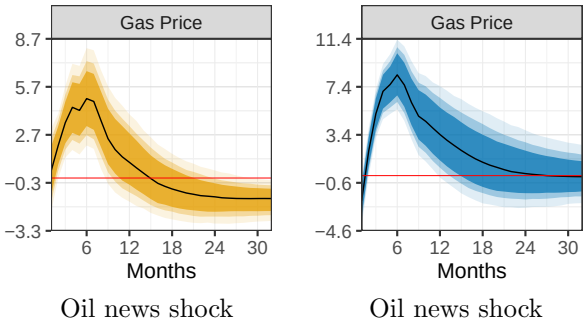
The sectoral analysis below examines whether these muted aggregate output responses conceal stronger effects in specific sectors and along the production chain. A central channel, shown in Figure XIII, is the strong pass-through from gas prices to electricity prices through marginal pricing in European power markets (Baget et al. 2024). The sectoral price responses

in Figure XIV show that these cost pressures are broader-based: producer prices rise first, cost increases are passed along the supply chain, although incompletely and with a delay, and consumer prices remain elevated even after producer-price pressures begin to ease. By contrast, output losses are concentrated in a few energy-intensive industries, while most sectors display only small responses. This pattern is consistent with firms passing higher input costs downstream while limiting output reductions where demand is relatively inelastic. Together, these results support the interpretation that Euro Area gas shocks transmit mainly through costs and prices, while aggregate real effects remain limited.

Interrelation between gas and oil markets. The last row of Figure VII shows how oil prices respond to the natural gas shocks identified in the baseline specification. Because gas and oil are linked through substitution and broader energy-market conditions, shocks to one commodity can affect the other, with transmission varying across shock types and regions.

Gas demand shocks raise oil prices in both regions, with peak responses of roughly 4–5% following a 10% increase in gas prices. By contrast, flow-supply shocks generate muted oil-price responses, consistent with the more localized nature of contemporaneous gas-supply disruptions. Euro Area supply-security news shocks also raise oil prices, suggesting that news about future gas availability can spill over to broader energy markets.

Figure XI: *Gas price responses to oil supply news shocks*



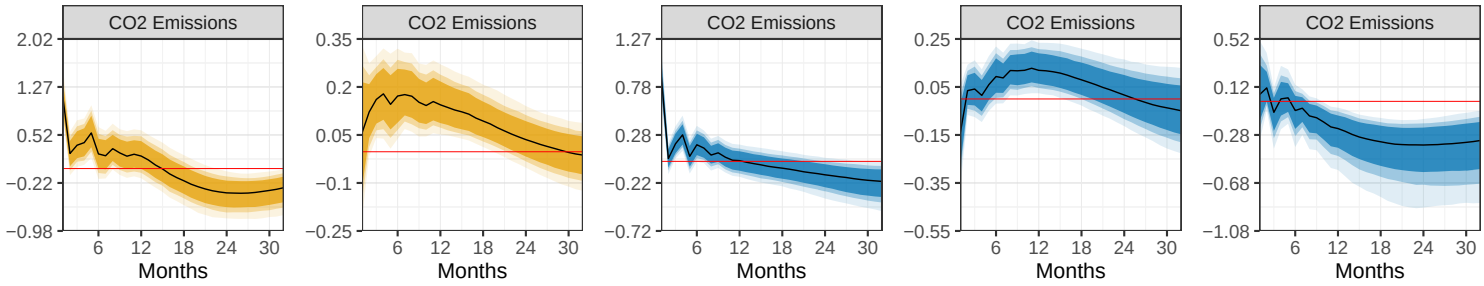
Notes: Responses of real gas prices to a 10% increase in real crude oil prices. Responses for the United States are shown in orange, while responses for the Euro Area are shown in blue. Oil supply news shocks are identified following Känzig (2021).

Figure XI complements this evidence by showing the reverse direction of pass-through: the response of gas prices to oil news shocks estimated in the same baseline specification. Oil news shocks raise gas prices in both regions, but the response is stronger and more persistent in the Euro Area than in the United States. The peak response is around one-half of the oil-price increase in the United States, compared with about 85% in the Euro Area, where the response also declines more gradually toward baseline.

This regional asymmetry is consistent with recent evidence that, as the LNG market has become more integrated, European and Asian gas prices have become more closely linked, whereas the North American market remains only partially integrated (Albrizio et al. 2023).[27] The evidence above also shows that Euro Area gas shocks linked to demand or supply-security concerns can spill over to oil prices. Previous studies have examined the relationship between crude oil and natural gas markets mainly in the U.S. context: some find no significant long-run relationship between the two prices (Bachmeier and Griffin 2006), while others focus on oil-to-gas transmission (Jadidzadeh and Serletis 2017). We add to this evidence by showing that oil-price shocks transmit to gas prices in the short run, while gas shocks can also have economically meaningful effects on oil prices.

Environmental impact of gas price shocks. The power-sector results above show that gas flow-supply shocks induce substitution away from natural gas toward more carbon-intensive fuels (see again Figure IX). To more directly evaluate the environmental consequences of gas price shocks, we draw on CO_2 emissions data from EDGAR (Emissions Database for Global Atmospheric Research). Figure XII shows the response of CO_2 emissions to natural gas price shocks.

Figure XII: *Responses of CO_2 emissions to natural gas shocks*



Notes: Impulse responses of CO_2 emissions to natural gas demand, flow-supply, and supply-security news shocks, estimated using the baseline specification augmented with the CO_2 emissions series. The first two columns report responses for the United States, while the last three columns report responses for the Euro Area. The sample period ends in 2022M12 due to data availability constraints.

A gas demand shock that raises the price of natural gas by 10% leads to an immediate increase in emissions in both the Euro Area and the United States, as more carbon-intensive fossil fuels are burned to satisfy demand. The increase is larger on impact in the United

[27] Infrastructure constraints, particularly the limited capacity of LNG export terminals, restrict the volume of gas that can be shipped abroad.

States, while in both regions the response fades over time and eventually falls below baseline. Following flow-supply shocks, emissions respond only modestly on impact but increase over the following months as higher gas prices induce substitution toward more carbon-intensive fuels; the response peaks at around 0.2% in the United States and around 0.1% in the Euro Area. The Euro Area supply-security news shock follows a different pattern: emissions do not show a clear positive impact response and then fall below baseline. Since these shocks convey information about future gas availability and can generate price increases perceived as more persistent, they are less likely to operate only through immediate interfuel substitution and may instead induce adjustment over a longer horizon.

These findings are in line with the distinction emphasized in the literature between short-run fuel substitution and longer-run adjustment. The responses to demand and flow-supply shocks mirror evidence that, in the short run, lower gas prices reduce emissions by encouraging coal-to-gas substitution (Knittel, Metaxoglou, and Trindade 2015; Cullen and Mansur 2017; Brehm 2019): when gas prices rise, the same substitution margin works in the opposite direction and emissions increase. The supply-news response is instead consistent with the longer-run channel emphasized by Stock and Zaragoza-Watkins (2024), who show that sustained increases in gas prices can promote investment in cleaner energy sources and reduce emissions over time. Taken together, these dynamics underscore the intertemporal trade-offs associated with the role of natural gas in the energy transition. As Harstad and Holtsmark (2024) argue, expanding gas production can initially reduce emissions by displacing coal through price-driven substitution. Nonetheless, such policies may also deter long-term investment in renewables, a dilemma they describe as the "gas trap" (see also Acemoglu et al. 2023). This distinction also matters for policy: a gas-specific price increase is not equivalent to a broad carbon price. When short-run gas demand is inelastic, higher gas prices may redirect energy use toward coal or oil rather than reduce fossil-fuel use overall.

V.C Robustness of the baseline results

We conclude the baseline analysis by assessing whether the main results depend on instrument construction, identifying assumptions, estimation method, lag length, sample period, or additional variables. Online Appendix E reports the corresponding figures and tables.

Instrument construction. We first consider alternative constructions of the instruments. Using the informationally robust instruments described in Section III.C, which remove observable confounding information from the instruments before estimation, yields responses close to the baseline. We then reconstruct the supply instruments using stricter event-selection thresholds, retaining only dates with absolute gas-price surprises above 3% and

4%, instead of the baseline 2% cutoff. The resulting estimates remain similar, indicating that the findings are not sensitive to the threshold choice. A leave-one-out jackknife likewise shows that no single nonzero instrument observation drives the main dynamic patterns. For the temperature-based demand instrument, we check whether the results depend on how weather-driven gas demand is measured. The responses are similar when we include warmer months, adjusting the sign of the index to account for cooling-related gas demand, or when we replace temperature deviations with heating degree days. Together, these exercises indicate that our findings are not an artifact of confounding information, a particular event sample, or the temperature measure.

Identifying assumptions. We next turn from instrument construction to the assumptions underlying identification. The external-instrument approach requires invertibility, meaning that the relevant structural gas-market shocks must be recoverable from the reduced-form VAR innovations. Following Plagborg-Møller and Wolf (2022) and Hamilton (forthcoming), we test whether lags of the instruments predict the VAR variables conditional on the baseline regressors; the joint tests provide no evidence against invertibility. As a complementary exercise, we estimate an internal-instrument specification (Ramey 2011; Plagborg-Møller and Wolf 2021), which orders the instrument first in a recursive VAR and does not require invertibility. The resulting responses remain close to the baseline. We also assess instrument relevance: the posterior distributions of the impact coefficients are well behaved and tightly concentrated (Caldara, Cavallo, and Iacoviello 2019; Giacomini, Kitagawa, and Read 2022), and frequentist first-stage diagnostics provide additional support.[28] Finally, estimating the model without orthogonalizing the identified shocks produces similar impulse responses, indicating that the results are not driven by the finite-sample orthogonalization procedure.

Estimation and specification. We then examine estimation and specification choices. Re-estimating the model by OLS and by the Bayesian local-projection approach of Ferreira, Miranda-Agrippino, and Ricco (2025), in which we adopt Minnesota-type priors, delivers similar responses, although some estimates are less precise. We also augment the VAR with electricity prices, electricity consumption, and renewable power generation to account for omitted power-market dynamics that could affect the gas-demand response estimates. The results remain close to the baseline. Lastly, we extend the lag length from six to twelve

[28]The heteroskedasticity-robust first-stage F-statistics from regressions of the gas-price reduced-form innovation on each instrument are 20.59, 16.88, and 16.22 for the Euro Area flow-supply, supply-security news, and demand instruments, and 10.01 and 25.99 for the U.S. supply and demand instruments, respectively; see Montiel Olea, Stock, and Watson (2021) on weak-instrument issues in SVARs identified with external instruments.

and, for the United States, expand the estimation sample back to 2001, the earliest date allowed by gas-consumption data. Neither modification materially alters the responses. The main results therefore do not hinge on the prior specification, the VAR representation of the dynamics, the lag structure, added power-market controls, or the common post-2004 sample.

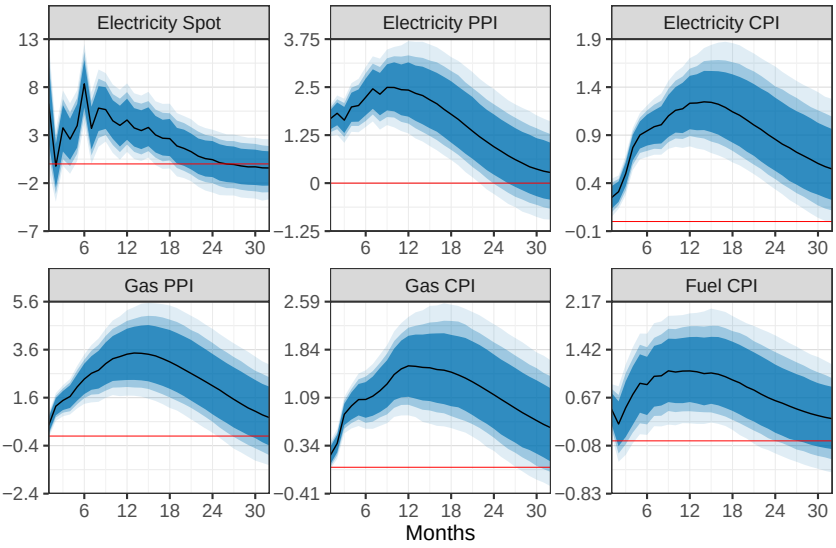
V.D Propagation of Euro Area gas supply shocks

The baseline results show that Euro Area gas supply shocks generate strong price effects but more limited real effects. In this section, we use a combined Euro Area supply instrument, pooling flow-supply events and supply-security news, to study the sectoral margins behind this aggregate response: how supply-driven gas price increases propagate across energy prices, sectoral consumer and producer prices, and real activity. This approach is useful for tracing the overall economic incidence of gas-supply risk, which is the relevant object for assessing the sectoral and policy implications of the Euro Area's exposure to supply disruptions. Consistent with the baseline responses in Figure VII, the separate sectoral responses reported in Online Appendix F.1 clarify the role of the two supply components: supply-security news shocks account for most of the price pass-through, whereas flow-supply shocks generate short-lived real effects and drive the immediate output losses.

Transmission to broader energy prices. We first examine how Euro Area gas supply shocks pass through to other energy prices. To do so, we augment the baseline specification with electricity, gas, and fuel price series, as shown in Figure XIII. Electricity spot prices rise sharply on impact and peak after about six months. This reflects merit-order pricing, under which natural gas sets the electricity price when gas-fired generation is the marginal source of power (Baget et al. 2024; Segarra, Atanasova, and Figuerola-Ferretti 2024). Producer prices for gas and electricity rise more than the corresponding consumer prices, consistent with firms' greater exposure to market rates and the protective measures available to households. Government interventions, primarily aimed at shielding consumers (Hemmerlé et al. 2023; Sgaravatti et al. 2023), mitigate the impact on household energy bills, while businesses absorb a larger share of the higher costs. Consumer prices for gas and electricity exhibit similar pass-through dynamics, with gas experiencing a slightly stronger response and peaking at about 1.5% around one year after the shock.

The broadly inverted-U shape of the producer and consumer energy-price responses highlights the gradual pass-through of energy costs, shaped by contractual rigidities in producer and retail markets (Ason 2022; Baget et al. 2024). These rigidities slow the adjustment of administered and contract-based prices, contributing to delayed peaks and a gradual return toward baseline. This pattern is consistent with the broader literature on price stickiness,

Figure XIII: *Euro Area energy-price responses to a gas supply shock*

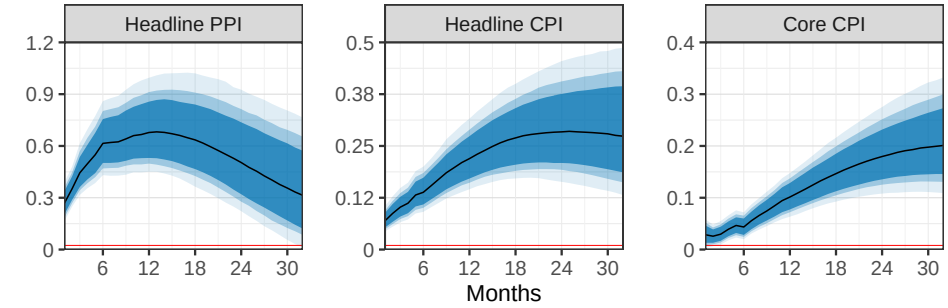


where mechanisms such as menu costs, information imperfections, and long-term agreements can slow price adjustment even after market-clearing prices have changed (e.g., Rotemberg 1982; Mankiw 1985; Borenstein and Shepard 2002). Fuel consumer prices also display a hump-shaped response, rising by around 0.5% on impact and peaking near 1%. Their relatively larger impact response reflects the closer link between retail fuel prices and wholesale energy markets, especially crude oil and refined petroleum products. This is consistent with Meyler (2009), who finds that oil price shocks in the Euro Area pass through rapidly to consumer fuel prices, with 90% of price changes transmitted within three to six weeks. Consumer gas and electricity prices adjust more gradually because of long-term contracting, regulatory frameworks, and weaker direct exposure to spot market fluctuations.

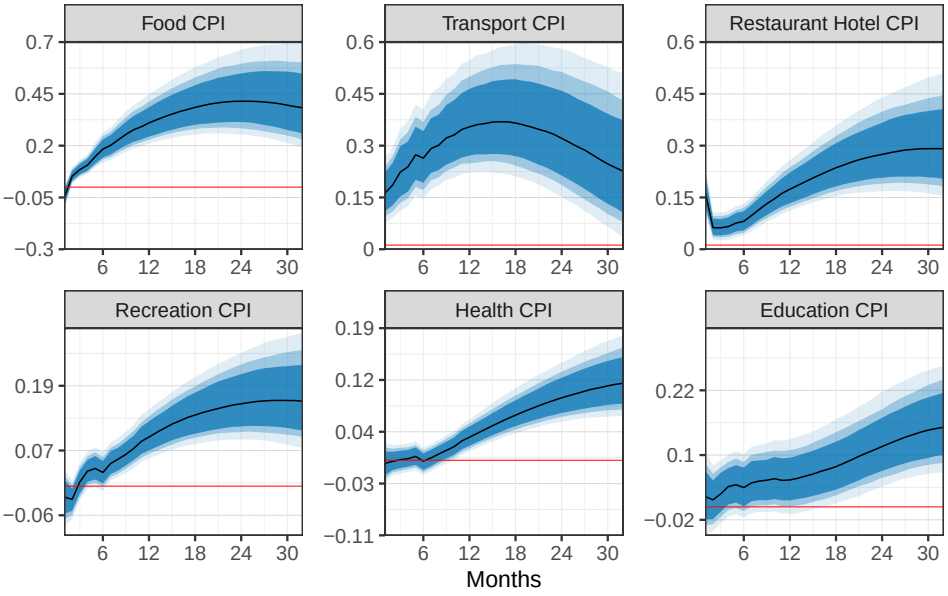
Sectoral effects on prices. Figure XIV reports aggregate consumer and producer price responses in the top panel and selected HICP consumer categories in the bottom panel. Producer prices respond earlier and more strongly than consumer prices, peaking around one year after the shock before gradually declining. Headline consumer prices rise more slowly and remain elevated, while core consumer prices show little initial response but rise gradually over the horizon. This pattern indicates that gas supply shocks propagate beyond the direct energy component, with cost pressures passing along the supply chain before reaching consumer prices.

Inflationary pressures are visible across most consumption categories, but their timing and magnitude differ substantially. Goods prices respond faster and more strongly than services. Transport reacts quickly, reflecting its direct exposure to fuel prices, while food

Figure XIV: *Euro Area price responses to a gas supply shock*



Aggregate consumer and producer prices



HICP consumer categories

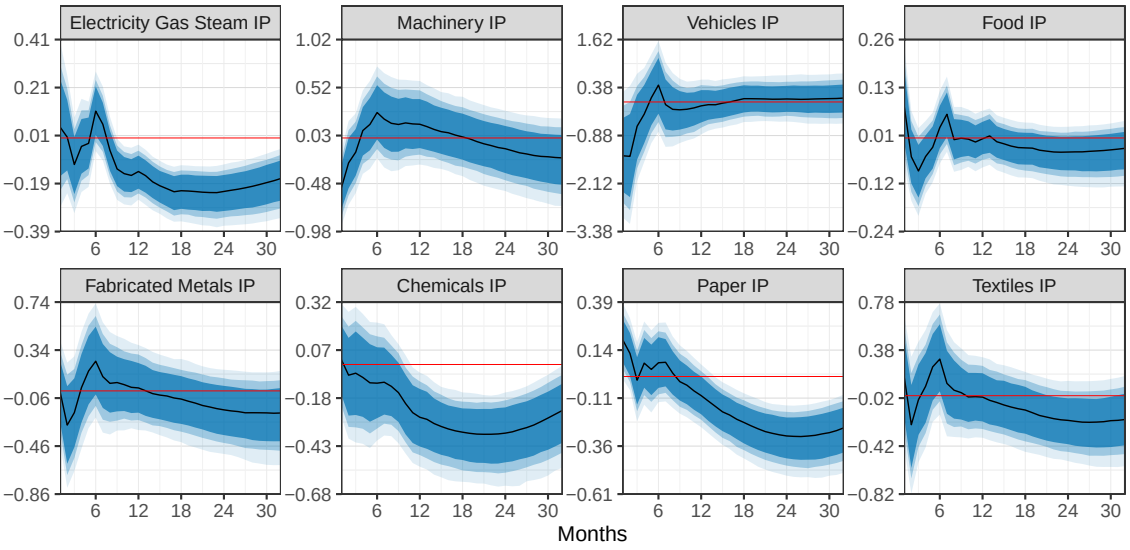
Notes: Impulse responses to a gas supply shock in the Euro Area. The top panel reports headline producer prices, headline consumer prices, and core consumer prices. The bottom panel reports six ECOICOP consumer-price categories, whose 2023 HICP weights, in per mille of the total basket, are: food (161), transport (150), restaurants and hotels (105), recreation and culture (89), health (49), and education (10); see Eurostat (2024b).

displays the largest response overall, consistent with natural gas raising input costs through fertilizer production (IEA, 2021). Together, these two divisions account for close to a third of the consumption basket, so their response is central to the aggregate consumer-price effect. Services react later and by less: restaurants and hotels rise gradually, while recreation, health, and education respond more modestly and with a longer delay. Overall, the timing of the responses is consistent with faster direct effects in energy-intensive goods categories and more gradual indirect effects where input-cost and supply-chain channels are likely to

matter.

Sectoral effects on quantities. Although the aggregate real effects documented above are limited, Figure XV shows how different subsectors of industrial production respond, ordered by value added. Machinery and vehicles display the clearest immediate declines in industrial production, whereas electricity, gas and steam, chemicals, and paper exhibit more delayed contractions.

Figure XV: *Euro Area industrial-production responses to a gas supply shock*



Notes: Sectors are ordered by value added in 2017, the latest year for which European value added is available for all sectors. Shares of total industrial value added are: electricity, gas, steam and air conditioning supply 10.0%; machinery and equipment 9.8%; motor vehicles, trailers and semi-trailers 9.5%; food products 8.6%; fabricated metal products 8.0%; chemicals and chemical products 6.1%; paper and paper products 2.1%; textiles 1.0%. Source: Eurostat (2024a).

The negative response in the electricity, gas, and steam sector, the largest in terms of value added, reflects the sector's heavy reliance on natural gas.[29] Within manufacturing, two-digit industries are affected unevenly. Chemicals and paper exhibit lagged but clear negative responses, consistent with their exposure to natural gas as a production input. In chemicals, gas is used as a feedstock for ammonia and other basic intermediates, while in paper manufacturing it is intensively used for process heat and steam generation, making substitution difficult in the short run. Other sectors, such as food and fabricated metals, show smaller or less persistent responses.

[29] See Gunnella et al. (2022) for sectoral gas intensity statistics in the Euro Area.

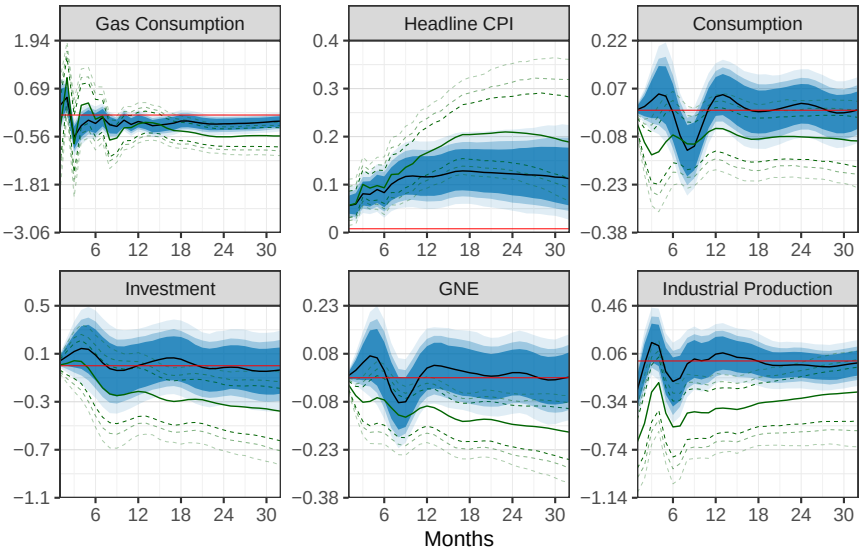
Considered alongside the sectoral HICP responses, these quantity responses are consistent with cost-push dynamics in energy-dependent sectors. Several exposed sectors display U-shaped or inverted-J output responses, with an initial decline followed by gradual recovery, while prices tend to follow an inverted-U pattern as higher production costs are passed on and then dissipate. The heterogeneity of these effects is a common feature of energy shocks (Davis and Haltiwanger 2001). At the same time, the fact that clear output losses are concentrated in a limited set of sectors suggests muted cascading effects, helping to explain the limited aggregate response of industrial production.

Real effects of gas supply shocks, Euro Area and Germany. Having shown how Euro Area gas supply shocks propagate across sectors, we now examine their aggregate real effects. We compare the Euro Area with Germany, the economy at the center of the debate on the costs of a Russian gas cutoff (Deutsche Bundesbank 2022; Bachmann et al. 2024; Di Bella et al. 2024; Albrizio et al. 2025). To this end, we augment the baseline specification with national expenditure variables, namely gross national expenditure (GNE), consumption, and investment.[30] We also estimate a Germany-specific version of the model, replacing the Euro Area macroeconomic and gas-quantity variables with their German counterparts. Figure XVI reports the corresponding impulse responses.

Following the gas supply shock, gas consumption falls more clearly in Germany than in the Euro Area, especially after the initial months. This is reflected in the estimated structural gas-demand elasticity, which is larger for Germany (-0.10) than for the Euro Area (-0.08). A larger decline in gas use, however, does not by itself reveal how costly the adjustment is. It may reflect alternative gas sourcing and other relatively low-cost substitution margins, in which case aggregate losses are limited. Adjustment can also be costly: when substitution is limited, the shock operates as a real-income loss, because agents either pay substantially more for gas or forgo valuable uses of it, and this propagates to spending and production. The real supply-side benchmark of Moll, Schularick, and Zachmann (2023) emphasizes the first type of adjustment: with small gas expenditure shares and scope for alternative sourcing or substitution, even sizable gas-import shortfalls imply limited aggregate losses and weak downstream propagation. Models with Keynesian demand amplification, such as Guerrieri et al. (2022), Auclert et al. (2023), and Pieroni (2023), instead emphasize how sectoral supply disturbances can propagate through lower real income, lower aggregate spending, and further reductions in output.

[30] As noted by Moll, Schularick, and Zachmann (2023), GNE is the welfare-relevant measure in this context, since GDP may not fully capture the terms-of-trade effects. However, the results for national expenditure should be interpreted with caution, as the data are originally available at a quarterly frequency and interpolated to monthly values for the analysis.

Figure XVI: *Euro Area and Germany: Real effects of a gas supply shock*



Notes: Impulse responses to a gas supply shock. Black solid lines with blue shaded credible bands denote Euro Area responses; green solid lines with green dashed credible bands denote German responses. Consumption, investment, and GNE are measured at quarterly frequency and interpolated to monthly frequency using cubic splines.

The German IRFs point to this amplification mechanism: lower gas use is accompanied by lower consumption, lower investment, and a persistent decline in industrial production. This suggests that part of the adjustment reflects lower effective availability and reduced real income rather than smooth adjustment through alternative sourcing or substitution, consistent with the view that the gas-supply disruption contributed to Germany's period of near-stagnation after the 2022 energy crisis. By contrast, Euro Area real effects are more muted and short-lived. This is consistent with lower exposure to imported natural gas: gas-import expenditure accounts for about 0.8% of nominal GNE in Germany, compared with 0.5% in the Euro Area.[31] It is also consistent with the sectoral evidence above: Euro Area output losses are concentrated in a few exposed industries, whereas German industrial production remains below baseline throughout the reported horizon. German GDP also declines by more than a rough industry-only benchmark obtained by weighting the industrial-production response by industry's share of value added (Online Appendix Figure F.4). Similarly, the German contraction extends more broadly across downstream manufacturing industries than in the Euro Area (Online Appendix Figure F.5).

To gauge the magnitude of these responses, we compare our estimates with the quantita-

[31] These pre-crisis values refer to 2019 and are based on own calculations using monthly bilateral gas-import volumes from the IEA, valued at the monthly TTF price, and nominal GNE from Eurostat.

tive predictions of the European gas-crisis models discussed above. This comparison is only a benchmark, since those models study counterfactual crisis scenarios whereas our estimates are impulse responses to an identified gas-price shock. Applying the real supply-side approximation in Moll, Schularick, and Zachmann (2023) to the observed crisis-period import declines—30% for Germany and 12% for the Euro Area—implies small GNE losses, about 0.7% and 0.1%. We calibrate the approximation with our gas-import expenditure shares and short-run elasticities.[32] Because imports are not part of the baseline specification, the empirical rescaling uses an auxiliary specification in which gas imports and domestic production enter separately and scales the responses so that the trough import response matches the observed import decline. This gives trough GNE losses of about 2.6% and 2.0%. Since our estimated elasticities are close to the conservative 0.1 calibration used in this literature, the gap between the supply-side benchmark and the empirical rescaling is unlikely to be explained by lower substitutability alone.[33]

The empirical magnitudes are instead closer to the demand-amplification benchmarks. Rescaling our 10% gas-price shock to the persistent energy-price doubling studied by Auclert et al. (2023) gives a German GNE decline of about 1.7%, close to their baseline contraction of about 2%; the corresponding Euro Area decline is about 0.8%. The German estimate also lies within the range reported by Pieroni (2023), between 0.8% and 2% in the main scenarios and up to 3.4% in a fossil-gas-only scenario. Taken together, these comparisons suggest that the empirical losses are too large to be rationalized by the real supply-side benchmark alone, and point to amplification beyond the direct supply-side effect, through real-income and Keynesian aggregate-demand channels.

V.E Contribution of gas price shocks to the inflation surge

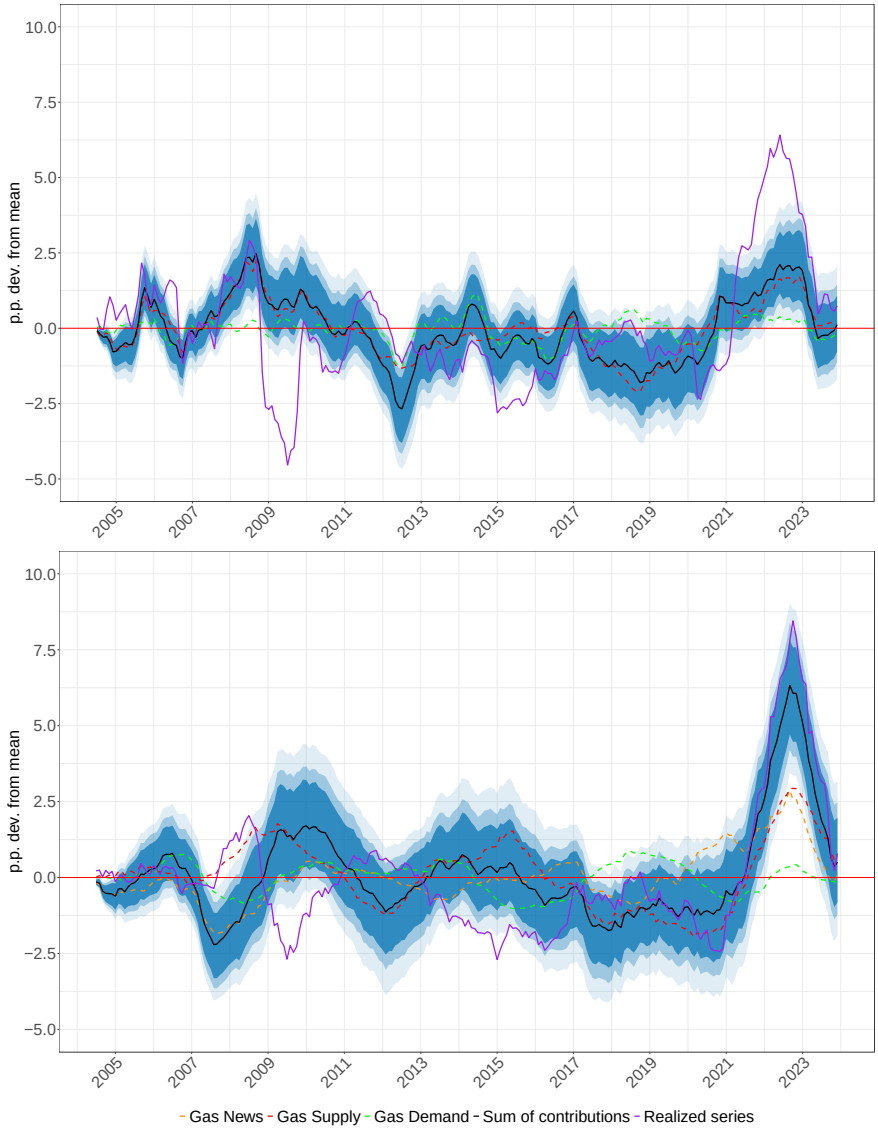
In the introduction, we noted that the surge in natural gas and broader energy prices coincided with rising inflation globally (see again Online Appendix Figure B.11). This raises two related questions: To what extent did gas price shocks contribute to the post-pandemic inflation surge? And how can we reconcile the fact that inflation followed a similar path in the United States and the Euro Area, despite stark differences in energy-price dynamics?

We provide a tentative answer to these questions through historical decompositions of headline inflation. Consistent with the impulse-response evidence discussed above, Fig-

[32]The import declines are measured over July–March windows relative to pre-crisis gas-import volumes. The supply-side benchmark is computed using the second-order approximation in Moll, Schularick, and Zachmann (2023), 2019 gas-import expenditure shares of nominal GNE, equal to 0.767% for Germany and 0.527% for the Euro Area, and the expenditure-share response implied by our estimated short-run elasticities.

[33]Equivalently, matching our estimated losses within the real supply-side framework would require post-shock gas expenditure shares well above those observed during the crisis.

Figure XVII: *Historical decomposition of year-on-year headline inflation in the United States (top panel) and the Euro Area (bottom panel)*



Notes: Estimated contributions of gas flow-supply shocks are shown by the red dashed line, gas demand shocks by the green dashed line, and their combined contribution by the solid black line with blue credible bands. In the Euro Area panel, supply-security news shocks are shown by the orange dashed line. The realized year-on-year headline inflation series is shown in purple. All series are expressed in percentage points, as deviations from their sample means.

ure XVII shows that gas price shocks, especially flow-supply and supply-security news shocks, were a key driver of the 2021–2023 inflation surge in the Euro Area. In contrast, supply shocks appear considerably less relevant in the United States. What, then, accounts for the inflationary pressures observed in the United States? A natural candidate is oil price

shocks, driven by global supply imbalances and heightened geopolitical uncertainty, which we can evaluate within our baseline specification. Indeed, the corresponding decomposition shows that oil shocks contributed more to the inflation surge in the United States than in the Euro Area (see Online Appendix Figure G.1), consistent with the comparative evidence in Baumeister (2023). This is consistent with Kilian and Zhou (2023), who show that gasoline-price shocks were the largest energy-related contributor to the 2021–2022 U.S. inflation surge, while also emphasizing that non-energy shocks played an important role. The greater contribution of gas supply shocks in the Euro Area also helps explain the higher inflation peak observed there compared with the United States.

Our findings suggest that energy-related supply shocks were major contributors to the recent inflation surge, consistent with several recent studies (see, e.g., Bańbura, Bobeica, and Martínez Hernández 2023; De Santis 2024). However, the fact that gas and oil shocks account for a sizable part of realized inflation dynamics does not imply that they were the only drivers. Other contributors likely include persistent supply-chain disruptions, such as component shortages and logistical bottlenecks (Benigno et al. 2022; Stiglitz and Regmi 2023; De Santis 2024), as well as the surge in aggregate demand during the reopening phase (Giannone and Primiceri 2024). At the same time, monetary tightening may have partly offset these inflationary pressures, as central banks responded through sharp increases in interest rates.

VI Conclusions

In this paper, we propose a novel strategy to identify demand and supply shocks in natural gas markets. Using exogenous temperature variation, we identify gas demand shocks, while using high-frequency movements in gas prices around gas-market events, we distinguish between flow-supply shocks and supply-security news shocks. This approach enables us to estimate gas demand and supply elasticities for both the United States and the Euro Area and to present detailed evidence on the macroeconomic and sectoral impacts of natural gas shocks, offering new insights into their transmission mechanisms.

Our findings show that natural gas shocks have economically significant effects beyond gas markets. Their transmission depends on the source of gas-price movements and the margins through which economies adjust. Demand is substantially less price-elastic than supply in both regions, limiting adjustment through gas consumption. Gas-demand shocks stimulate domestic production and industrial activity in the United States, whereas the Euro Area adjusts through imports and experiences larger and more persistent consumer-price increases. Flow-supply shocks have modest consumer-price effects, with little aggregate

industrial-production response in the United States and only a temporary decline in the Euro Area. For the same initial gas-price increase, Euro Area supply-security news shocks generate larger and more persistent consumer-price effects despite little aggregate production response. Inventory accumulation and heightened financial volatility point to precautionary demand and uncertainty. Across Euro Area sectors, output losses remain concentrated in exposed industries, while price pressures spread more broadly, including to core inflation. In Germany, lower consumption and investment accompany broader and more persistent output losses, consistent with aggregate-demand amplification beyond directly exposed industries. Together, these findings help reconcile the Euro Area's aggregate resilience to the Russian gas cutoff with persistent inflation and concentrated industrial losses across the region, alongside prolonged economic weakness in Germany. Finally, historical decompositions identify gas-supply shocks as a key driver of the Euro Area's 2021–2023 inflation surge, while oil shocks contributed more in the United States.

Our findings suggest several policy implications. First, energy-security policies should address both current shortages and uncertainty about future availability. Supplier diversification, reliable access to stored gas, and credible and timely communication about how future gas needs will be met may help contain precautionary price pressures even after lost deliveries have been replaced. Second, for monetary policy, the distinction between flow disruptions and supply-security news matters when assessing the persistence and breadth of consumer-price transmission and the scope for looking through an energy-price increase. Finally, our results speak to the green transition. Gas-specific price increases can redirect power generation toward more carbon-intensive fuels in the short run, raising CO_2 emissions. This implies that policies aimed at reducing gas dependence need to be paired with measures that expand clean-energy substitutes or limit switching toward coal and oil; otherwise, lower gas use may come at the cost of higher emissions.

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Online Appendix

ENERGY SHOCKS AND THE MACROECONOMY: EVIDENCE FROM NATURAL GAS MARKETS

Daniele Colombo[†] Francesco Toni[‡]

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[†]Corresponding author. Department of Economics, London Business School, United Kingdom. Email: dcolombo@london.edu.

[‡]Université Côte d'Azur, CNRS, GREDEG, France. Institute of Economics, Sant'Anna School of Advanced Studies, Italy. Email: francesco.toni@unice.fr

A VAR econometrics

Consider the structural VAR(p) model:

$$\mathbf{B}_0\mathbf{y}_t = \mathbf{B}_1\mathbf{y}_{t-1} + \cdots + \mathbf{B}_p\mathbf{y}_{t-p} + \mathbf{w}_t,$$

where $\mathbf{y}_t$ is a $K \times 1$ vector of endogenous variables, assumed to have zero mean without loss of generality. The vector $\mathbf{w}_t$ is a $K \times 1$ vector of structural shocks. Following the standard definition of structural shocks as primitive exogenous disturbances, we assume that they are mutually uncorrelated, so that $\mathbb{E}(\mathbf{w}_t\mathbf{w}_t') = \mathbf{\Sigma}_w$ is diagonal (Ramey 2016, Section 2.1). Since the matrices $\mathbf{B}_0$ and $\mathbf{w}_t$ are generally unobserved, we rely on the reduced-form representation to estimate the model:

$$\begin{aligned}\mathbf{y}_t &= \mathbf{B}_0^{-1}\mathbf{B}_1\mathbf{y}_{t-1} + \cdots + \mathbf{B}_0^{-1}\mathbf{B}_p\mathbf{y}_{t-p} + \mathbf{B}_0^{-1}\mathbf{w}_t \\ &= \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{u}_t, \end{aligned} \tag{15}$$

where $\mathbf{A}_i = \mathbf{B}_0^{-1}\mathbf{B}_i$ for $i = 1, \ldots, p$, and $\mathbf{u}_t = \mathbf{B}_0^{-1}\mathbf{w}_t$. We estimate the reduced-form parameters $\mathbf{A}_1, \ldots, \mathbf{A}_p$, $\mathbf{\Sigma}_u = \mathbb{E}(\mathbf{u}_t\mathbf{u}_t')$, and the reduced-form residuals $\mathbf{u}_t$ using Bayesian methods (see below). The key equation linking the reduced-form innovations to the structural shocks is:

$$\mathbf{u}_t = \mathbf{B}_0^{-1}\mathbf{w}_t, \tag{16}$$

where the matrix $\mathbf{B}_0^{-1}$ has to be retrieved.

A.1 Impulse response functions

Impulse response functions (IRFs) describe the responses of each element of $\mathbf{y}_t = (y_{1t}, \ldots, y_{Kt})'$ to a one-time impulse in each element of $\mathbf{w}_t = (w_{1t}, \ldots, w_{Kt})'$:

$$\frac{\partial \mathbf{y}_{t+h-1}}{\partial \mathbf{w}_t'} = \mathbf{\Theta}_h, \quad h = 1, 2, \ldots, H,$$

Thus, $h = 1$ denotes the impact response, and subsequent values of h denote months after impact. This is a $(K \times K)$ matrix whose elements are given by

$$\theta_{jk,h} = \frac{\partial y_{j,t+h-1}}{\partial w_{kt}}.$$

In order to recover the IRFs, we first resort to the VAR(1) representation of the VAR(p) process:

$$\mathbf{Y}_t = \mathbf{A}\mathbf{Y}_{t-1} + \mathbf{U}_t,$$

with

$$\mathbf{Y}_t \equiv \begin{bmatrix} \mathbf{y}_t \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix} \quad \mathbf{A} \equiv \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \dots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_K & \mathbf{0} & & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_K & & \mathbf{0} & \mathbf{0} \\ \vdots & & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I}_K & \mathbf{0} \end{bmatrix} \quad \mathbf{U}_t \equiv \begin{bmatrix} \mathbf{u}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}.$$

By recursive substitution, it can be shown that the response of variable $j = 1, \dots, K$ to a unit reduced-form innovation in component k, $h-1$ periods in the past, for $k = 1, \dots, K$, is given by $\mathbf{\Phi}_h = \begin{bmatrix} \phi_{jk,h} \end{bmatrix} \equiv \mathbf{J}\mathbf{A}^{h-1}\mathbf{J}'$, where $\mathbf{J} \equiv \begin{bmatrix} \mathbf{I}_K & \mathbf{0}_{K \times K(p-1)} \end{bmatrix}$ is a selector matrix. These are sometimes called dynamic multipliers or reduced-form impulse responses.

Under covariance stationarity of $\mathbf{y}_t$, the system can be expressed as a weighted average of current and past shocks (multivariate MA(∞) representation), with weights $\mathbf{\Phi}_h$:

$$\mathbf{y}_t = \sum_{s=1}^{\infty} \mathbf{\Phi}_s \mathbf{u}_{t-s+1} = \sum_{s=1}^{\infty} \mathbf{\Phi}_s \mathbf{B}_0^{-1} \mathbf{w}_{t-s+1} = \sum_{s=1}^{\infty} \mathbf{\Theta}_s \mathbf{w}_{t-s+1}, \tag{17}$$

where $\mathbf{\Theta}_h \equiv \mathbf{\Phi}_h \mathbf{B}_0^{-1}$. It follows that:

$$\frac{\partial \mathbf{y}_t}{\partial \mathbf{w}'_{t-h+1}} = \frac{\partial \mathbf{y}_{t+h-1}}{\partial \mathbf{w}'_t} = \mathbf{\Theta}_h, \quad h = 1, 2, \dots, H$$

These structural impulse responses can be obtained simply by post-multiplying $\mathbf{\Phi}_h$ by $\mathbf{B}_0^{-1}$.

A.2 Structural shocks

In the partially identified case, where not all columns of $\mathbf{B}_0^{-1}$ are recovered, the matrix cannot be directly inverted to obtain the full vector of structural shocks from equation (16). To recover an individual identified shock, we use the unit-variance representation

$$\mathbf{u}_t = \tilde{\mathbf{B}}_0^{-1} \tilde{\mathbf{w}}_t, \qquad \tilde{\mathbf{w}}_t = \mathbf{\Sigma}_w^{-1/2} \mathbf{w}_t, \qquad \tilde{\mathbf{B}}_0^{-1} = \mathbf{B}_0^{-1} \mathbf{\Sigma}_w^{1/2},$$

where $\mathbb{E}(\tilde{\mathbf{w}}_t \tilde{\mathbf{w}}'_t) = \mathbf{I}_K$. Following Stock and Watson (2018), if the j-th column of $\tilde{\mathbf{B}}_0^{-1}$ is identified, and using $\mathbf{\Sigma}_u = \tilde{\mathbf{B}}_0^{-1} (\tilde{\mathbf{B}}_0^{-1})'$, the corresponding structural shock can be recovered as

$$\left[(\tilde{\mathbf{B}}_0^{-1})_{:,j} \right]' \mathbf{\Sigma}_u^{-1} \mathbf{u}_t = \mathbf{e}'_j \tilde{\mathbf{B}}_0 \mathbf{u}_t = \mathbf{e}'_j \tilde{\mathbf{B}}_0 \mathbf{B}_0^{-1} \mathbf{w}_t = \mathbf{e}'_j \tilde{\mathbf{B}}_0 \tilde{\mathbf{B}}_0^{-1} \tilde{\mathbf{w}}_t = \mathbf{e}'_j \tilde{\mathbf{w}}_t = \tilde{w}_{j,t},$$

where $\mathbf{e}_j$ denotes the j-th standard basis vector.

A.3 Forecast error variance decomposition

Under the unit-variance representation above, the forecast error at IRF horizon h, conditional on information available through $t-1$, is given by

$$\mathbf{y}_{t+h-1} - \mathbb{E}_{t-1}[\mathbf{y}_{t+h-1}] = \sum_{s=1}^{h} \tilde{\mathbf{\Theta}}_s \tilde{\mathbf{w}}_{t+h-s},$$

with forecast error variance:

$$\mathbf{\Sigma}_h \equiv \mathbb{E}_{t-1}\left[(\mathbf{y}_{t+h-1} - \mathbb{E}_{t-1}[\mathbf{y}_{t+h-1}])(\mathbf{y}_{t+h-1} - \mathbb{E}_{t-1}[\mathbf{y}_{t+h-1}])'\right] = \sum_{s=1}^{h} \tilde{\mathbf{\Theta}}_s \tilde{\mathbf{\Theta}}_s'. \tag{18}$$

The jth entry on the diagonal of $\mathbf{\Sigma}_h$ is therefore the variance of the forecast error of variable j due to all shocks $1, \ldots, K$. This is:

$$[\mathbf{\Sigma}_h]_{j,j} = \sum_{s=1}^{h} \sum_{\ell=1}^{K} \left([\tilde{\mathbf{\Theta}}_s]_{j,\ell}\right)^2$$

while the contribution of a single shock k at horizon h is

$$\sum_{s=1}^{h} \left([\tilde{\mathbf{\Theta}}_s]_{j,k}\right)^2$$

It follows that the percentage contribution of shock k to the total error variance of variable j at horizon h is:

$$\text{FEVD}_{j,k}(h) = \frac{\sum_{s=1}^{h} \left([\tilde{\mathbf{\Theta}}_s]_{j,k}\right)^2}{\sum_{s=1}^{h} \sum_{\ell=1}^{K} \left([\tilde{\mathbf{\Theta}}_s]_{j,\ell}\right)^2}$$

Partially identified case. When not all columns of $\tilde{\mathbf{\Theta}}_h$ are identified, we cannot compute $\mathbf{\Sigma}_h$ directly using equation (18). Instead, leveraging the identity $\tilde{\mathbf{\Theta}}_h \tilde{\mathbf{\Theta}}_h' = \mathbf{\Phi}_h \tilde{\mathbf{B}}_0^{-1} (\tilde{\mathbf{B}}_0^{-1})' \mathbf{\Phi}_h' = \mathbf{\Phi}_h \mathbf{\Sigma}_u \mathbf{\Phi}_h'$, we compute:

$$\text{FEVD}_{j,k}(h) = \frac{\sum_{s=1}^{h} \left([\tilde{\mathbf{\Theta}}_s]_{j,k}\right)^2}{\sum_{s=1}^{h} [\mathbf{\Phi}_s \mathbf{\Sigma}_u \mathbf{\Phi}_s']_{j,j}}.$$

This expression uses the reduced-form forecast-error variance in the denominator and therefore does not require all columns of $\tilde{\mathbf{\Theta}}_h$ to be identified.

A.4 Historical decomposition

Structural impulse responses describe average movements in the data. Historical decompositions instead quantify how much a given identified structural shock contributes to the realized fluctuations of the variables in the VAR. Under the unit-variance representation above, with $\tilde{\mathbf{\Theta}}_s \equiv \mathbf{\Phi}_s \tilde{\mathbf{B}}_0^{-1}$, equation (17) can be written as

$$\mathbf{y}_t = \sum_{s=1}^{t} \tilde{\mathbf{\Theta}}_s \tilde{\mathbf{w}}_{t-s+1} + \sum_{s=t+1}^{\infty} \tilde{\mathbf{\Theta}}_s \tilde{\mathbf{w}}_{t-s+1}.$$

Since covariance stationarity implies that the MA coefficients decay over time,

$$\mathbf{y}_t \approx \sum_{s=1}^{t} \tilde{\mathbf{\Theta}}_s \tilde{\mathbf{w}}_{t-s+1}.$$

In the partially identified case, we compute the historical contribution of the identified shock only, using the corresponding column of $\tilde{\mathbf{\Theta}}_s$ and the recovered series $\tilde{w}_{j,t}$. This approximation can only be computed from $t = p+1$ onward and becomes less sensitive to omitted pre-sample shocks as t increases.

A.5 Bayesian estimation

The Bayesian estimation procedure we implement employs Minnesota (Doan, Litterman, and Sims 1984), sum-of-coefficients (see also Sims and Zha 1998; Robertson and Tallman 1999), and dummy-initial-observation (Sims 1993) priors. These priors are incorporated through dummy observations, with their informativeness selected by maximizing the marginal likelihood, following Bańbura, Giannone, and Reichlin (2007) and the hierarchical procedure of Giannone, Lenza, and Primiceri (2015). Parameter estimation is conducted via a standard Gibbs sampling algorithm.

Prior distribution. Following Kadiyala and Karlsson (1997), we specify the prior distributions as

$$\text{vec}(\beta) \mid \Psi \sim \mathcal{N}(\text{vec}(\beta_0), \Psi \otimes \Omega_0) \quad \text{and} \quad \Psi \sim \mathcal{IW}(S_0, \alpha_0),$$

where $\beta = [\mathbf{A}_1', \ldots, \mathbf{A}_p', \mathbf{c}']'$ is the $(Kp+1) \times K$ matrix of reduced-form coefficients, with $\mathbf{c}$ an intercept, and Ψ is the covariance matrix of reduced-form residuals, corresponding to $\mathbf{\Sigma}_u$ in (15). The prior parameters β_0, Ω_0, S_0 and α_0 are chosen so that prior expectations and variances coincide with those implied by Minnesota, sum-of-coefficients and dummy-initial observation priors (more below). We implement the prior by adding dummy observations: Bańbura, Giannone, and Reichlin (2007) have shown that adding T_d dummy observations

$\mathbf{y}_d$ and $\mathbf{x}_d$ to the VAR(p) induces the Normal-Inverted Wishart prior:

$$\begin{aligned}\beta_0 &= (\mathbf{x}_d'\mathbf{x}_d)^{-1}(\mathbf{x}_d'\mathbf{y}_d);\\ \Omega_0 &= (\mathbf{x}_d'\mathbf{x}_d)^{-1};\\ S_0 &= (\mathbf{y}_d - \mathbf{x}_d\beta_0)'(\mathbf{y}_d - \mathbf{x}_d\beta_0);\\ \alpha_0 &= T_d,\end{aligned}$$

where S_0 is a $(K \times K)$ scale matrix and α_0 denotes the degrees of freedom of the Inverted Wishart distribution. Conveniently, this prior distribution is conjugate, implying that the posterior distribution has a known analytical form.

Minnesota prior. The Minnesota prior (Doan, Litterman, and Sims 1984; Litterman 1986) formalizes the belief that, ex ante, each variable in the system follows a random walk with drift, represented by the process $y_t = c + y_{t-1} + u_t$. This prior encodes the assumption that more recent lags contain greater predictive content than more distant ones, and that a variable's own history is more informative for forecasting its future values than the histories of other variables in the system. Accordingly, the prior shrinks the diagonal elements of $\mathbf{A}_1$ toward one, while shrinking the off-diagonal elements of $\mathbf{A}_1$ and all elements of $\mathbf{A}_2, \ldots, \mathbf{A}_p$ toward zero. The mean of the prior distribution is given by:

$$\mathbb{E}[(\mathbf{A}_s)_{ij}] = \begin{cases}\delta_i & \text{if } i = j \text{ and } s = 1\\ 0 & \text{otherwise}\end{cases}$$

Originally, $\delta_i = 1$ was imposed for all i, based on the notion that most variables exhibit strong persistence. However, this prior is not suitable for variables that display substantial mean reversion. In such cases, we adopt the belief that the variable behaves like white noise by setting $\delta_i = 0$. The prior variance of the parameters is

$$\mathbb{V}[(\mathbf{A}_s)_{ij}] = \begin{cases}\frac{\lambda^2}{s^2} & \text{if } i = j\\ \vartheta\frac{\lambda^2}{s^2} \cdot \frac{\sigma_i^2}{\sigma_j^2} & \text{otherwise}\end{cases}$$

where the hyperparameter λ controls the relative importance of prior and data. As $\lambda \to 0$, the prior dominates; as $\lambda \to \infty$, the prior becomes diffuse. The factor $1/s^2$ is the rate at which prior variance decreases with increasing lag length, and σ_i^2/σ_j^2 accounts for the different scale and variability of the data. The coefficient $\vartheta \in (0, 1)$ governs the degree of shrinkage applied to the lags of other variables relative to a variable's own lags.

The original formulation of the Minnesota prior assumes that the covariance matrix of the

residuals is diagonal and known: $\boldsymbol{\Sigma}_u = \text{diag}(\sigma_1^2, \ldots, \sigma_K^2)$. However, for structural analysis, it is important to account for potential correlations among residuals across equations. To address this, we follow the approach of Kadiyala and Karlsson (1997) and Robertson and Tallman (1999), who propose a Normal-inverted Wishart prior that preserves the key features of the Minnesota prior. The conjugate Normal-inverted Wishart formulation requires setting $\vartheta = 1$, which we impose. The prior on the intercept c is assumed to be diffuse. The dummy observations implementing these prior components are:

$$\underset{(Kp+K+1)\times K}{\mathbf{y}^+} = \begin{bmatrix} \frac{\text{diag}(\delta_1\sigma_1,\ldots,\delta_K\sigma_K)}{\lambda} \\ \hline \mathbf{0}_{K(p-1)\times K} \\ \hline \text{diag}(\sigma_1, \ldots, \sigma_K) \\ \hline \mathbf{0}_{1\times K} \end{bmatrix} \qquad \underset{(Kp+K+1)\times(Kp+1)}{\mathbf{x}^+} = \begin{bmatrix} J_p \otimes \frac{\text{diag}(\sigma_1, \ldots, \sigma_K)}{\lambda} & \mathbf{0}_{Kp\times 1} \\ \hline \mathbf{0}_{K\times Kp} & \mathbf{0}_{K\times 1} \\ \hline \mathbf{0}_{1\times Kp} & \underset{1\times 1}{\mathbf{1}}\,\varepsilon \end{bmatrix}$$

where the first block of dummies imposes prior beliefs on the autoregressive coefficients, the second block implements the prior for the covariance matrix, the third block reflects the uninformative prior for the intercept (ε is a very small number).

Refinements of the Minnesota prior have been introduced to accommodate unit roots and cointegration, reflecting common practices in applied research. These refinements involve additional priors designed to diminish the influence of the deterministic component.

Sum-of-coefficients prior. This prior shrinks the sum of autoregressive coefficients toward one, favoring persistent dynamics. We can rewrite the VAR in error-correction form:

$$\Delta \mathbf{y}_t = \mathbf{c} - (\mathbf{I}_K - \mathbf{A}_1 - \cdots - \mathbf{A}_p)\mathbf{y}_{t-1} + \tilde{\mathbf{A}}_1 \Delta \mathbf{y}_{t-1} + \cdots + \tilde{\mathbf{A}}_{p-1} \Delta \mathbf{y}_{t-p+1} + \mathbf{u}_t.$$

Let $\Pi := \mathbf{I}_K - \mathbf{A}_1 - \cdots - \mathbf{A}_p$. If $\Pi = 0$, the levels VAR has unit roots rather than being stationary; its first differences are stationary provided the dynamics in differences are stable. If Π has reduced but positive rank, the variables can instead be cointegrated, so stationary linear combinations may coexist with nonstationary levels. The sum-of-coefficients prior shrinks Π toward zero, favoring persistent dynamics and a no-change forecast at the beginning of the sample. It is implemented by adding the following dummy observations at the beginning of the sample:

$$\underset{K\times K}{\mathbf{y}^{++}} = \begin{bmatrix} \frac{\mu_1\delta_1}{\tau} & 0 & \cdots & 0 \\ 0 & \frac{\mu_2\delta_2}{\tau} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \frac{\mu_K\delta_K}{\tau} \end{bmatrix} \qquad \underset{K\times(Kp+1)}{\mathbf{x}^{++}} = \left[\mathbf{y}^{++}, \cdots, \mathbf{y}^{++}, \underset{K\times 1}{\mathbf{0}}\right],$$

where μ_i captures the unconditional mean of variable y_{it}, δ_i is the term previously defined, and the hyperparameter τ controls the variance of these prior beliefs: as $\tau \to \infty$, the prior becomes uninformative, while $\tau \to 0$ enforces the presence of a unit root in each equation and rules out cointegration entirely. This is because we can have cointegration (individual series display unit-roots but some linear combination of the series is stationary) only if Π has reduced rank but is not zero, otherwise the VAR becomes a purely differenced system.

Dummy-initial-observation. Since the sum-of-coefficients prior rules out cointegration in the limit, the dummy-initial-observation prior is used to favor cointegrating relationships while still limiting overfitting of deterministic components. This is achieved by adding dummy observations at the beginning of the sample, constructed as follows:

$$\underset{K\times K}{\mathbf{y}^{+++}} = \begin{bmatrix} \frac{\mu_1}{\gamma} & 0 & \cdots & 0 \\ 0 & \frac{\mu_2}{\gamma} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \frac{\mu_K}{\gamma} \end{bmatrix} \qquad \underset{K\times(Kp+1)}{\mathbf{x}^{+++}} = \left[\mathbf{y}^{+++}, \cdots, \mathbf{y}^{+++}, \underset{K\times 1}{\mathbf{1}}/\gamma\right]$$

The hyperparameter γ controls the tightness of the prior implied by these artificial observations. As $\gamma \to \infty$, the prior becomes uninformative. As $\gamma \to 0$, the model tends to a form in which either all variables are stationary with means equal to the sample averages of the initial conditions, or there are unit root components without drift terms.

Posterior distribution. When we augment the data with the dummy observations

$$\mathbf{y}_* = \begin{bmatrix} \mathbf{y} \\ \mathbf{y}^{+} \\ \mathbf{y}^{++} \\ \mathbf{y}^{+++} \end{bmatrix}, \quad \mathbf{x}_* = \begin{bmatrix} \mathbf{x} \\ \mathbf{x}^{+} \\ \mathbf{x}^{++} \\ \mathbf{x}^{+++} \end{bmatrix}$$

we can rewrite the VAR(p) as

$$\mathbf{y}_* = \mathbf{x}_*\beta + \mathbf{u}_*$$

and it can be shown that the posterior distribution has the form

$$\text{vec}(\beta) \mid \Psi, \mathbf{y}_*, \mathbf{x}_* \sim \mathcal{N}(\text{vec}(\tilde{\beta}), \Psi \otimes (\mathbf{x}_*'\mathbf{x}_*)^{-1}) \quad \text{and} \quad \Psi \mid \mathbf{y}_*, \mathbf{x}_* \sim \mathcal{IW}(\tilde{S}, T_d + T),$$

where the posterior mean of the coefficients corresponds to the OLS estimate of the regression model based on the appended data, with:

$$\tilde{\beta} = (\mathbf{x}_*'\mathbf{x}_*)^{-1}(\mathbf{x}_*'\mathbf{y}_*) \qquad \tilde{S} = (\mathbf{y}_* - \mathbf{x}_*\tilde{\beta})'(\mathbf{y}_* - \mathbf{x}_*\tilde{\beta})$$

Choosing the informativeness of the priors. The prior distribution of our Bayesian VAR is governed by prior targets and scale parameters. The indicator δ_i takes the value 1 or 0, depending on whether the ith variable is shrunk toward a highly persistent process (random walk, $\delta_i = 1$) or toward a stationary process (white noise, $\delta_i = 0$). The scale parameter σ_i is the standard deviation of the residuals for variable i under the prior and accounts for differences in variable volatility. The hyperparameter λ controls the overall tightness of the Minnesota prior: as $\lambda \to 0$, the prior dominates, while as $\lambda \to \infty$, the posterior approaches the OLS estimate. The hyperparameters τ and γ govern the tightness of the sum-of-coefficients and dummy-initial-observation priors, respectively.

Following the hierarchical approach of Giannone, Lenza, and Primiceri (2015), we choose the informativeness of the priors by selecting the hyperparameters that maximize the marginal likelihood of the data. Intuitively, the marginal likelihood balances fit and parsimony, favoring priors that regularize the high-dimensional VAR without imposing excessive shrinkage.

To combine posterior uncertainty in the reduced-form VAR with external-instrument identification, for each retained posterior draw, we recover reduced-form residuals from the observed sample, excluding the appended dummy observations, and apply the standard external-instrument moment conditions to the fixed instrument series and draw-specific residuals to obtain the identified impact vectors. We then orthogonalize these initial impact vectors at each iteration as described in Section IV; neither the instruments nor the orthogonalization step enters the BVAR likelihood or priors.

B Data and descriptive statistics

This appendix documents the data and provides descriptive evidence on the natural gas markets studied in the paper. It first reports the sources, transformations, and coverage of the main empirical variables, then compares the energy market structures of the Euro Area and the United States, documents the role of the TTF as the benchmark price for European natural gas, and finally reports the joint evolution of energy prices and headline inflation in the two regions.

B.1 Data sources

Table B.1 reports the sources, sample coverage, and transformations for the main data used in the empirical analysis.

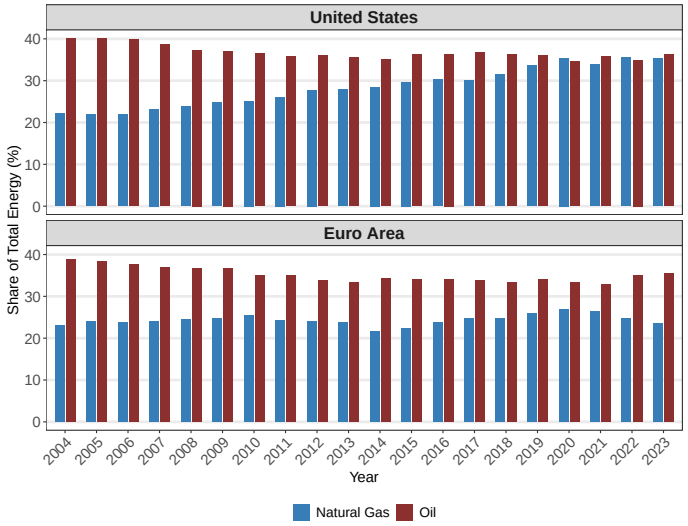
Table B.1: *Data description, sources, and coverage*

Variable	Description	Source	Time Coverage	Trans.
EA Instruments				
TRNLTTFc-h (RIC)	TTF natural gas futures contracts (settlement price): monthly (1M-4M), quarterly (1Q-4Q), seasonal (1S-2S), and calendar-year (1Y)	Datastream	05/01/2004-01/01/2024	100Δ log
LCOc-h (RIC)	Brent crude oil futures h-month contracts from 1M to 12M (settlement price)	Datastream	18/03/1980-01/01/2024	100Δ log
ERA5 surface temperature	2m daily mean temperature on a 0.25° grid, aggregated to country level (GADM0) and weighted by lagged national gas consumption	ERA5, EUROSTAT	01/01/1950-01/01/2024	None
Heating and cooling degree days	Monthly HDD and CDD, aggregated to country level and weighted by lagged national gas consumption	EUROSTAT	1979M1-2024M1	None
EA Variables (baseline)				
Gas Consumption	EA gas consumption (GDINCTROAD), SA	IEA	1984M1-2024M1	100 ∗ log
Gas Supply	EA dry marketable production (INDPROD) plus gas imports from non-EA countries (IMPORTS) minus gas exports to non-EA countries (EXPORTS), SA	IEA	1984M1-2024M1	100 ∗ log
Gas Inventories	EA closing stock levels (CSNATTER), SA	IEA	1984M1-2024M1	100 ∗ log
Gas Price	Dutch TTF spot price at close (TRNLTTFD1) in Euro per MWh, deflated by the EA HICP all-items index	Datastream	2004M1-2024M1	100 ∗ log
Headline CPI	HICP all-items index (000000), SA	EUROSTAT	2000M12-2024M1	100 ∗ log
Industrial Production	Total industry excluding construction (B-D), SA	EUROSTAT	1995M1-2024M1	100 ∗ log
Oil Price	Brent spot price at close (DCOILBRENTEU) in Euro per barrel, deflated by the EA HICP all-items index	FRED	1999M1-2024M1	100 ∗ log
US Instruments				
NGc-h (RIC)	NYMEX HH natural gas futures h-month contracts from 1M to 12M (settlement price)	Datastream	03/04/1990-01/01/2024	100Δ log
CLc-h (RIC)	WTI crude oil futures h-month contract from 1M to 12M (settlement price)	Datastream	18/03/1980-01/01/2024	100Δ log
ERA5 surface temperature	2m daily mean temperature on a 0.25° grid, aggregated to state level (GADM1) and weighted by lagged state gas consumption	ERA5, EIA	01/01/1940-01/01/2024	None
Heating and cooling degree days	Monthly HDD and CDD by state, aggregated to the national level and weighted by lagged state gas consumption	NOAA, EIA	1985M1-2024M1	None
US Variables (baseline)				
Gas Consumption	U.S. total gas consumption (N9140US1) SA	EIA	2001M1-2024M1	100 ∗ log
Gas Supply	U.S. dry gas production (N9070US1) plus gas imports (N9100US2) minus gas exports (N9130US2), SA	EIA	1984M1-2024M1	100 ∗ log
Gas Inventories	U.S. Total Natural Gas in Underground Storage Working Gas (N5020US2), SA	EIA	1975M9-2024M1	100 ∗ log
Gas Price	NYMEX HH spot price at close (NG-W-HH-SNL) in Dollar per Million BTU, deflated by the US CPI all-items index	Datastream	1991M8-2024M1	100 ∗ log
Headline CPI	Headline CPI index (CPIAUCSL), SA	FRED	1960M1-2024M1	100 ∗ log
Industrial Production	Total index (INDPRO), SA	FRED	1974M1-2024M1	100 ∗ log
Oil Price	WTI spot price at close (DCOILWTICO) in Dollar per barrel, deflated by the US CPI all-items index	FRED	1986M1-2024M1	100 ∗ log
Additional Variables				
Gas Production (EA)	EA dry marketable production (INDPROD), SA	IEA	1984M1-2024M1	100 ∗ log
Gas Net Imports (EA)	EA gas imports from non-EA countries (IMPORTS) minus EA gas exports to non-EA countries (EXPORTS), SA	IEA	1984M1-2024M1	100 ∗ log
Gas, Oil, Coal, and Renewable Power (EA)	Monthly electricity generation by source in GWh, SA	ENTSO-E	2007M1-2024M1	100 ∗ log
Financial Volatility (EA)	Euro Stoxx 50 Volatility Index (VSTOXX)	STOXX	1999M1-2024M1	100 ∗ log
Core CPI (EA)	HICP index excluding energy and food seasonally adjusted (TOT_X_NRG_FOOD)	EUROSTAT	2000M12-2024M1	100 ∗ log
Headline PPI (EA)	Total industry producer price index, SA (PRON.NS0020)	EUROSTAT	2000M1-2024M1	100 ∗ log
Nominal Exchange Rate (EA)	Nominal Broad Effective Exchange Rate for EA (NBXMBIS)	FRED	1994M1-2024M1	100 ∗ log
IP C10-C33 (EA)	NACE2 2-digit level Manufacturing Industrial Production indexes, SA	EUROSTAT	1991M1-2024M1	100 ∗ log
PPI D35-10/20 (EA)	NACE2 4-digit level Electricity and Gas Producer Price indexes, SA	EUROSTAT	2000M1-2024M1	100 ∗ log
HICP CP01-CP12 (EA)	ECOICOP 1-digit level indexes, SA	EUROSTAT	2000M1-2024M1	100 ∗ log
Electricity Price (EA)	Consumption-weighted average of euro-area day-ahead spot prices, deflated by the EA HICP	Refinitiv, EUROSTAT	2004M1-2024M1	100 ∗ log
Electricity Consumption (EA)	Monthly electricity demand, EA19 aggregate, SA	ENTSO-E	2007M1-2024M1	100 ∗ log
German macro variables (DE)	Headline HICP (CP00_DE) and industrial production (B-D), defined and transformed as the corresponding euro-area series	EUROSTAT	2000M12-2024M1	100 ∗ log
German gas variables (DE)	German gas consumption, supply and inventories, defined as the corresponding euro-area series	IEA	1984M1-2024M1	100 ∗ log
Consumption, Investment, GNE (EA, DE)	National accounts aggregates, chain-linked volumes, SA; quarterly, interpolated to monthly by cubic spline	EUROSTAT	1995Q1-2023Q4	100 ∗ log
Gas Production (US)	U.S. dry Natural Gas Production (N9070US1) in Bcf, SA	EIA	1973M1-2024M1	100 ∗ log
Financial Volatility (US)	CBOE Volatility Index (VIXCLS)	FRED	1990M1-2024M1	100 ∗ log
Gas, Oil, and Coal Power (US)	Fossil fuel consumption by the electric power sector, SA	EIA	2004M1-2024M1	100 ∗ log
Nominal Exchange Rate (US)	Nominal Broad Effective Exchange Rate for US (NBUSBIS)	FRED	1994M1-2024M1	100 ∗ log
Electricity Price (US)	On-peak wholesale electricity spot price, deflated by the US CPI	EIA, ICE	2001M1-2024M1	100 ∗ log
Electricity Consumption (US)	Electricity retail sales, SA	EIA	2001M1-2024M1	100 ∗ log
Renewable Power (US)	Renewable power generation in thousand MWh, SA	EIA	2001M1-2024M1	100 ∗ log
CO2 Emissions (EA, US)	Total CO_2 emissions from fossil fuel combustion and industrial processes, measured in gigagrams	EDGAR CO2	1970M1-2022M12	100 ∗ log
Weather controls (EA, US)	Maximum wind gust, precipitation, SPEI drought index and solar radiation, aggregated to the region with fixed 2015 technology-specific generation weights, seasonally adjusted and standardized	ERA5, ERA5-Land, OWID	1940M1-2024M1	Standardized

B.2 Energy statistics in the United States and the Euro Area

Figures B.1 and B.2 place natural gas within the broader energy mix. Natural gas accounts for a substantial share of energy supply in both regions, but the composition of that supply and the exposure to external producers differ markedly across regions.

Figure B.1: *Natural gas and crude oil and petroleum products as a share of total energy supply*



Notes: Share of natural gas and crude oil and petroleum products in total energy supply (2004–2023); United States in the top panel, Euro Area in the bottom panel. Energy supply is defined as production plus imports minus exports minus stock changes. Source: IEA.

Figure B.3 shows that gas is used across residential, industrial, and power-generation activities, with residential use accounting for a larger share in the Euro Area. Figure B.4 highlights the sharper difference on the supply side: the Euro Area relies heavily on imported gas, whereas the United States has become increasingly self-sufficient.

Table B.2 and Figure B.5 report the gas-market quantities behind these supply-side differences. They show that the Euro Area combines declining domestic production with continued import dependence, whereas U.S. production has expanded steadily.

Figures B.6 and B.7 provide a longer-run comparison of external energy trade. The Euro Area remains a large net importer of both fuels, while rising U.S. production has reversed the gas trade balance and substantially narrowed net crude-oil imports.

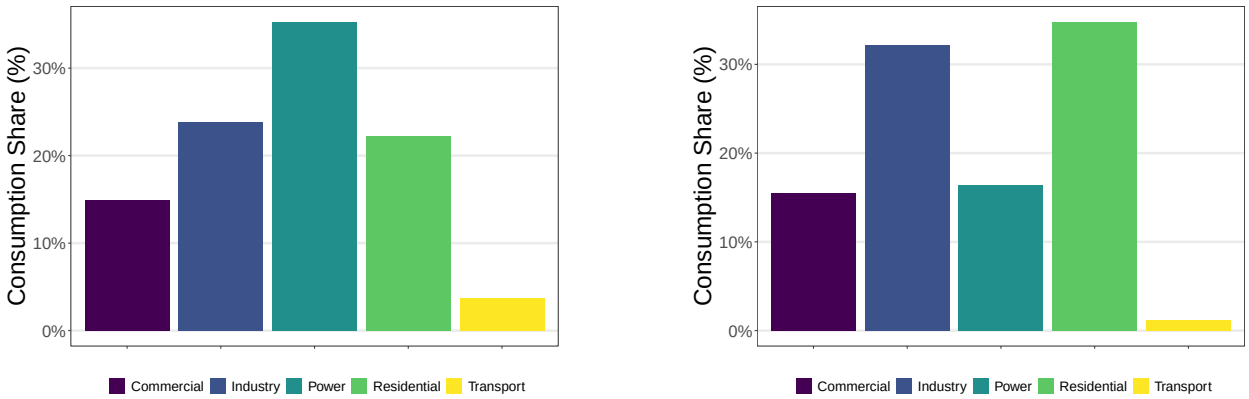
Figure B.2: *Energy supply by source*



United States Euro Area

Notes: Composition of total energy supply by source, averaged over 2004–2023; United States in the left panel, Euro Area in the right panel. Energy supply is defined as domestic production plus imports minus exports minus stock changes. Source: IEA.

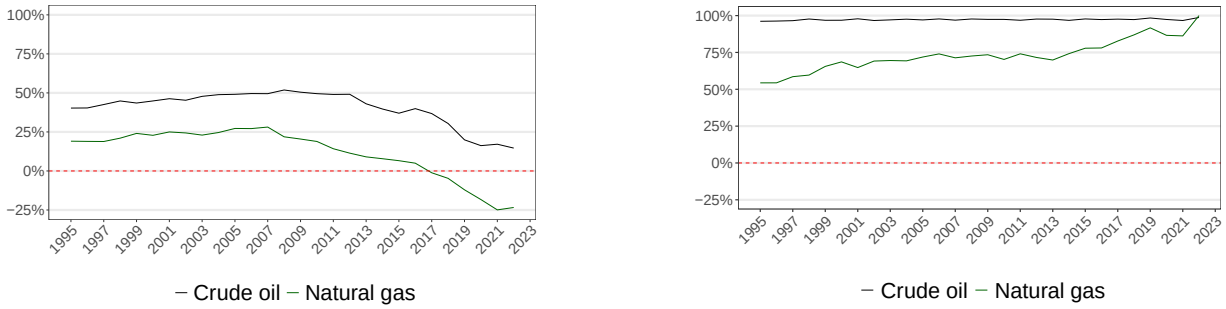
Figure B.3: *Natural gas consumption share by sector*



United States Euro Area

Notes: Sectoral shares of natural gas consumption, averaged over 2004–2022; United States in the left panel, Euro Area in the right panel. Total consumption encompasses deliveries to the power generation, industrial, commercial, residential, and transportation sectors. Source: IEA.

Figure B.4: *Energy import dependency*



United States Euro Area

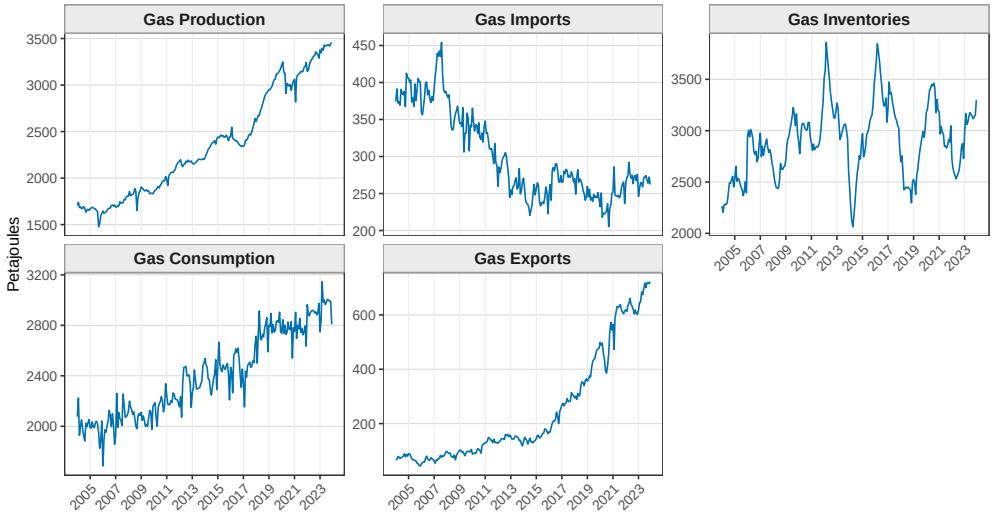
Notes: Import dependency on natural gas and oil, 1995–2022, measured as the share of net imports in total consumption of each fuel; United States in the left panel, Euro Area in the right panel. Sources: Eurostat, EIA, Energy Institute.

Table B.2: *Gas market variables for the United States and the Euro Area*

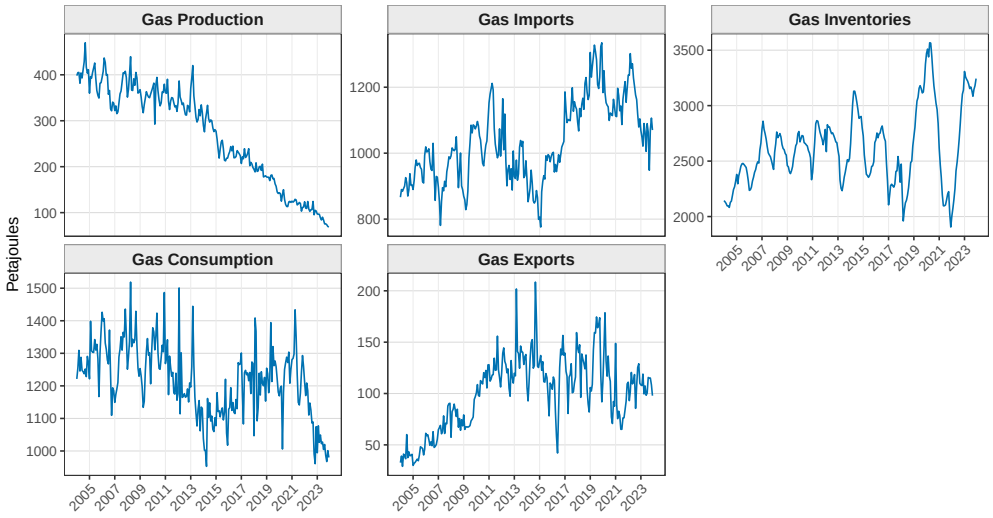
Region	Production	Imports	Exports	Consumption	Inventories	Supply
United States	2,354	301	246	2,399	2,897	2,409
Euro Area	271	1,033	100	1,216	2,616	1,204

Notes: Monthly averages in petajoules over 2004–2023. Supply is defined as production plus imports minus exports. Inventories are end-of-month levels. Sources: IEA and EIA.

Figure B.5: *Gas market variables for the United States and the Euro Area*



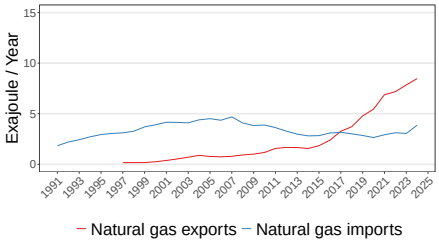
United States



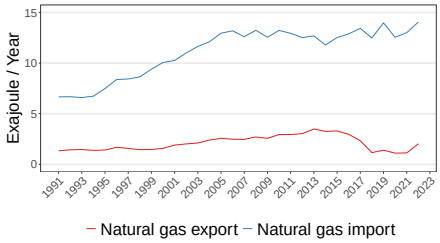
Euro Area

Notes: All series are expressed in petajoules and seasonally adjusted. Sources: IEA and EIA.

Figure B.6: *Natural gas imports and exports*



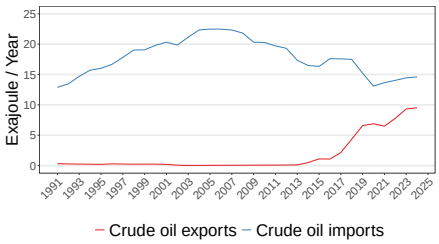
United States



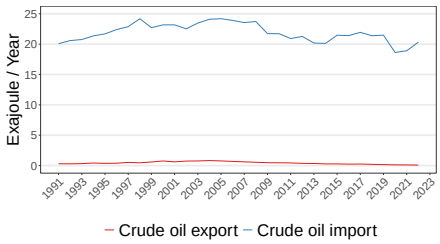
European Union

Notes: Natural gas imports and exports in exajoules per year; United States in the left panel (1991–2024), European Union in the right panel (1991–2022). Sources: Eurostat and EIA.

Figure B.7: *Crude oil imports and exports*



United States



European Union

Notes: Crude oil imports and exports in exajoules; United States in the left panel (1990–2024), European Union in the right panel (1990–2022). Sources: Eurostat and EIA.

B.3 The TTF as the European benchmark gas price

This subsection documents the role of the Dutch TTF as the benchmark price for European natural gas markets. Historically, natural gas pricing in Europe has been predominantly linked to oil products. However, over the last two decades, gas prices have moved away from oil indexation to spot pricing through a series of regulatory reforms aimed at liberalizing the European gas market. The European Union began its liberalization process in 1992 with the EU energy market regulatory framework. However, meaningful developments only started in the late 1990s, leading to the issuance of three European Directives designed to foster competition and create a single market for natural gas. This reform process culminated in the "Gas Regulation" of 2009, which further strengthened market integration efforts.[1] These regulatory reforms led to the development of trading hubs across Europe, with 11 main active hubs as of 2021, though varying significantly in liquidity and infrastructure (Heather 2021).[2]

While full liberalization of the European gas market remains incomplete (Cardinale 2019; ACER 2022), regional integration has nonetheless advanced significantly and has been a defining feature of the market for over two decades. The Dutch Title Transfer Facility (TTF) gas hub, recognized as the most liquid trading hub, has emerged as the benchmark for European gas prices. The TTF, listed on the ICE ENDEX futures exchange in Amsterdam, was established in 2003, whereas the first gas hub in the region, the National Balancing Point (NBP), was created in the United Kingdom in 1996.

Additionally, the share of hub-indexed imported gas relative to fixed contracted prices has grown significantly over time, representing approximately 80% of total gas imports in the European Union in 2021 (IEA, 2021). As part of this trend, oil-indexed contracts, which constituted over 90% of European gas imports in 2005, declined sharply to only 25% by 2019 (IEA, 2020). These developments support our use of the TTF as the European benchmark gas price. TTF-linked futures are the most liquid natural-gas contracts in Europe, with trading volume reaching 5.7 million contracts in May 2023 (ICE 2023). Indeed, most studies that examine the role of gas prices in Europe focus on the TTF price (e.g., Adolfsen et al. 2024; Boeck and Zörner 2025; López et al. 2026). Jotanovic and D'Ecclesia (2021) provide detailed evidence of a high level of integration among the European trading hubs, with the TTF playing the role of the reference trading hub.

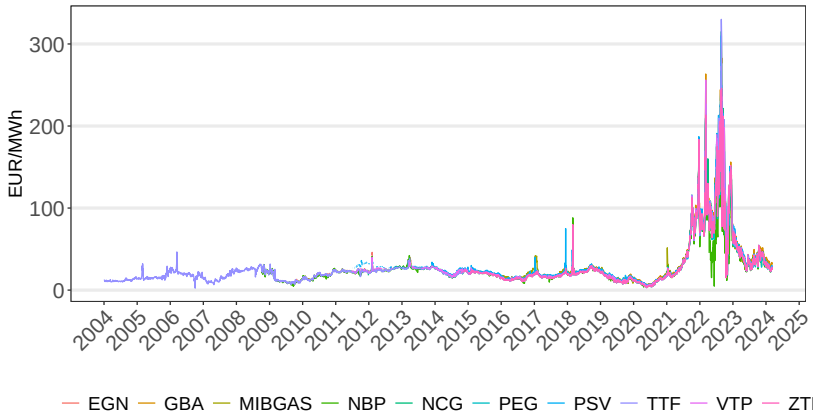
This integration is reflected in Figure B.8 and Table B.3, which show that price dynamics across European hubs are highly correlated. Figure B.8 shows the natural gas spot prices at selected European trading hubs: the Italian European Gas Network (EGN), the British

[1] Regulation (EC) No 715/2009 (the "Gas Regulation").

[2] While there are approximately 30 gas trading hubs in Europe, not all of them are actively operational.

Greater Buchan Area (GBA), the Spanish *Mercado Ibérico del GAS* (MIBGAS), the British National Balancing Point (NBP), the German NetConnect Germany (NCG), the French *Point d'échange de Gaz* (PEG), the Italian *Punto di Scambio Virtuale* (PSV), the Austrian Virtual Trading Point (VTP), and the Belgian Zeebrugge Trade Point (ZTP). These prices closely followed the TTF not only in the period before the pandemic but also amid the subsequent market disruptions. Table B.3 quantifies the comovement between TTF and these gas prices. The correlations are very high, ranging from 0.93 for the British NBP to 1.00 for NCG.

Figure B.8: *TTF and other European gas prices*



Notes: Daily Dutch TTF spot price alongside spot prices at other European trading hubs. Source: Datastream.

Table B.3: *Correlation between TTF and other European gas prices*

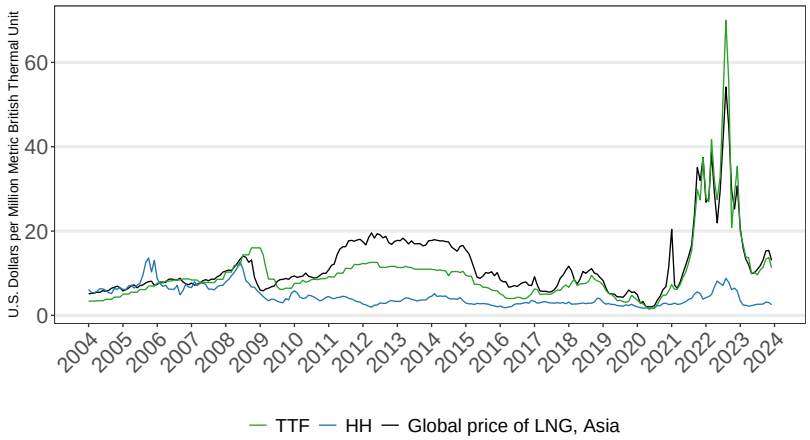
Hub price	TTF
NCG	1.00
VTP	1.00
PSV	1.00
ZTP	0.97
EGN	0.98
NBP	0.93
GBA	1.00
PEG	0.97
MIBGAS	0.97

Notes: Correlations between the Dutch TTF spot price and natural gas spot prices at other European trading hubs. Source: Datastream.

Crucially, while LNG has become increasingly important in the European market, its expansion has not resulted in greater price divergence within the region. On the contrary, the integration between European and global LNG markets has strengthened significantly,

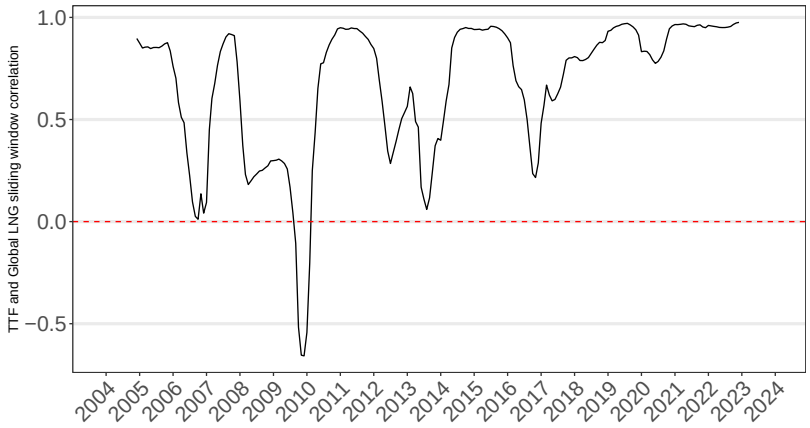
driven by the substantial growth of seaborne gas trade (Albrizio et al. 2023). Historically, LNG prices did not closely track the TTF, reflecting its limited role in the European market at the time. However, as LNG has gained prominence, its price has become more closely aligned with the TTF, reducing divergence. This convergence is documented in Figures B.9 and B.10.

Figure B.9: *TTF, HH, and Global LNG gas prices comparison*



Notes: The figure plots the monthly TTF spot price, converted to U.S. dollars for illustrative purposes, alongside the Henry Hub and the JKM LNG price. Source: Datastream.

Figure B.10: *TTF and Global LNG gas prices correlation*

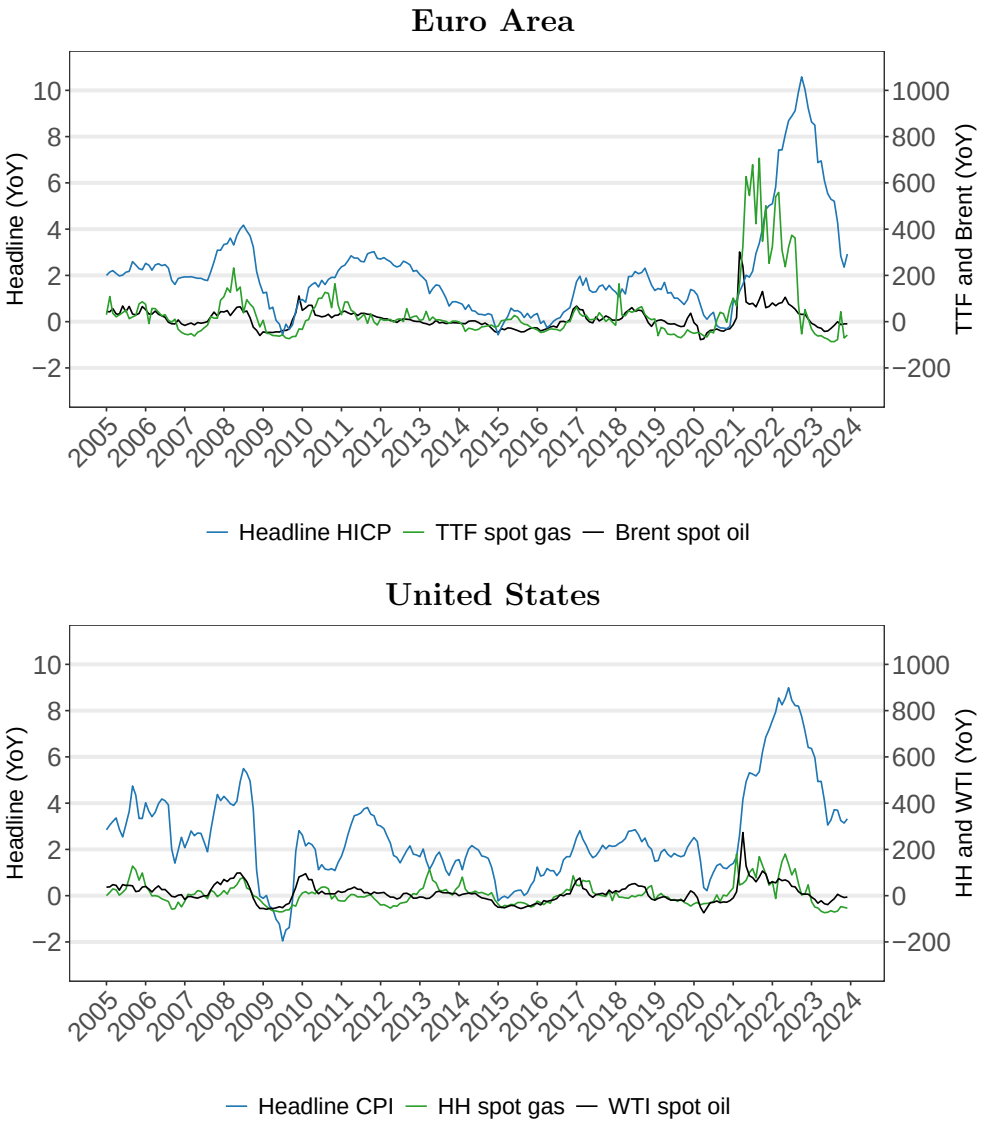


Notes: Rolling correlation between the TTF and global LNG spot prices, computed over a 24-month window. Source: Datastream.

B.4 Energy prices and inflation

Figure B.11 plots year-on-year headline inflation together with year-on-year changes in benchmark natural gas and oil spot prices. Gas prices rose far more in the Euro Area than in the United States in 2021–22, while headline inflation followed a broadly similar path in the two regions.

Figure B.11: *Inflation and energy prices*



Notes: Year-on-year (YoY) headline inflation and YoY changes in benchmark natural gas and oil spot prices, 2005–2023. The benchmarks are the Dutch TTF and Brent for the Euro Area (top panel) and the NYMEX Henry Hub and WTI for the United States (bottom panel). Gas and oil spot prices are referenced to the right-hand scale. Sources: Eurostat, FRED, and Datastream.

C Additional details on natural gas instruments

This appendix provides additional information on the construction and validation of the instruments. It first describes the gas demand instrument and the evidence supporting the heating-demand channel, then documents gas-supply event selection, an LLM-based sensitivity check, and additional diagnostics for the instrument series.

C.1 Gas demand instrument

C.1.1 Construction of gas demand instrument

ERA5 surface temperature data. The daily temperature data are taken from the ERA5 single-levels reanalysis (Hersbach et al. 2020), produced by the European Centre for Medium-Range Weather Forecasts, on a regular latitude-longitude grid of $0.25° \times 0.25°$. Average daily temperature corresponds to the *2m temperature* daily mean variable. The series are aggregated from daily to monthly by taking temporal averages. To aggregate grid-level data to regional units, we use the Database of Global Administrative Areas (GADM):[3] GADM0 for countries and GADM1 for U.S. states.

Natural gas consumption data. Annual natural gas consumption data are used to construct the aggregation weights for the aggregate temperature series. For Europe, the data are taken from Eurostat's *Supply, transformation and consumption of gas* dataset, from which we use inland consumption of natural gas at the country-year level. For the United States, we use state-level annual natural gas consumption data from the U.S. Energy Information Administration's *Natural Gas Consumption* database. In both cases, these data are used only to define predetermined aggregation weights rather than to measure contemporaneous demand conditions directly.

C.1.2 Validation of the demand channel

Alternative computations. We have explored several alternative constructions of the temperature index, all of which yield very similar series and leave the main results unchanged. In particular, we consider detrending the temperature series before seasonal adjustment, removing seasonality using calendar-day rather than calendar-month averages, constructing the index from daily maximum or minimum temperatures rather than daily averages, using alternative cross-sectional weights based on demographic or geographic factors such as night lights (Gortan et al. 2024), and computing seasonal norms over rolling windows. We also

[3] https://gadm.org/.

verify that the results are robust to using fixed gas-consumption weights computed over the full sample and to omitting cross-sectional weights altogether.

Heating and cooling needs. If temperature deviations move gas prices through heating demand, their association with gas prices should be strongest when heating needs are largest. Table C.1 confirms this pattern. In the top tercile of heating degree days, the correlation between the temperature index and the reduced-form gas-price innovation is −0.54 in the United States and −0.44 in the Euro Area, compared with 0.30 and 0.09 in the bottom tercile. Sorting months by cooling degree days yields the opposite pattern: the negative association is concentrated in low-CDD months and disappears in high-CDD months. The seasonal split is similar, with winter correlations of about −0.41 in both regions, a positive warmer-month correlation in the United States, and essentially no warmer-month correlation in the Euro Area. The correlation remains negative after residualizing both series on the informationally robust control set.

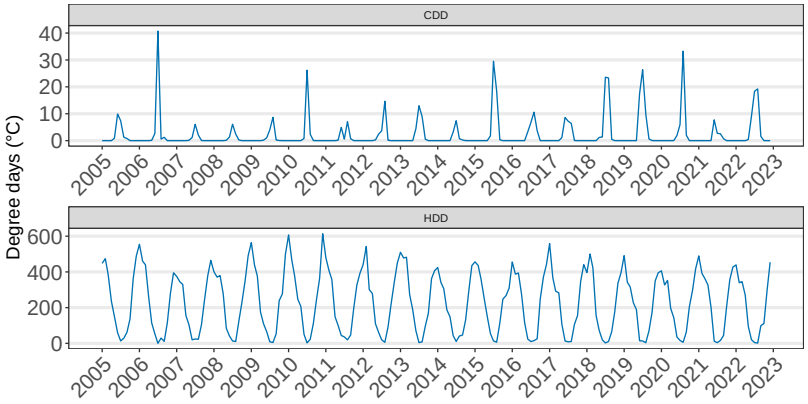
Table C.1: *Temperature index and gas prices, by heating and cooling need*

	United States	Euro Area
All months	−0.214	−0.269
Low heating degree days	0.299	0.090
High heating degree days	−0.543	−0.436
Low cooling degree days	−0.511	−0.538
High cooling degree days	0.179	0.077
Winter	−0.406	−0.419
Warmer months	0.257	−0.012
Residualized	−0.311	−0.291

Notes: Entries are correlations between the temperature index and the reduced-form innovation to the gas price from the baseline VAR, 2004–2023. Low and high are the bottom and top terciles of monthly heating and cooling degree days; Winter is October–March. The index is the version that includes warmer months, since the baseline instrument is zero from April to September by construction. The residualized row residualizes both series on the informationally robust control set of Section C.3.2. Sources: authors' calculations, Eurostat, EIA, ERA5.

Figure C.1 shows that heating needs are concentrated in winter, while cooling needs are concentrated in summer, in the European countries used for the degree-day comparison.

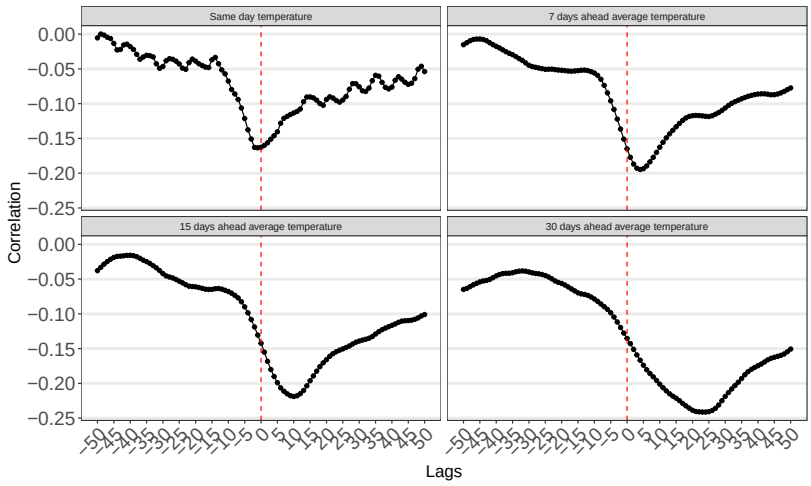
Figure C.1: *Cooling and heating degree days*



Notes: Monthly cooling degree days (CDD) and heating degree days (HDD) from Eurostat for Belgium, France, Germany, Luxembourg, and the Netherlands, aggregated using weights given by a three-year moving average of lagged gas consumption.

Anticipation effects. Figure C.2 provides evidence that temperature anticipation effects on natural gas prices are likely minimal. The top-left panel illustrates the correlation between daily temperatures and daily natural gas prices, revealing that the strongest (negative) correlation occurs contemporaneously. This indicates that price movements are primarily driven by actual temperature realizations rather than expectations about future temperatures. Additionally, the temporal correlation of daily temperatures can explain the presence of some lead correlations without necessarily implying anticipation effects. In practice, correctly forecasting average temperatures over a period may be more common than predicting daily temperatures with precision. However, the other panels show that even when considering the average temperature over several subsequent days, the strongest correlations remain contemporaneous.

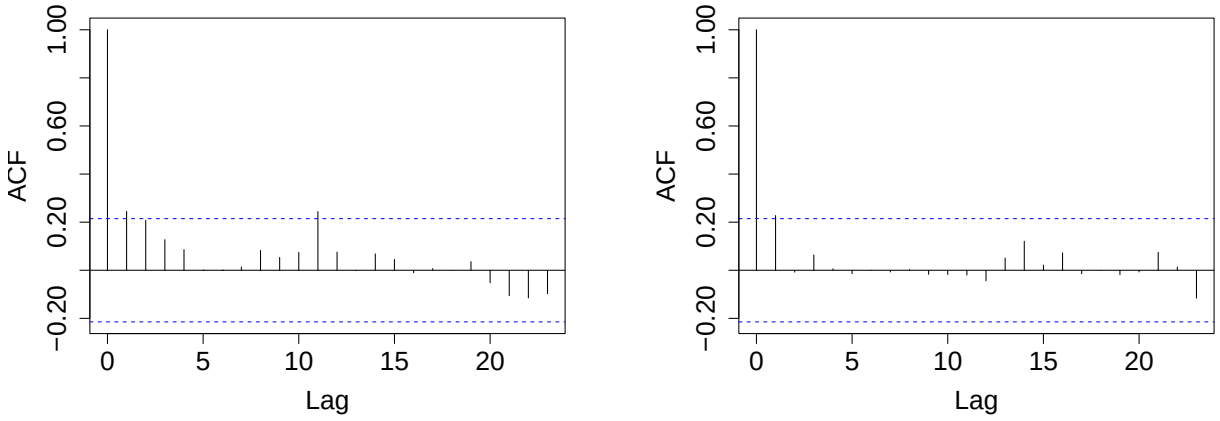
Figure C.2: *Temperature and gas-price correlations*



Notes: The figure plots correlations between the TTF spot price of natural gas and average temperatures at several leads and lags. Different panels use temperature averages computed over different time spans.

Figure C.3 reports the autocorrelation functions of the temperature indices.

Figure C.3: *Autocorrelation functions of the temperature indices*



United States Euro Area

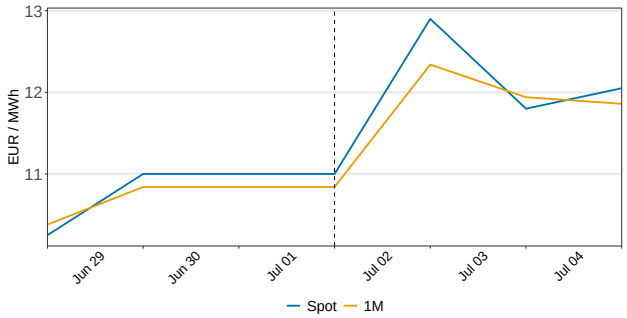
C.2 Gas supply instruments

This subsection documents the gas-supply event-selection procedure. We first discuss illustrative retained and excluded events, then present the event ledger and an LLM-based sensitivity check on event selection.

C.2.1 Illustrative retained and excluded supply events

Figure C.4 provides an additional example of a Euro Area flow-supply event. On Sunday, July 1, 2007, BP shut the U.K.'s Central Area Transmission System (CATS) pipeline as a precaution after a vessel's anchor damaged the pipeline's concrete casing, preventing J-Block North Sea gas fields from restarting exports after maintenance. The shutdown reduced short-run physical gas availability and was first priced on Monday, July 2, when TTF spot and front-month prices rose sharply.

Figure C.4: *Illustrative Euro Area flow-supply event*



Notes: TTF spot and front-month prices around the July 2, 2007, market pricing of BP's July 1 shutdown of the U.K.'s Central Area Transmission System (CATS) pipeline. The dashed vertical line marks the beginning of the price-relevant event day. Event-window surprises are measured from the previous trading-day close to the close of that day, equal to 15.9% for spot and 13.0% for front-month prices.

Illustrative exclusions. The Factiva search and manual review identify many candidate events that are clearly related to natural gas markets but are nonetheless excluded for failing one or more of the criteria detailed in Section III.B. In particular, a retained event must be new, discrete, gas-supply-specific, correctly timed, and not dominated by confounding news. For the United States, an instructive case is Hurricane Katrina. Coverage on August 29, 2005, the day of landfall on the northern Gulf Coast, reported that the storm had already shut in more than 15 Bcf of Gulf production, with the extent of damage to offshore platforms, pipelines, and treatment plants still unknown. Although the episode was clearly relevant for gas supply, it is not a clean, isolated supply event: the same coverage emphasized extreme summer demand, tightening storage, oil and gasoline disruptions, refinery damage, and potentially broad macroeconomic effects. We therefore exclude it.

Other flow-supply candidates are excluded when the supply signal is endogenous, anticipated, or dominated by demand-side and weather-related developments. Chesapeake's

announcement of voluntary output curtailments on April 17, 2009, is excluded as endogenous: although it implied lower production, the company explicitly framed the curtailments as a response to prevailing low prices, rather than as an exogenous disruption to gas availability. For Europe, the reduced Langeled nominations of April 18, 2019, are excluded because the article attributed the price move chiefly to warm weather, weak residential demand, and high wind generation, and noted that the lower nominations might reflect commercial decisions rather than an unplanned outage.

For supply-security news, we retain only events whose price response can be traced to specific information about expected European gas availability or supply security. We therefore exclude both broad geopolitical, financial, or commodity-market news and gas-specific announcements whose event-window price movements are dominated by such confounding developments. One such case is the February 28, 2022 announcement of an earlier-than-planned opening of the Poland-Lithuania Gas Interconnector, which would improve Baltic access to alternative gas supplies. Although the announcement carried favorable implications for future regional gas availability, markets were simultaneously processing major war-related financial and geopolitical news, including new restrictions on the Russian Central Bank, the removal of selected Russian banks from SWIFT, and intensified fighting in Ukraine. European gas prices rose despite the favorable interconnector news, indicating that the event-window price movement reflected these broader developments rather than the announcement itself. Another example is Gazprom's reduction in planned Power of Siberia capacity, announced on November 12, 2018. Although the news was gas-specific and involved Gazprom, the route serves Asian markets, with no direct channel established to European supply.

C.2.2 The event ledger

After the mechanical screens, selecting supply events requires judgment about whether the articles report new, gas-supply-specific information and whether competing news could explain the price movement. We document this process in two linked datasets. The *day ledger* records the screening and review outcome for each of the 7,305 calendar days in each region. The *event ledger* records each candidate identified during article review, its supporting articles and price-relevant date, and the reason it was retained or excluded. The machine-readable ledgers, a guide to their variables, and the scripts used to construct them are included in the replication files. Table C.2 summarizes the progression from calendar days to retained events. Its final rows count candidates rather than dates, since a date can carry more than one candidate.

Table C.2: *From calendar days to retained supply events*

	United States		Euro Area	
	Removed	Remaining	Removed	Remaining
Calendar days, 2004–2023		7,305		7,305
Dates without a usable gas-price change	25	7,280	80	7,225
Dates excluded by confounder screens	2,093	5,187	1,867	5,358
Dates below the 2% gas-price threshold	1,619	3,568	2,337	3,021
Dates without retrieved gas-related articles	978	2,590	587	2,434
Dates with no candidate recorded	2,476	114	2,176	258
Candidate events recorded		121		285
Candidates excluded from instruments	96	**25**	222	**63**

Notes: Date exclusions are sequential: a date meeting multiple criteria is counted only at the first criterion it fails. "Without a usable gas-price change" includes dates on which neither the spot nor the front-month change can be computed and dates whose price change would span more than four trading days. "Excluded by confounder screens" includes temperature-driven months, OPEC and monetary-policy announcement days, March–May 2020, and, for the United States, dates on which the oil-price surprise exceeds one standard deviation at one or more of the maturities used and carries the same sign as the gas-price surprise at every maturity. A date falls below the 2% threshold when neither the absolute spot nor front-month price change exceeds 2%. The upper block counts dates; the final two rows count candidate events. "Dates with no candidate recorded" have retrieved gas-related articles but no discrete event plausibly relevant to gas supply. A date may contain more than one candidate, including candidates subsequently excluded at review.

Table C.3 shows why the candidates counted in the final rows of Table C.2 were retained or excluded. Weather-related episodes account for more than half of U.S. exclusions, chiefly Gulf of Mexico storms and winter freeze-offs, whose supply effects are difficult to separate from demand and broader disruptions. In the Euro Area, the most frequent reason for exclusion is the absence of a material supply implication, including pipeline, agreement, and project announcements too remote or too small to alter current or expected gas availability.

Table C.3: *Outcomes of article review for candidate events*

Review outcome	U.S.	Euro Area
Retained		
Flow-supply event	25	28
Supply-security news		35
Excluded: reason documented		
No material supply implication	8	59
Weather-related confounding	53	4
No new or unexpected supply information	12	46
Other same-day news dominates	5	32
Duplicate of a recorded event	6	26
Inconsistent price response	5	24
Not a gas-supply event	1	16
No link to regional gas availability	2	5
Information date uncertain	0	6
Price-driven supply action	2	2
News priced on an excluded date	2	2
Total candidates	**121**	**285**

Notes: Each candidate is counted once; an exclusion is coded by its primary reason when several criteria apply. Flow-supply events affect current or near-term gas availability; supply-security news concerns expected future availability. Exclusion codes are: "No material supply implication" (too remote or small to affect current or expected regional availability); "Weather-related confounding" (weather also affects demand or activity); "No new or unexpected supply information" (follow-up or continuing coverage, or a planned or previously priced development); "Other same-day news dominates" (competing news explains the price move); "Duplicate of a recorded event" (same disruption recorded elsewhere); "Inconsistent price response" (wrong-signed price move); "Not a gas-supply event" (demand, storage, price, or commercial news); "No link to regional gas availability" (no established regional channel); "Information date uncertain" (market-information date unknown); "Price-driven supply action" (voluntary response to prices); "News priced on an excluded date" (price-relevant date failed a mechanical screen).

C.2.3 Large-language-model checks

The preceding section documents our event-selection decisions in the day and event ledgers. To assess whether the results depend on those decisions, we use a large language model to repeat the event-selection exercise on the same Factiva articles under the criteria of Section III.B, without access to our retained-event list.[4] We then rebuild the instruments from alternative event sets and re-estimate the impulse responses.

[4]We use the OpenAI API model identifier `gpt-5.4` with temperature set to zero; no seed or `top_p` value is specified. The U.S. and Euro Area classification runs began on 8 and 10 April 2026, respectively. The replication materials provide the prompts, settings, code, and saved classifications.

The classification has two stages. First, the model screens Factiva articles for each eligible date for possible supply events and same-day confounders. We deduplicate candidates identified on the same date and select up to three supporting articles published within one day of each candidate's provisional date. Using those articles, the second pass classifies each candidate as keep, exclude, or uncertain and records its event type, price-relevant date, and expected price direction. Box C.2.3 summarizes the principal decision rules rather than reproducing the prompts verbatim.

To assess sensitivity to event selection, we compare three alternatives to the baseline. Here, classifier-selected events are candidates labeled `keep`. The *LLM-filtered baseline* contains events selected by both our manual procedure and the classifier; the *augmented baseline* contains events selected by either; and the *LLM-selected set* contains the classifier's selections alone. Each set applies the baseline mechanical and price-sign screens.[5] Table C.4 reports the event counts, and Figures C.5 to C.7 compare the resulting responses with the baseline.

Table C.4: *Event counts under alternative selection rules*

	Euro Area		United States
Event set	Flow supply	Supply-security news	Flow supply
Baseline	28	35	25
LLM-filtered baseline	27	20	19
Augmented baseline	81	80	62
LLM-selected	80	65	56

The LLM-filtered baseline excludes 22 of the 88 baseline events: one Euro Area flow event, 15 Euro Area supply-security news events, and six U.S. flow events. Across all three alternative event sets, the impulse responses remain broadly similar to the baseline. This suggests that the main response patterns do not hinge on whether we retain only events selected by both procedures, supplement the baseline with classifier-selected events, or use classifier selections alone.

[5] Some U.S. flow events concern hurricanes or tropical storms, which can generate reports on several dates. For each named storm, the alternative sets retain the first report that passes the screens, so continuing coverage is not coded as separate supply shocks.

Summary of the classification prompts

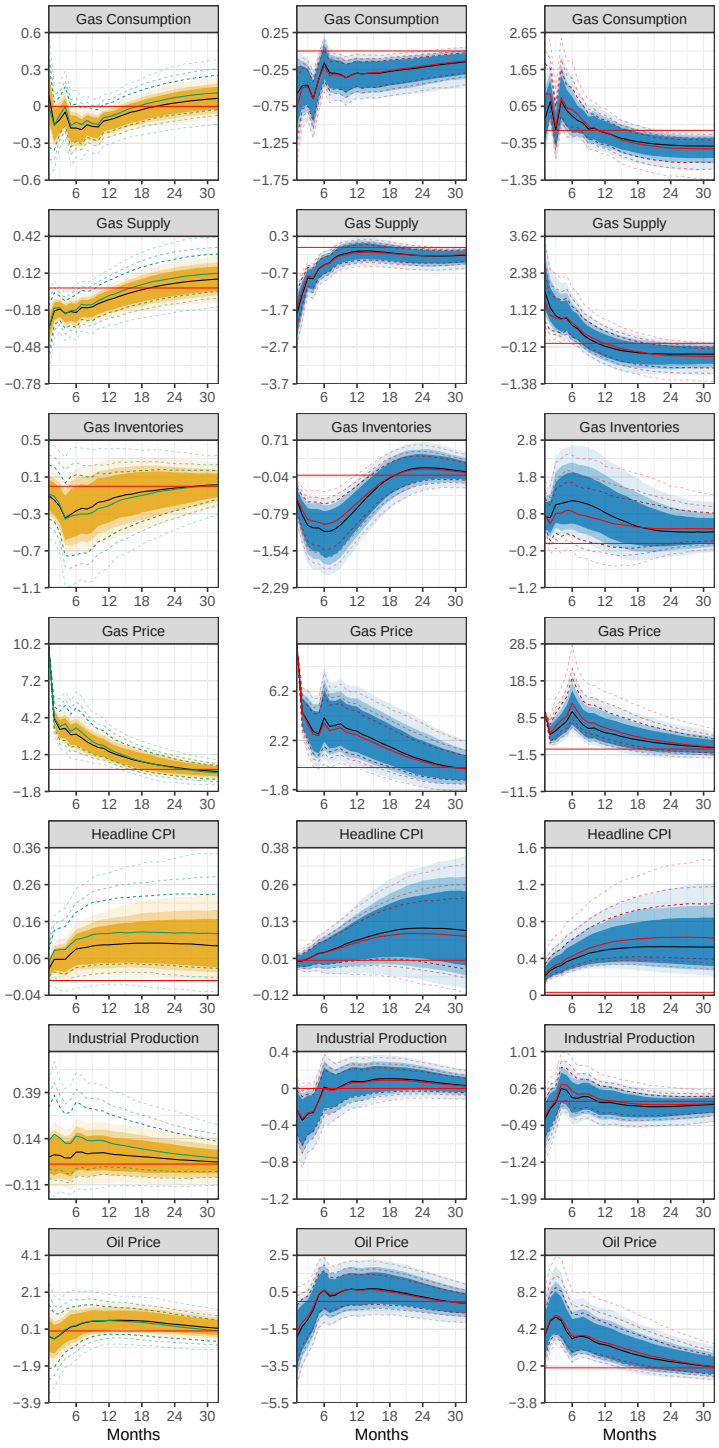
First pass, one call per date. You are screening the Factiva articles published on one day for candidate exogenous natural-gas supply events relevant for the region. This is a permissive prescreen for supply events and a selective prescreen for confounders; it is not a final decision. Most days contain no such event: return none when the material does not describe one, and at most two candidate supply events and two candidate confounders otherwise.
A candidate supply event is a discrete, plausibly exogenous development that could change current or expected future gas availability for the region. For Europe, classify it as exactly one of:

- *Flow disruption*: an unexpected change in current or very near-term physical gas availability, such as an unplanned outage, production shut-in, pipeline or LNG disruption, or unexpected restoration;
- *Supply news*: unexpected, credible news that changes expected future gas availability or supply security before physical flows change.

For the United States, consider flow events only. At the screening stage, consider a development outside the region only if the article explicitly links it to regional gas availability or gas prices; final inclusion requires a concrete regional gas-supply channel. Report as confounders only same-day developments that could plausibly account for the price move.

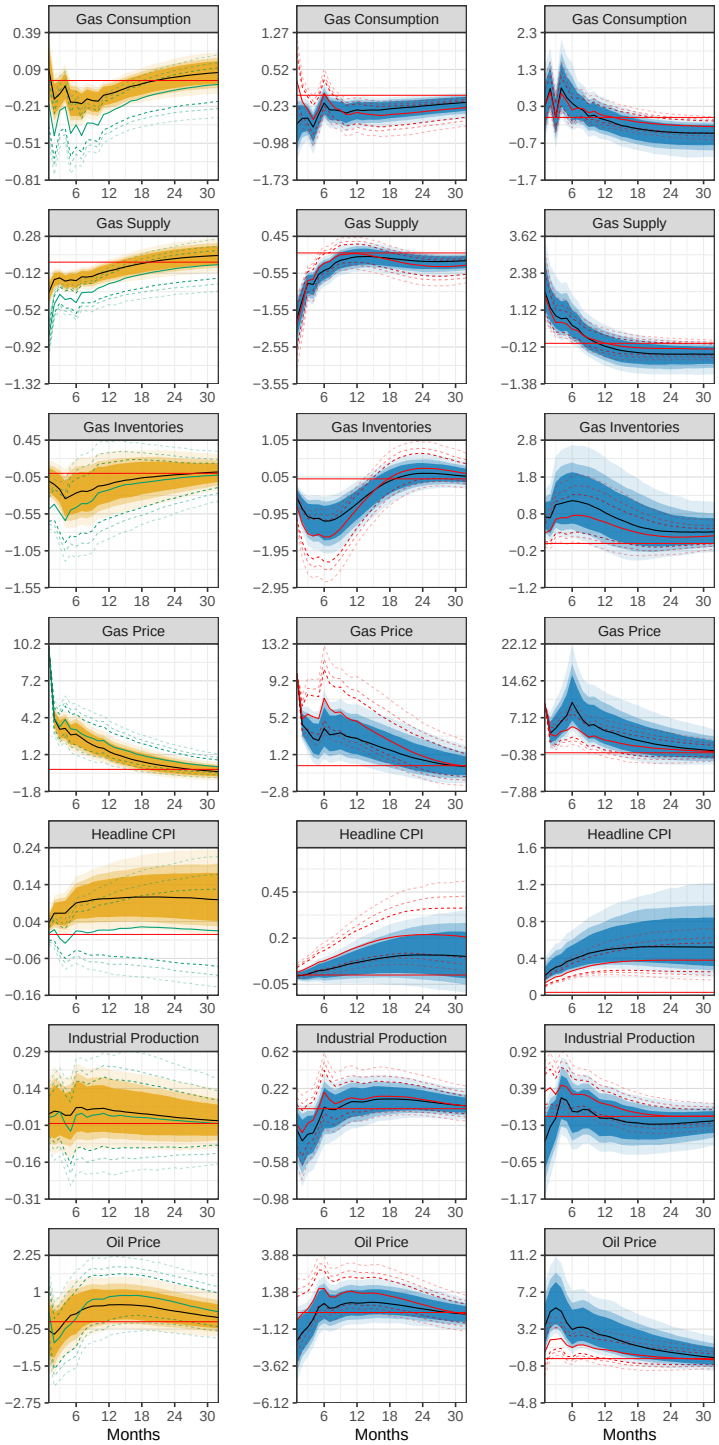
Final review of each candidate. You are reviewing the retrieved articles for one candidate event. Use only these articles; do not use outside knowledge. Return `keep`, `exclude`, or `uncertain`, with the event type, the price-relevant date, the announcement and event dates, the direction in which the event should move prices, and a short summary.
Keep the event only if the articles support a real, discrete, plausibly exogenous supply event that represents new information reaching the market. Exclude when the articles mainly describe demand, prices, positioning, routine storage or inventory movements, macroeconomic or oil news; when maintenance has no unexpected extension or new supply effect; when the development is stale or speculative; or when the link to gas supply in the region is not established. Mark as uncertain the cases in which the evidence points to a target event but is too mixed or incomplete to decide cleanly.

Figure C.5: *Comparison of baseline responses with those obtained from the LLM-filtered baseline*



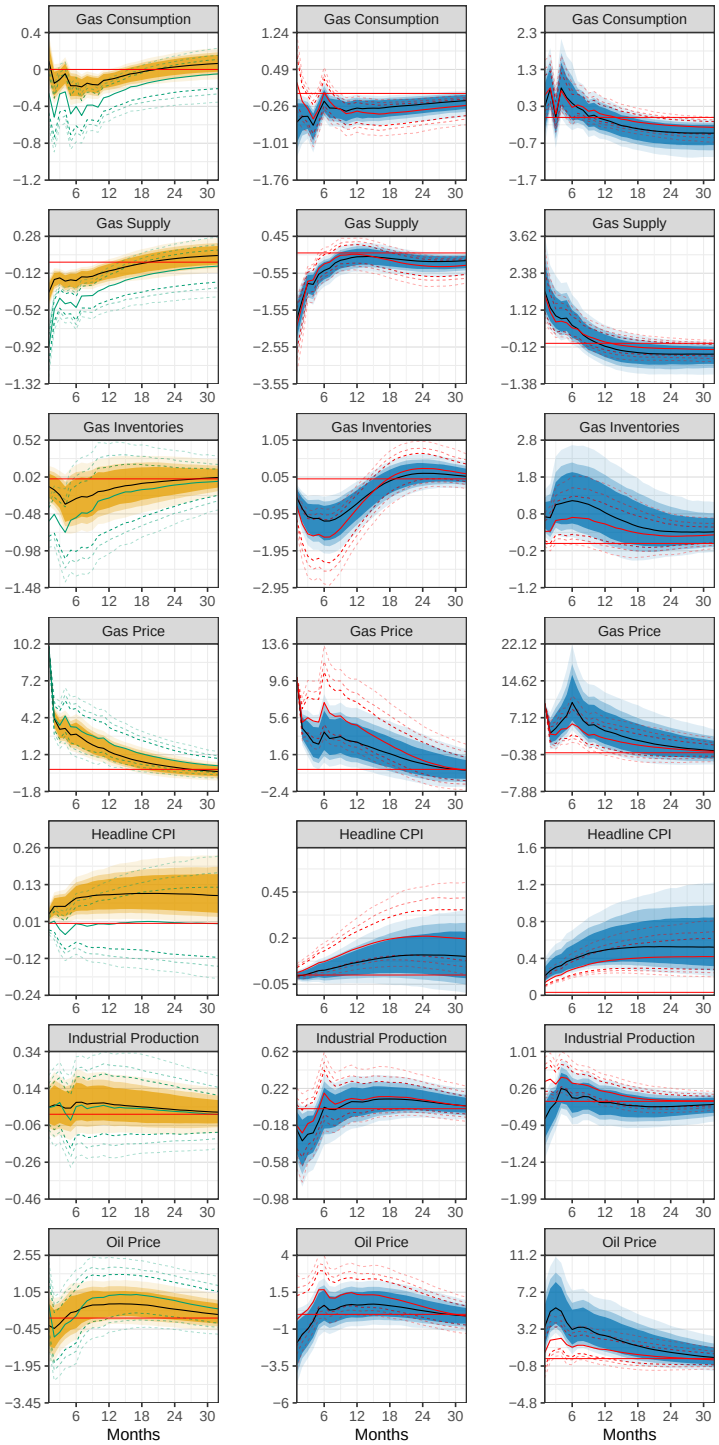
Notes: Black solid lines show baseline median responses, with orange (United States) and blue (Euro Area) shaded credible bands. Green lines show the alternative U.S. responses, and red lines the alternative Euro Area responses.

Figure C.6: *Comparison of baseline responses with those obtained from the augmented baseline*



Notes: Black solid lines show baseline median responses, with orange (United States) and blue (Euro Area) shaded credible bands. Green lines show the alternative U.S. responses, and red lines the alternative Euro Area responses.

Figure C.7: *Comparison of baseline responses with those obtained from the LLM-selected event set*



Notes: Black solid lines show baseline median responses, with orange (United States) and blue (Euro Area) shaded credible bands. Green lines show the alternative U.S. responses, and red lines the alternative Euro Area responses.

C.3 Additional instrument diagnostics

This subsection reports three complementary validity checks. We examine whether the instruments comove with other structural shocks, construct informationally robust versions that account for observable confounding information, and test whether the supply surprises are serially correlated or predictable from lagged macroeconomic and financial variables.

C.3.1 Correlations with other shocks

Table C.5 compares the instruments with established measures of oil, monetary-policy, uncertainty, financial, and geopolitical shocks. Across the 60 entries, the largest absolute correlation is 0.18, and only two are different from zero at the 5% level, fewer than the three false positives expected under 60 independent tests. Moreover, these isolated correlations do not point to a common economic channel that would systematically confound the instruments.

Table C.5: *Correlations between instruments and external shock measures and indicators*

Source	Indicator	US demand		US flow-supply		EA demand		EA flow-supply		EA news		n
		ρ	p-value	ρ	p-value	ρ	p-value	ρ	p-value	ρ	p-value	
Kilian (2009)**	Oil supply	0.02	0.76	0.01	0.84	0.03	0.68	-0.04	0.58	-0.03	0.64	237
Kilian (2009)**	Aggregate demand	-0.08	0.25	0.06	0.34	0.02	0.72	-0.06	0.35	-0.04	0.57	237
Kilian (2009)**	Oil-specific demand	-0.10	0.11	0.00	0.97	-0.10	0.14	-0.05	0.43	0.07	0.29	237
Baumeister and Hamilton (2019)*	Oil supply	0.06	0.39	-0.03	0.67	0.06	0.36	0.01	0.94	-0.07	0.29	237
Baumeister and Hamilton (2019)*	Oil demand	-0.01	0.84	0.02	0.73	-0.05	0.43	0.01	0.89	-0.05	0.44	237
Känzig (2021)**	Oil supply expectations	-0.03	0.68	-0.01	0.89	-0.04	0.58	-0.05	0.41	0.02	0.75	237
Känzig (2023)	Carbon-policy shock	0.17	0.01	0.05	0.46	0.02	0.71	0.01	0.86	0.05	0.45	237
Altavilla et al. (2019)*	Target monetary policy (EA)	0.10	0.13	0.00	0.98	0.08	0.20	0.01	0.93	0.07	0.31	237
Bauer and Swanson (2023)*	Monetary policy surprise (US)	-0.09	0.16	-0.11	0.10	-0.05	0.42	0.04	0.54	0.18	0.00	237
Bloom (2009)**	VXO-VIX	0.10	0.14	0.07	0.29	0.08	0.19	0.00	0.95	0.06	0.38	237
Gilchrist and Zakrajšek (2012)*	Corporate credit spread index	-0.06	0.35	-0.02	0.77	-0.10	0.13	-0.01	0.89	-0.07	0.31	237
Caldara and Iacoviello (2022)	Geopolitical risk index	0.11	0.10	0.00	0.98	0.11	0.13	0.07	0.31	0.08	0.28	201

Notes: The table reports Pearson correlation coefficients (ρ) and p-values for the U.S. demand and flow-supply instruments and for the Euro Area demand, flow-supply, and supply-security news instruments with external shock measures and indicators from the literature. The p-values correspond to two-sided tests for the null hypothesis of no correlation. The sample size (n) varies across shocks, as some are unavailable or not easily extendable to the full sample (2004–2023).

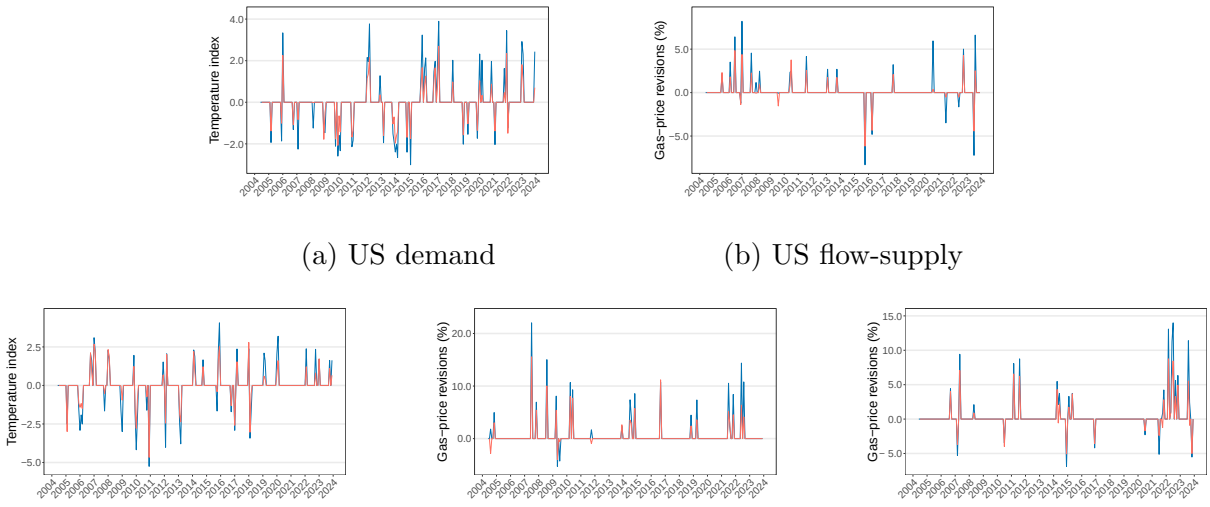
*Extended by the original authors beyond the original sample used in the published paper.
**Extended by the authors of this study.

C.3.2 Informationally robust instruments

Figure C.8 compares the baseline instruments with the informationally robust series. These series are obtained by residualizing all instruments with respect to monthly energy, macro-financial, and weather-related controls, and the supply instruments also with respect to daily oil and food-price surprises. The refined series closely track the baseline series. As

shown in Figure E.2, they also produce impulse responses close to the baseline, suggesting that informational confounding is not material for the results.

Figure C.8: *Informationally robust gas surprise series*



(a) US demand (b) US flow-supply

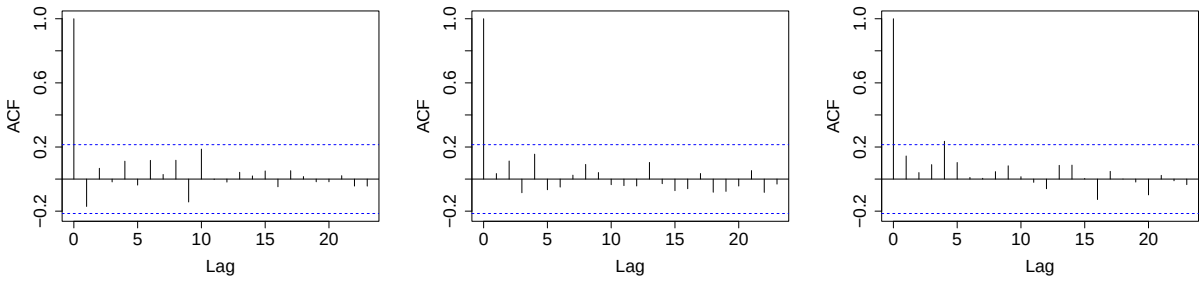
(c) EA demand (d) EA flow-supply (e) EA news

Notes: This figure shows the baseline gas surprise series (blue line) alongside the informationally robust gas surprise series (red line). Panels (a)–(b) report the US demand and flow-supply instruments. Panels (c)–(e) report the Euro Area demand, flow-supply, and supply-security news instruments.

C.3.3 Serial correlation and forecastability of supply surprises

Figure C.9 shows little serial dependence in the supply surprise series, with autocorrelations at nonzero lags remaining close to zero. We then assess predictability using Granger's causality tests. Table C.6 shows that none of these series is forecastable from its own lags or from the lagged gas-market, macroeconomic, or financial variables considered, either individually or jointly.

Figure C.9: *Sample autocorrelation functions of supply surprise series*



United States flow-supply | Euro Area flow-supply | Euro Area supply-security news

Table C.6: *Granger causality tests*

Variable	US flow-supply	EA flow-supply	EA news
Instrument Lags	0.13	0.81	0.67
Gas price	0.79	0.75	0.96
Oil price	0.81	0.95	0.71
Gas consumption	0.87	0.76	0.77
Gas total available supply	0.97	0.88	0.95
Gas inventories	0.95	0.81	0.66
Headline CPI	0.11	0.75	0.10
Industrial production	0.40	0.87	0.53
Interest rate	0.61	0.35	0.99
Nominal exchange rate	0.40	0.17	0.52
Stock market (SP500/STOXX50E)	0.36	0.59	0.64
Financial volatility	0.53	0.67	0.23
Real economic activity	0.53	0.96	0.94
Joint Test	0.85	0.99	0.54

Notes: The table reports p-values from Granger causality tests for the gas supply surprise series. Each row tests whether lags of the indicated variable help predict the instrument; the final row reports the joint test across all predictors. Series are rendered stationary by taking first or second differences as required. The tests include 12 lags and a constant term. The interest rate is EURIBOR for the Euro Area and the effective FFR for the United States; the stock market indices are the EURO STOXX 50 and the S&P 500; real economic activity is the index of Kilian (2009).

D Short-run gas market elasticities

This appendix reports the estimated structural price elasticities of gas demand and supply and compares them with previous estimates. Short-run gas demand is relatively inelastic in both regions, with estimates of -0.04 for the United States and -0.08 for the Euro Area; the corresponding 68% credible bands are $[-0.07, -0.01]$ and $[-0.11, -0.05]$. Gas supply is more responsive, with elasticities of 0.23 and 0.31, respectively; the corresponding credible bands are $[0.15, 0.40]$ and $[0.21, 0.52]$.

Table D.1 summarizes selected estimates from the literature. Our demand elasticities are

at the less elastic end of the reported range and are smaller in absolute value than the short-run elasticity of −0.18 reported in the meta-analysis of Labandeira, Labeaga, and López-Otero (2017). On the supply side, our estimates are close to the higher estimates reported by Casoli, Manera, and Valenti (2024) and Wiggins and Etienne (2017), and well above the near-zero elasticities reported by Rubaszek, Szafranek, and Uddin (2021) and Farag (2024). These comparisons should nevertheless be interpreted cautiously, as the studies differ in their measures of gas quantities, sample periods, data frequencies, and identification strategies.

Table D.1: *Selected previous estimates of short-run gas market elasticities*

Study	**Market and sample**	**Demand**	**Supply**	**Identification**
Wiggins and Etienne (2017)	United States 1976Q1–2015Q2	Prod. $[-0.18, -0.08]^{**}$	Prod. $[0.15, 0.30]^{**}$	Sign restrictions
Rubaszek, Szafranek, and Uddin (2021)	United States 1978Q1–2020Q3	Cons. −0.42	Prod. 0.01	Priors and sign restrictions*
Farag (2024)	United States 1992M1–2023M10	Cons. −0.18	Prod. + imp. 0.02	Priors and sign restrictions*
Casoli, Manera, and Valenti (2024)	Europe 2010M1–2022M7	Cons. −0.47	Prod. 0.34	Priors and sign restrictions*

Notes: Entries report the quantity used in the original study and the corresponding short-run price elasticity. Cons., Prod., and imp. denote consumption, production, and imports, respectively.

*Baumeister–Hamilton framework (Baumeister and Hamilton 2015; Baumeister and Hamilton 2019).
**Wiggins and Etienne (2017) report a range from a time-varying parameter model.

E Robustness of the baseline results

This appendix reports sensitivity exercises for the baseline results. We first consider alternative constructions of the instruments, including informationally robust refinements, alternative event-selection thresholds, leave-one-out exercises, and alternative temperature-based demand instruments. We then assess the identifying assumptions by testing invertibility, estimating an internal-instrument specification, examining instrument relevance, and relaxing the orthogonality restriction across shocks. Finally, we consider alternative estimators, lag lengths, specifications, and samples, which leave the baseline shock construction unchanged but alter the empirical system or estimation approach.

Unless stated otherwise, each robustness figure compares an alternative exercise with the

baseline model, uses the same sample and variables as the baseline, and normalizes each shock to raise the real price of natural gas by 10% on impact. The baseline median response is drawn as a black solid line, with credible bands shaded orange for the United States and blue for the Euro Area. Alternative estimates are shown in red or green.

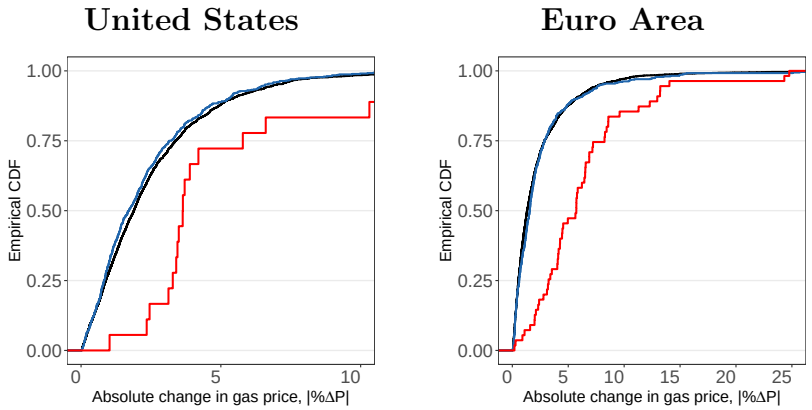
E.1 Alternative instrument construction

Informationally robust instruments. The baseline instruments could contain information about observable confounders that affect the outcome variables through channels unrelated to natural gas. We therefore apply the informationally robust refinement described in Section III.C, which residualizes the instrument series with respect to daily oil and food-price surprises, where applicable, and monthly energy, macro-financial, weather, and power-market controls. Figure C.8 compares the baseline and refined instruments, while Figure E.2 reports the associated impulse responses. The refined instruments remain close to the baseline series and yield similar responses.

Event selection and influential observations. The baseline supply instruments retain events associated with absolute gas-price surprises above 2%. Figure E.3 reports responses obtained after reconstructing the instruments using stricter 3% and 4% thresholds. Figure E.4 reports a leave-one-out jackknife, in which one nonzero instrument observation is set to zero at a time. Across both exercises, the responses remain close to the baseline, indicating that neither the screening cutoff nor any single instrument observation drives the results.

We also assess whether scheduled macroeconomic releases contaminate the event set. Such releases may move gas prices through channels unrelated to gas supply. Excluding all such dates, however, would remove 311 of the 3,021 Euro Area dates (10.3%) and 463 of the 3,568 U.S. dates (13.0%) surviving the confounder and 2% price screens. Figure E.1 compares gas-price changes on release, control, and retained supply-event days. For release and control days, two-sided tests do not reject equality of mean absolute changes at the 5% level; equivalence tests reject differences of at least 0.5 percentage points in either region. Four of the 63 retained Euro Area events—two flow disruptions and two supply-security news events—and six of the 25 U.S. events coincide with a release. Figure E.5 shows that excluding these events yields impulse responses similar to the baseline.

Figure E.1: *Gas-price changes on supply-event (red), macroeconomic-release (blue), and control days (black)*

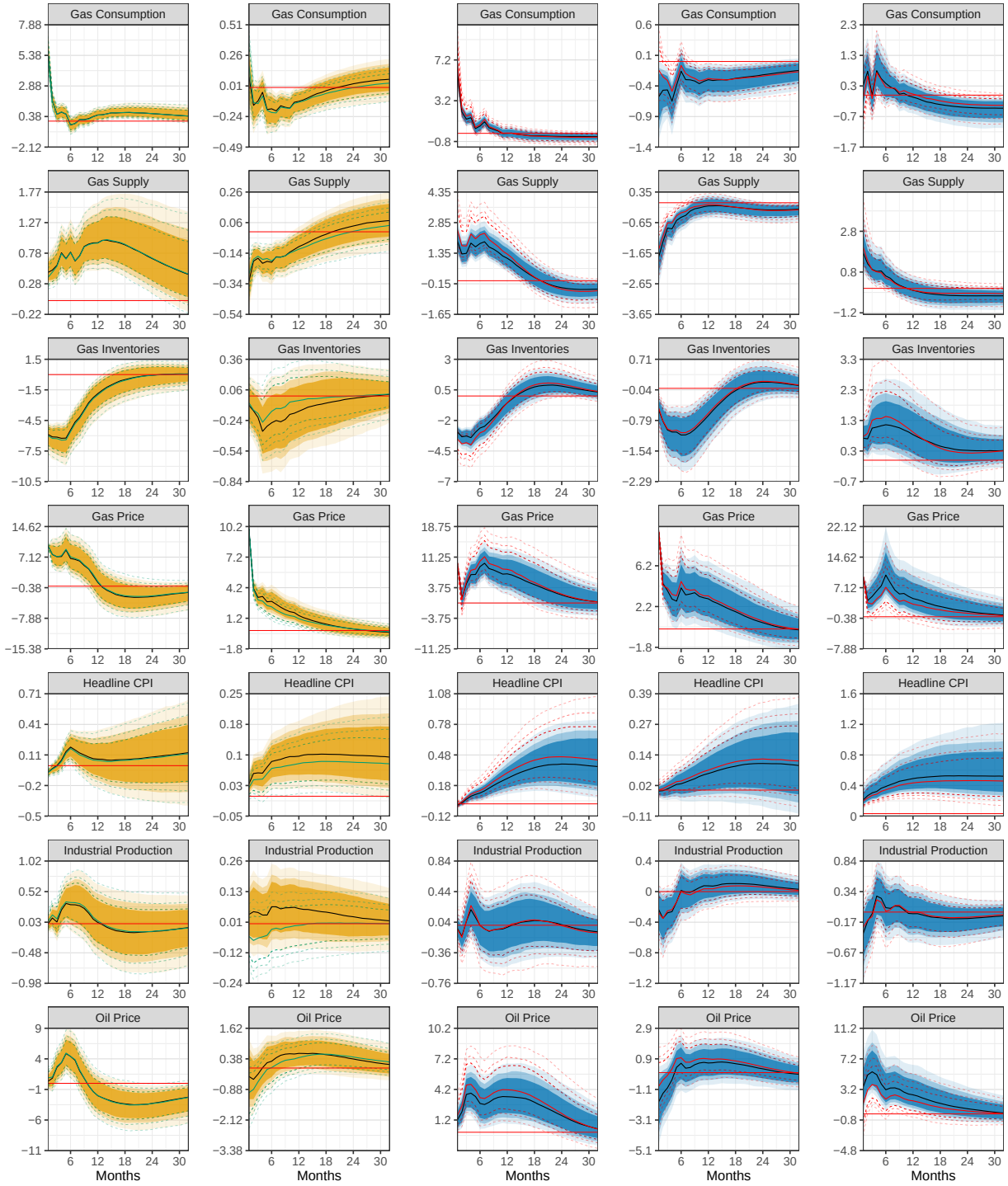


Notes: Empirical CDFs of absolute daily changes in the one-month gas-futures price. Among trading days with a usable, nonzero one-month gas-futures-price change that pass the confounder screens, we distinguish days on each region's macroeconomic-release calendar from control days with neither a release nor a retained supply event. Retained supply-event days are plotted separately, excluding those that coincide with a release. The 2% event-selection threshold is not applied to the release and control groups. Mean absolute changes on release and control days are 2.53% and 2.37% in the Euro Area, and 2.33% and 2.51% in the United States. A two-sided resampling test of equality in mean absolute price changes, based on 10,000 control-day samples drawn with replacement and matched in size to the release-day group, yields $p = 0.26$ in the Euro Area and $p = 0.06$ in the United States. Two one-sided tests (TOST) for equivalence, using bounds of ± 0.5 percentage points, reject mean differences at least that large ($p = 0.021$ and $p < 0.001$, respectively).

Alternative gas demand instruments. The baseline demand instrument excludes warmer months to isolate heating-related variation. Figure E.6, panel (a), reports responses obtained when observations from warmer months are retained, with the sign of the temperature index adjusted in those months to account for cooling-related gas demand. Panel (b) replaces temperature deviations with heating degree days. Both alternatives yield responses similar to the baseline.

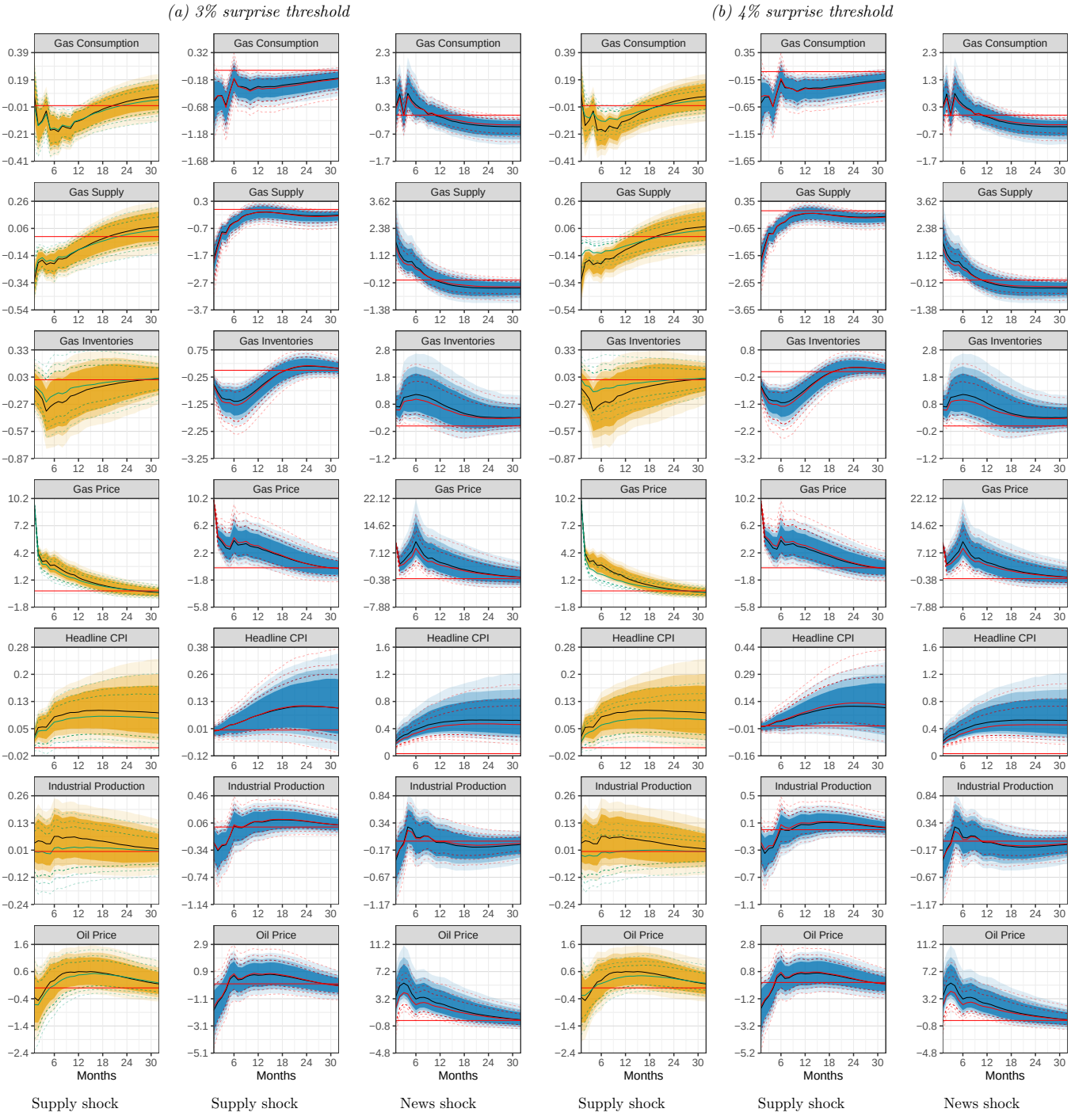
Informationally robust instruments

Figure E.2: *Comparison with baseline responses: informationally robust instruments*



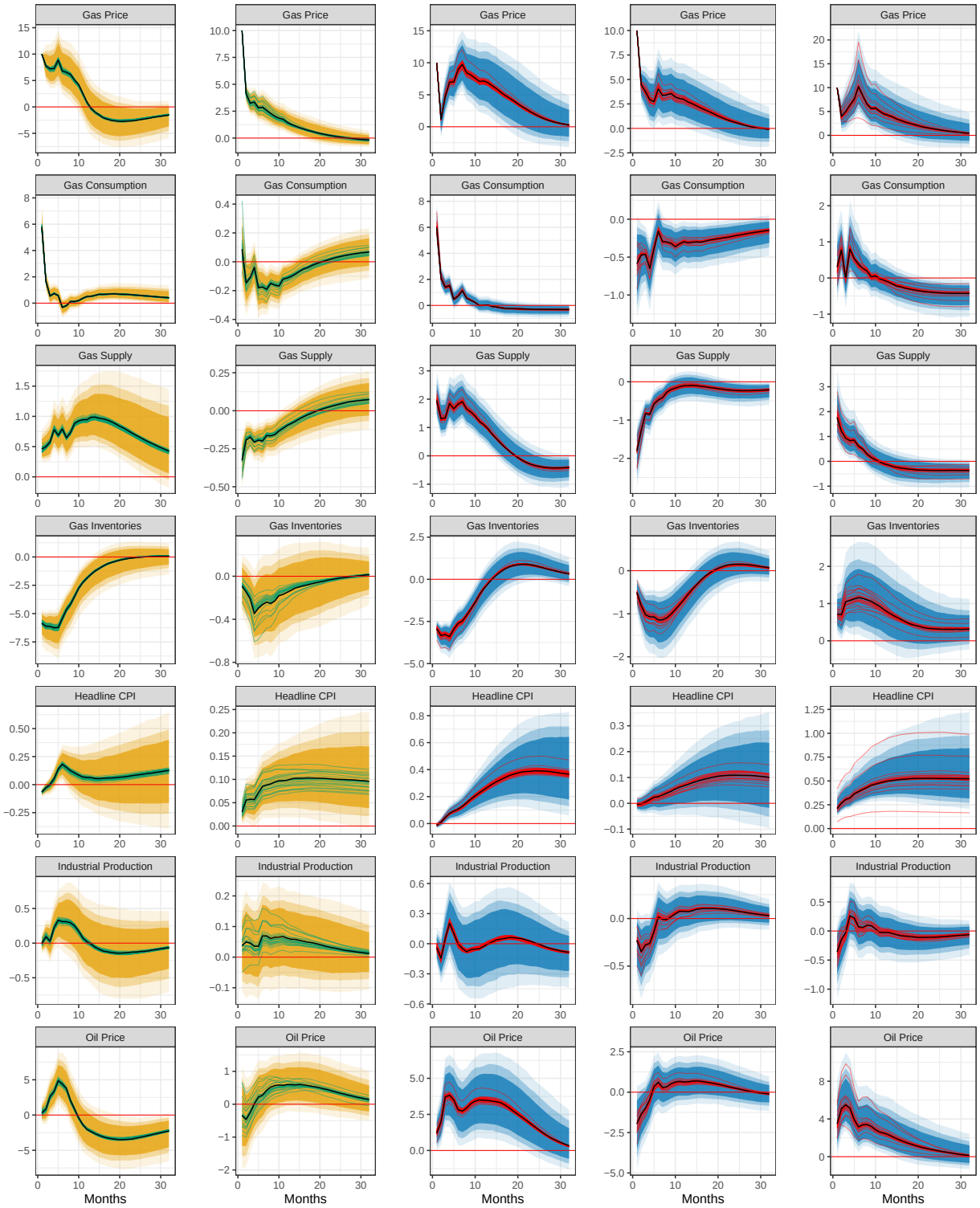
Alternative event-selection thresholds

Figure E.3: *Comparison with baseline responses: supply instruments constructed using 3% and 4% event-selection thresholds*



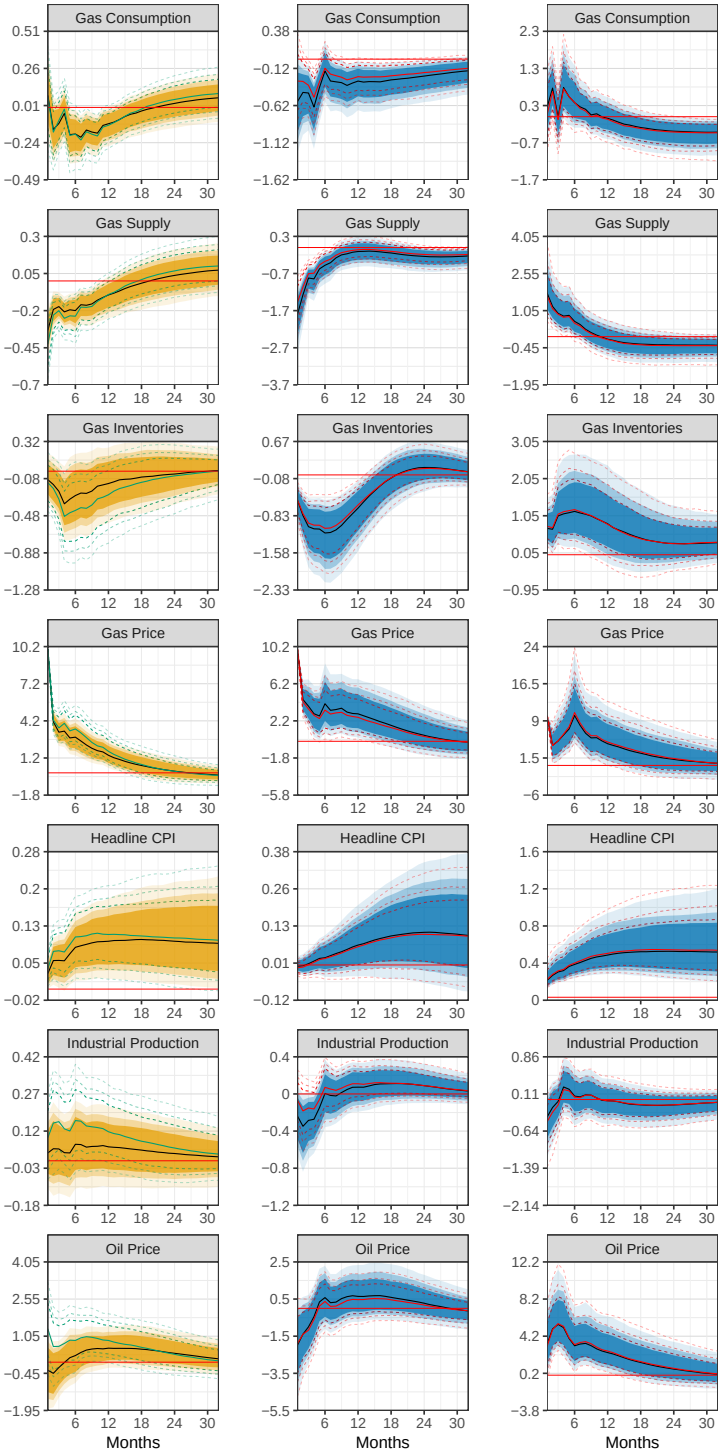
Jackknife exercise

Figure E.4: *Comparison with baseline responses: leave-one-out jackknife over nonzero instrument observations*



Instruments without events on macroeconomic release days

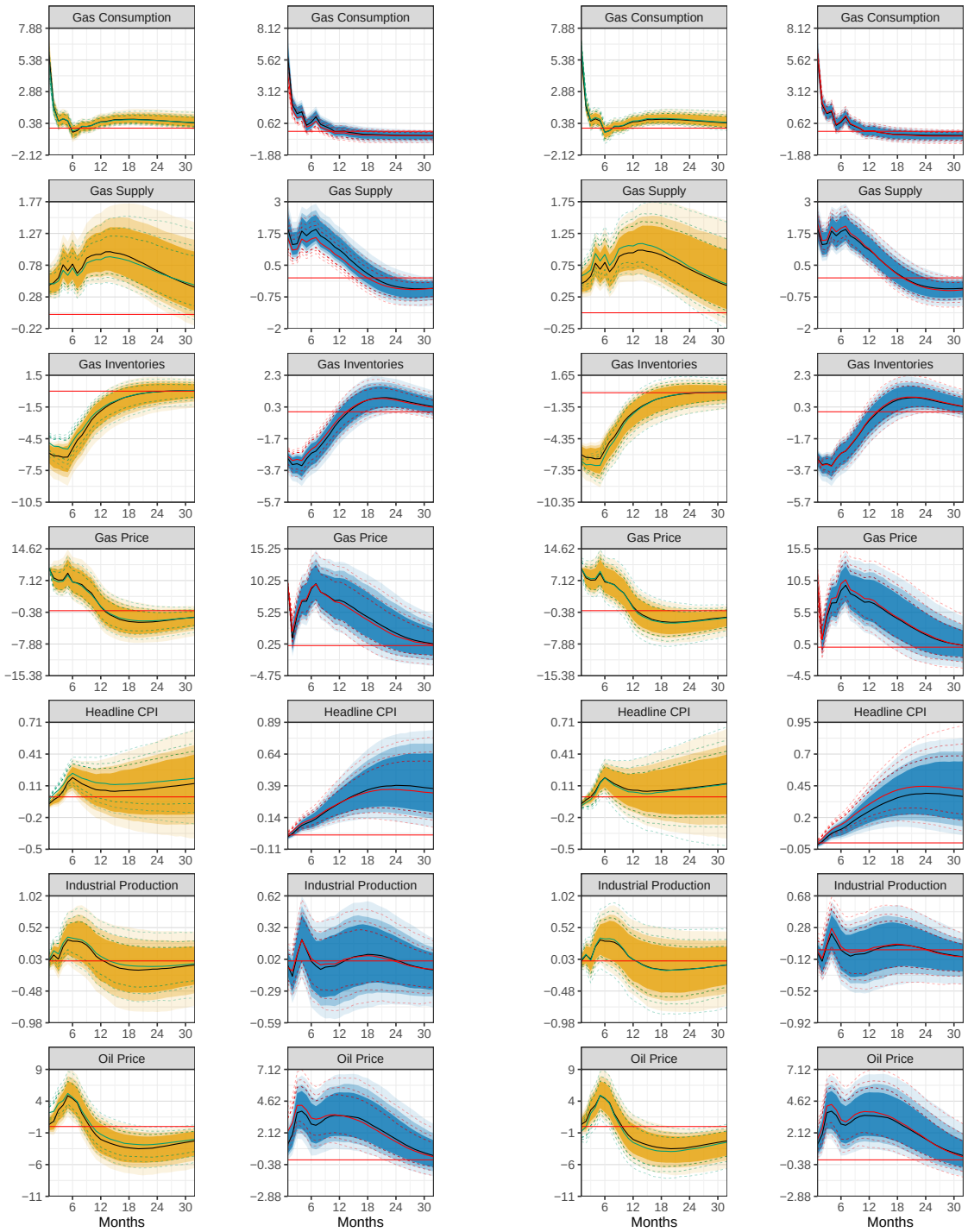
Figure E.5: *Comparison with baseline responses: excluding events on macroeconomic-release days*



Alternative demand instruments: warmer months and heating degree days

Figure E.6: *Comparison with baseline responses: alternative gas demand instruments. Panel (a) includes warmer months; panel (b) uses heating degree days.*

(a) Including warmer months *(b) Heating degree days*



E.2 Identifying assumptions

Invertibility. Table E.1 reports the invertibility tests of Plagborg-Møller and Wolf (2022) and Hamilton (forthcoming). We test whether lags of each instrument predict the VAR variables after conditioning on the baseline regressors, using specifications with six to twelve lags of the instrument and reporting both equation-by-equation and joint p-values. The joint tests provide no evidence against invertibility for any instrument. Figure E.7 reports a complementary internal-instrument specification (Ramey 2011; Plagborg-Møller and Wolf 2021). This specification orders the instrument first in the VAR, followed by the baseline variables, and avoids the invertibility requirement imposed by the external-instrument approach. The resulting responses remain close to the baseline.

Instrument relevance. Figure E.8 reports the posterior distributions of the identified impact coefficients (Caldara, Cavallo, and Iacoviello 2019; Giacomini, Kitagawa, and Read 2022), while Figure E.9 plots the relationship between each instrument and the relevant reduced-form innovation. The posterior distributions are well behaved, and the instruments display clear comovement with the reduced-form innovations used for identification. The heteroskedasticity-robust first-stage F-statistics from regressions of the gas-price reduced-form innovation on each instrument are 20.59, 16.88, and 16.22 for the Euro Area flow-supply, supply-security news, and demand instruments, and 10.01 and 25.99 for the U.S. supply and demand instruments, respectively. Together, these diagnostics support instrument relevance.

Finite-sample orthogonalization. The baseline estimates impose exact orthogonality among the identified shocks in finite samples. Figure E.10 reports responses obtained after relaxing this restriction. The responses are broadly unchanged, indicating that the finite-sample orthogonalization step is not driving the baseline results.

Invertibility tests

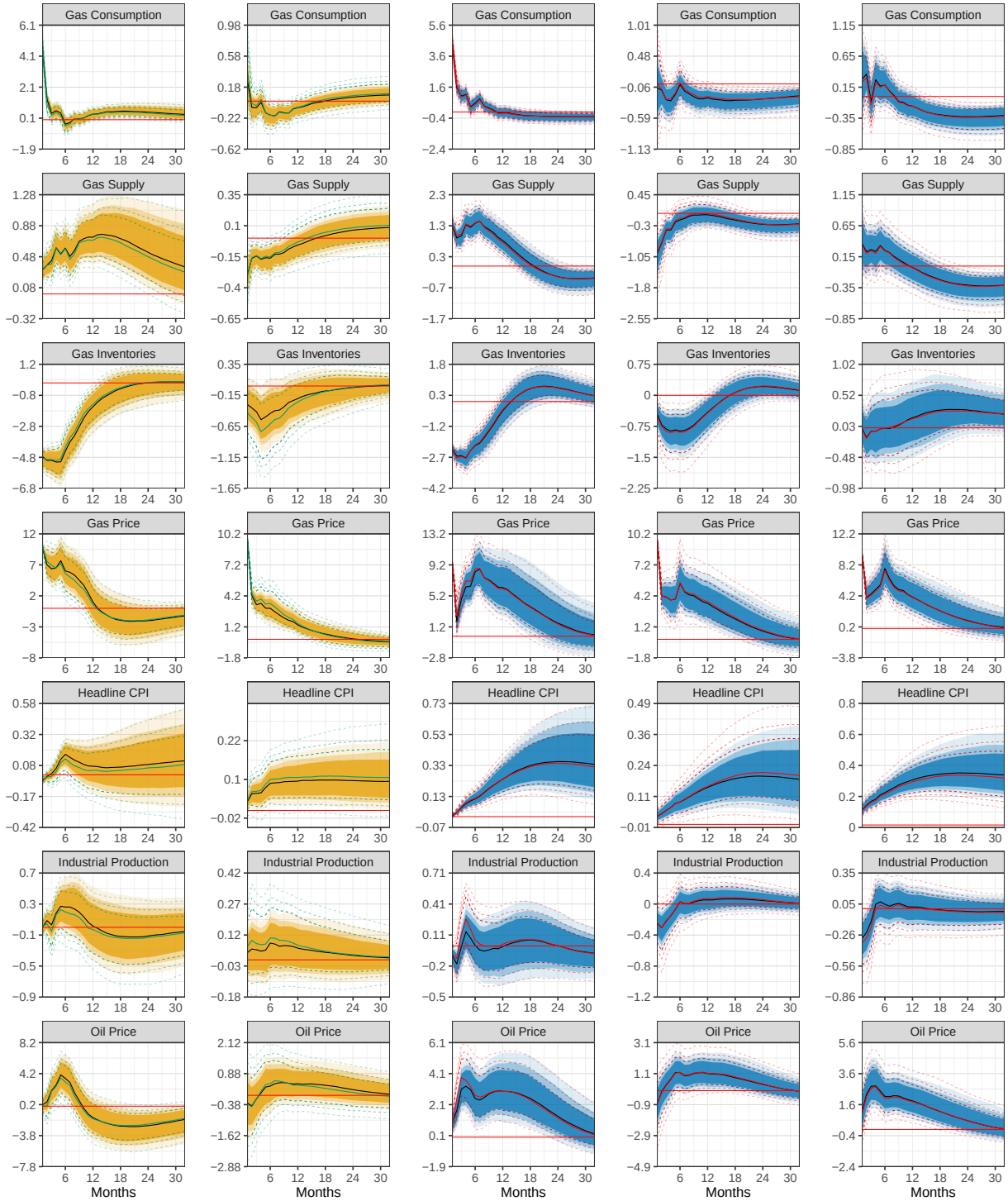
Table E.1: *Testing the invertibility assumption*

US demand								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	*Joint*
6	0.430	0.033	0.900	0.345	0.292	0.190	0.529	0.353
7	0.472	0.056	0.831	0.447	0.358	0.214	0.589	0.502
8	0.582	0.047	0.882	0.393	0.293	0.292	0.649	0.524
9	0.476	0.048	0.930	0.498	0.187	0.388	0.651	0.334
10	0.550	0.083	0.899	0.578	0.242	0.474	0.581	0.502
11	0.457	0.088	0.926	0.562	0.244	0.417	0.666	0.523
12	0.384	0.120	0.874	0.602	0.197	0.459	0.798	0.534
US flow-supply								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	*Joint*
6	0.798	0.299	0.633	0.831	0.303	0.310	0.925	0.640
7	0.794	0.238	0.301	0.890	0.280	0.244	0.817	0.526
8	0.839	0.148	0.337	0.836	0.099	0.328	0.641	0.434
9	0.827	0.213	0.290	0.870	0.121	0.429	0.657	0.542
10	0.805	0.290	0.270	0.920	0.113	0.507	0.736	0.532
11	0.726	0.334	0.326	0.953	0.135	0.591	0.756	0.511
12	0.789	0.414	0.217	0.958	0.156	0.470	0.842	0.553
EA demand								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	*Joint*
6	0.145	0.353	0.998	0.579	0.422	0.698	0.542	0.630
7	0.162	0.369	0.999	0.680	0.427	0.766	0.642	0.721
8	0.213	0.417	1.000	0.759	0.238	0.842	0.558	0.785
9	0.196	0.523	0.998	0.823	0.306	0.852	0.641	0.852
10	0.230	0.585	0.858	0.877	0.365	0.677	0.500	0.684
11	0.134	0.596	0.848	0.904	0.364	0.749	0.567	0.566
12	0.178	0.063	0.890	0.165	0.265	0.797	0.646	0.249
EA flow-supply								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	*Joint*
6	0.181	0.858	0.868	0.591	0.548	0.312	0.646	0.203
7	0.259	0.825	0.922	0.544	0.644	0.411	0.708	0.377
8	0.355	0.764	0.735	0.635	0.531	0.328	0.782	0.352
9	0.455	0.767	0.821	0.714	0.615	0.261	0.848	0.511
10	0.421	0.832	0.497	0.719	0.521	0.290	0.822	0.453
11	0.283	0.869	0.142	0.462	0.587	0.246	0.653	0.223
12	0.246	0.893	0.188	0.559	0.640	0.243	0.773	0.329
EA news								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	*Joint*
6	0.217	0.294	0.075	0.405	0.564	0.976	0.877	0.593
7	0.147	0.390	0.108	0.501	0.333	0.987	0.794	0.493
8	0.182	0.470	0.150	0.613	0.221	0.978	0.837	0.580
9	0.230	0.474	0.176	0.700	0.135	0.990	0.889	0.667
10	0.294	0.583	0.189	0.748	0.148	0.994	0.936	0.796
11	0.339	0.327	0.143	0.760	0.200	0.978	0.961	0.784
12	0.299	0.238	0.194	0.756	0.177	0.958	0.745	0.605

Notes: The table reports p-values from F-tests that the coefficients $\alpha_1, \ldots, \alpha_m$ are zero in the regression $y_{it} = \pi_i' x_{t-1} + \alpha_1 z_{t-1} + \cdots + \alpha_m z_{t-m} + \eta_{it}$. The test is conducted for each series included in the baseline VAR model and for different numbers of instrument lags, reported in the first column. The VAR includes seven lags, one more than in the baseline specification; using six lags gives similar results. Only a constant is included as a deterministic term. The last column reports the p-value of the joint system test that lags of z do not enter any equation.

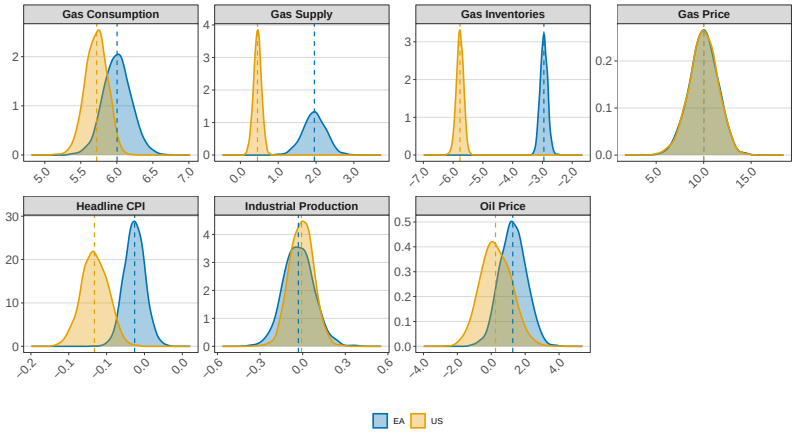
Estimation by internal instrument

Figure E.7: *Comparison with baseline responses: internal-instrument specification*

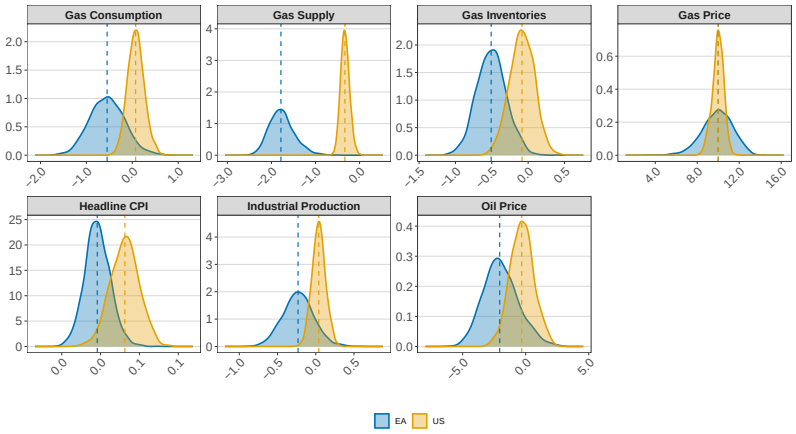


Instrument relevance

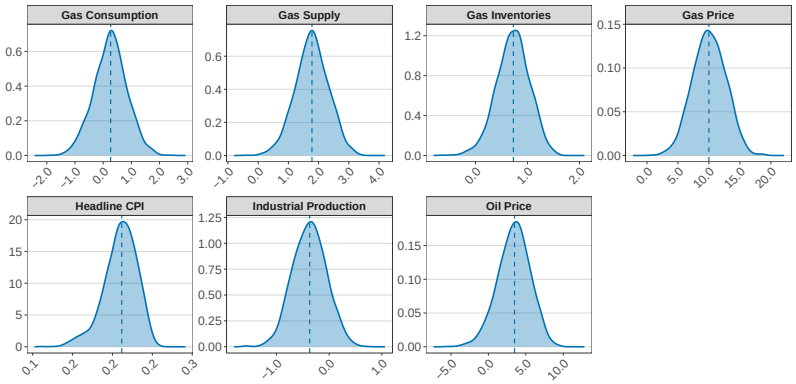
Figure E.8: *Posterior distributions of impact coefficients*



Demand shock



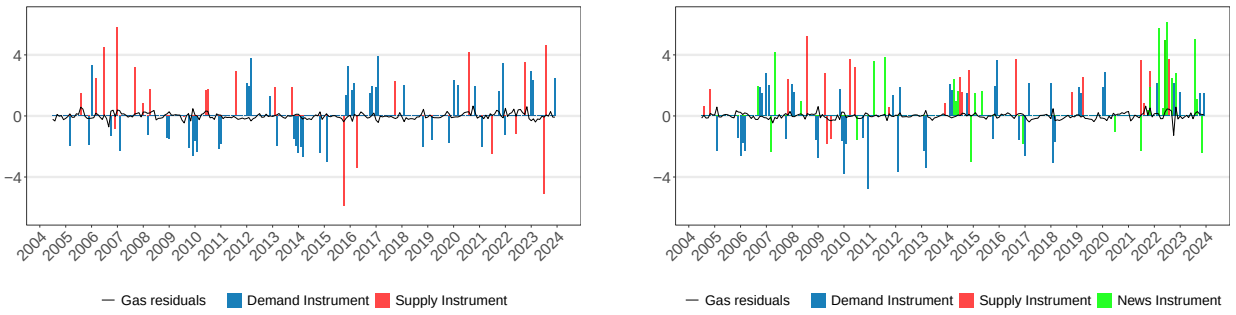
Supply shock



News shock

Notes: This figure shows the posterior distributions of the impact coefficients identified for gas demand, flow-supply, and supply-security news shocks. Dashed vertical lines indicate the median of each distribution.

Figure E.9: *Instrument strength*

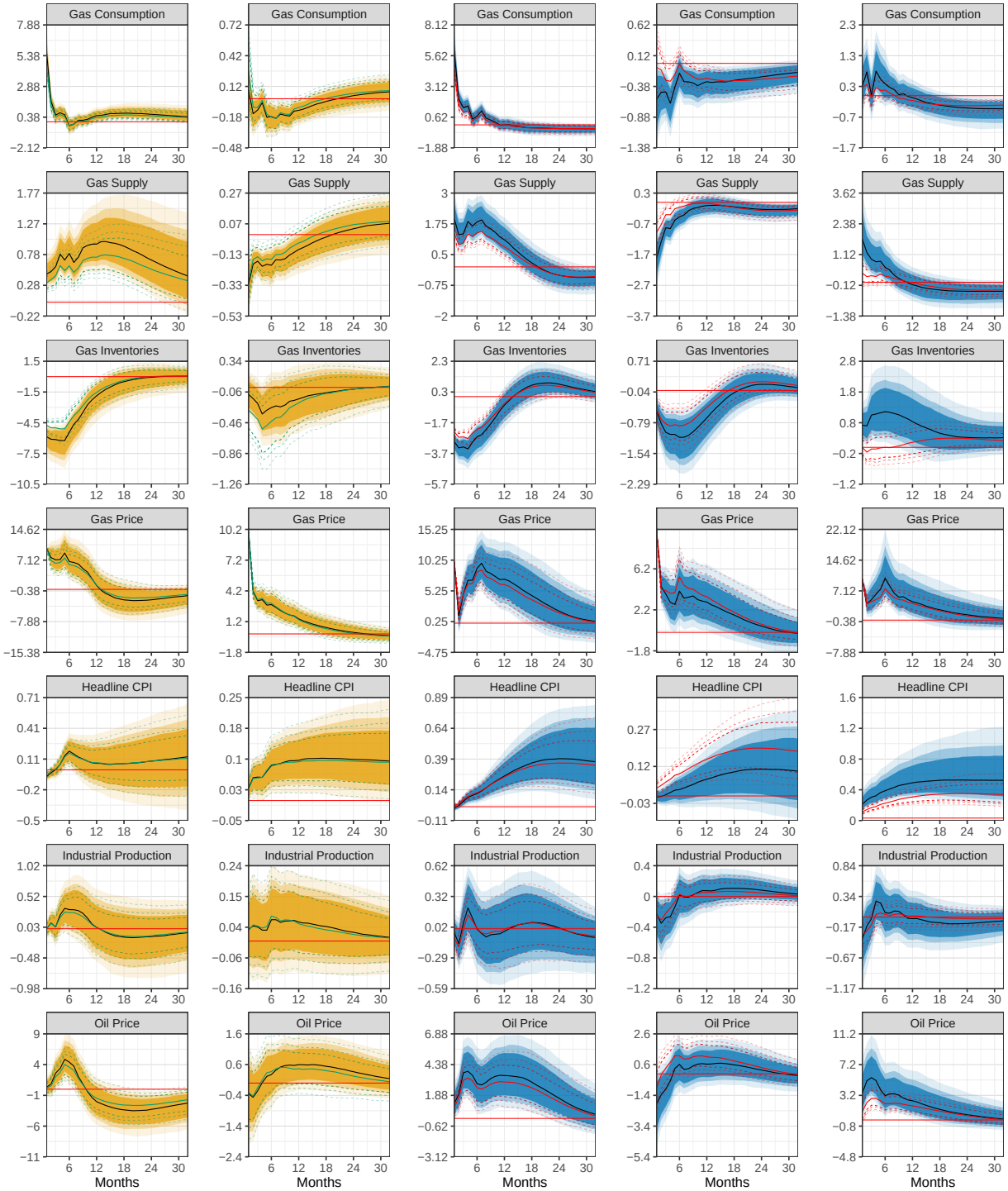


United States Euro Area

Notes: The figure shows how the gas-market instruments are related to the relevant gas-price reduced-form residuals from the baseline specification. All series are rescaled to have unit variance for comparability. The left panel reports the U.S. instruments and gas-price residuals, and the right panel reports the corresponding Euro Area series.

Baseline without orthogonalization

Figure E.10: *Comparison with baseline responses: relaxing the orthogonality restriction across shocks*



E.3 Estimation, specification, lag length, and sample

Alternative estimators. To assess the role of Bayesian regularization and the VAR representation, we re-estimate the model by OLS and with the Bayesian local-projection approach of Ferreira, Miranda-Agrippino, and Ricco (2025), using Minnesota-type priors. Figures E.11 and E.12 show similar dynamic patterns, although the alternative estimates are sometimes less precise.

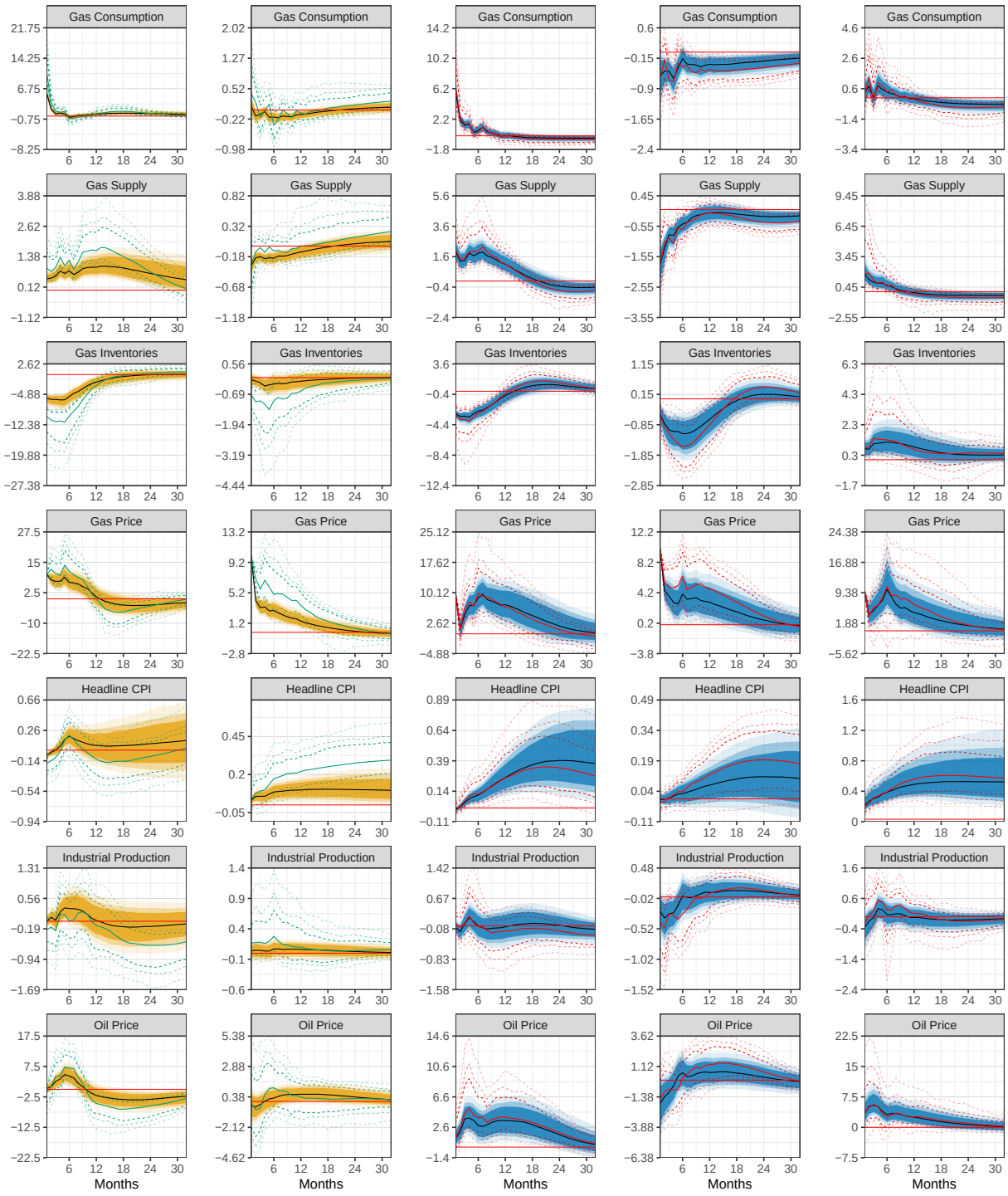
Additional variables. We also augment the VAR with the power-market variables listed in Table B.1: real wholesale electricity prices, electricity consumption, and aggregate renewable power generation. Figure E.13 shows that the responses remain close to the baseline.

Lag length. The baseline VAR contains six monthly lags. Figure E.14 reports estimates from an otherwise identical specification with twelve lags; the responses are similar.

Extended U.S. sample. The common baseline sample begins in 2004, the earliest date for which TTF prices are available. For the United States, the data permit estimation from 2001. Figure E.15 shows that extending the U.S. sample to 2001–2023 does not materially alter the demand- or supply-shock responses.

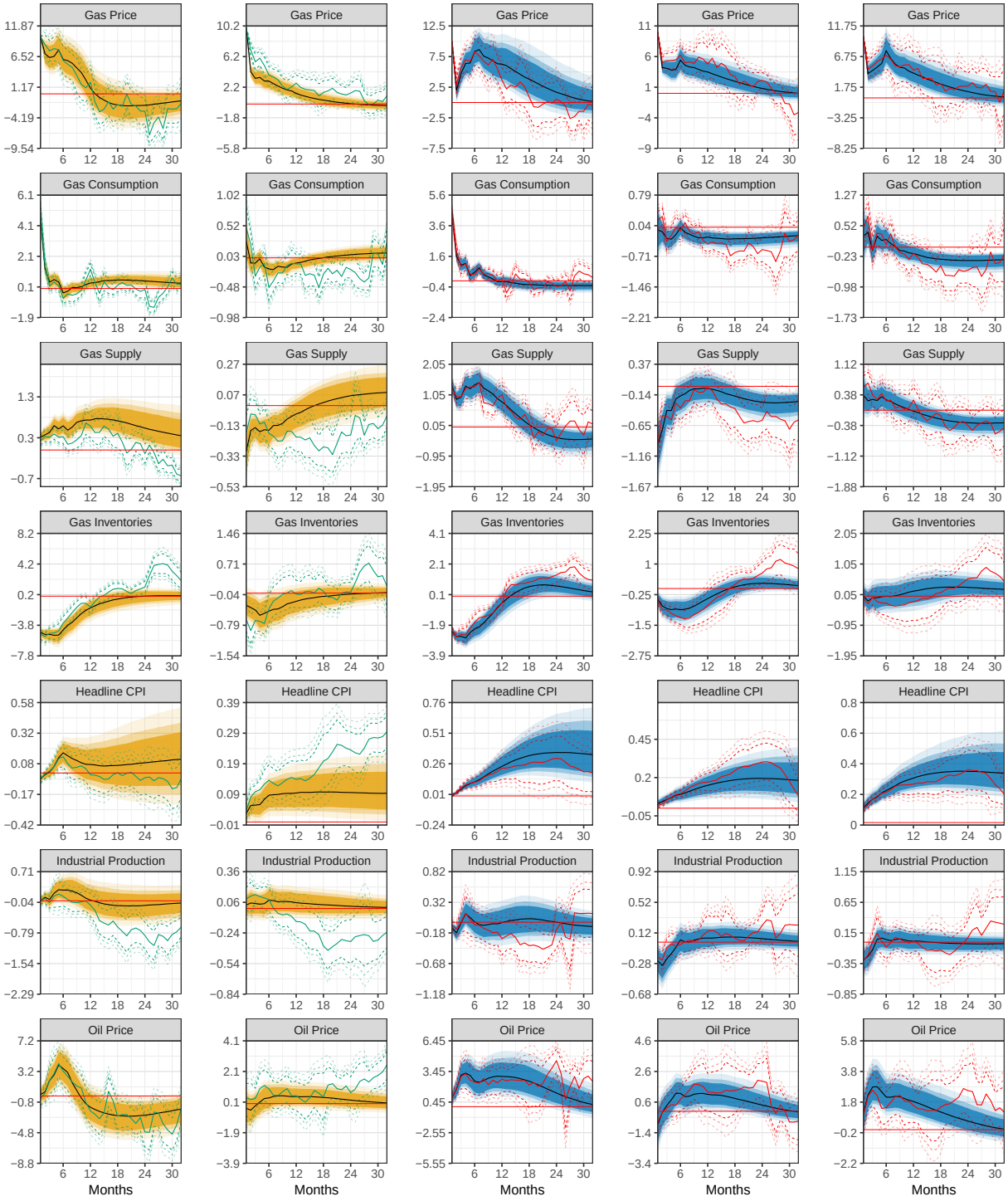
Estimation by VAR-OLS

Figure E.11: *Comparison with baseline responses: VAR estimated by OLS*



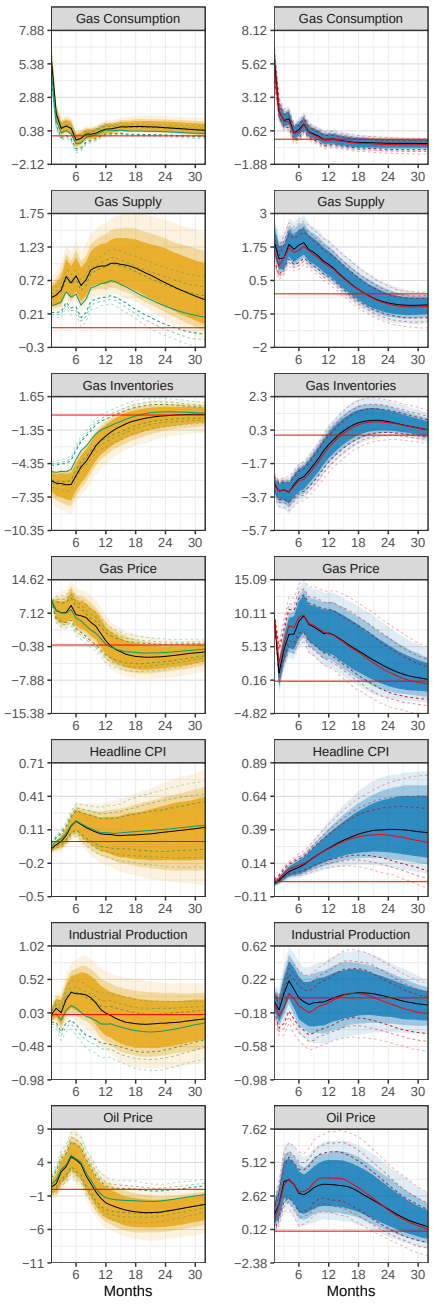
Estimation by Bayesian Local Projections

Figure E.12: *Comparison with baseline responses: Bayesian local projections*



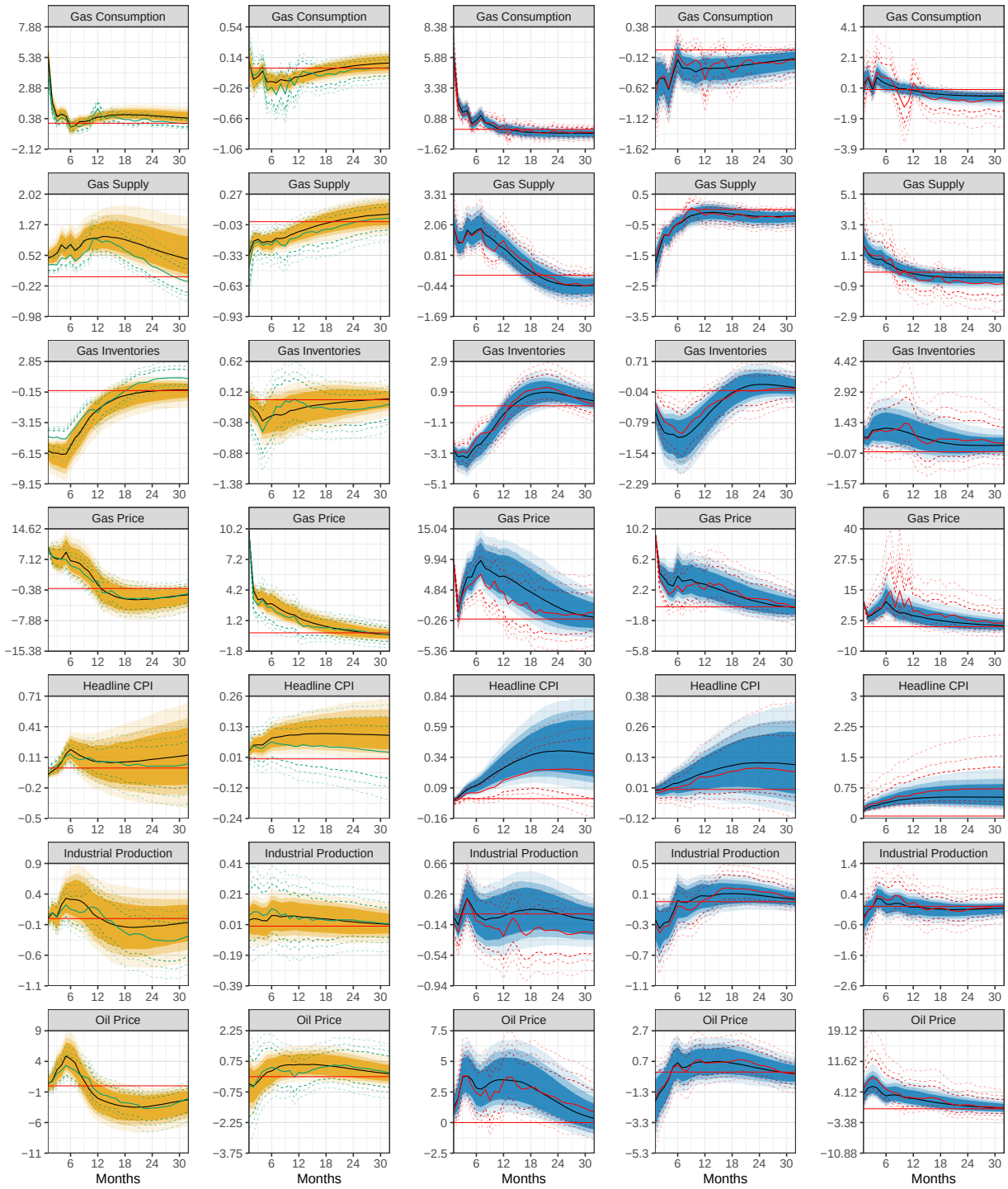
Specification with power and renewable-energy variables

Figure E.13: *Comparison with baseline responses to gas demand shocks: VAR augmented with electricity prices, electricity consumption, and renewable power generation*



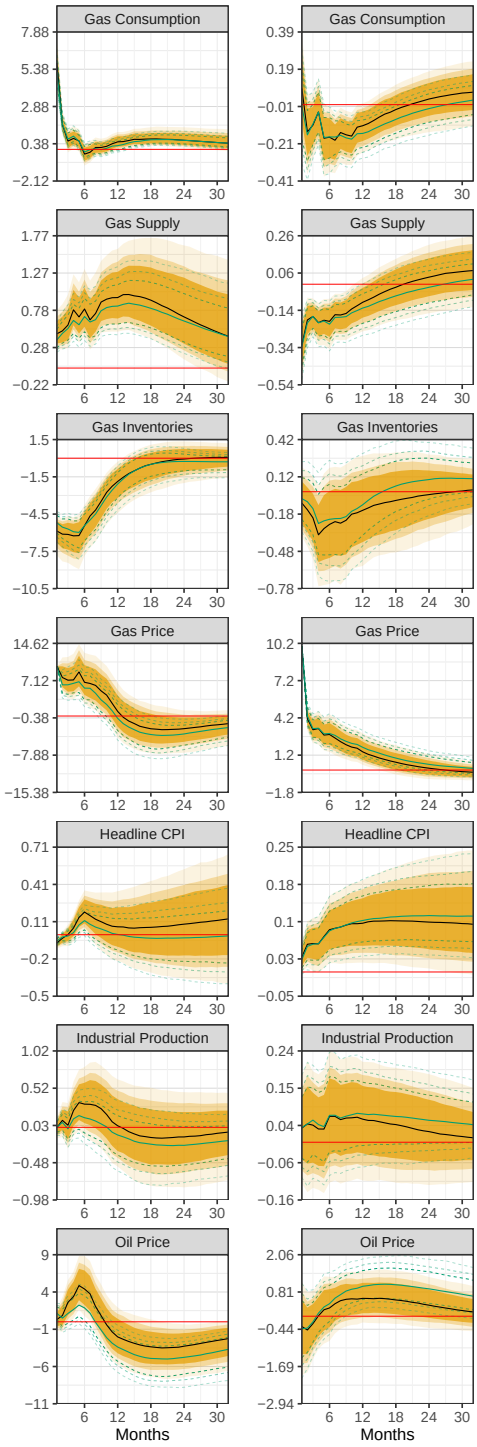
Specification with 12 lags

Figure E.14: *Comparison with baseline responses: VAR with twelve lags*



Extended estimation sample for the U.S. (2001–2023)

Figure E.15: *Comparison with baseline responses: extended U.S. sample, 2001–2023*



F Sectoral and aggregate effects of gas shocks

F.1 Euro Area sectoral effects of gas supply shocks

This subsection complements the combined Euro Area supply-shock estimates in Section V.D by reporting separate sectoral responses to flow-supply and supply-security news shocks. Both shocks are normalized to raise the real price of natural gas by 10% on impact. In Figures F.1–F.3, black solid lines with blue shaded credible bands denote responses to flow-supply shocks, while red solid lines with red dashed credible bands denote responses to supply-security news shocks. The comparison clarifies the composition of the main sectoral results: supply-security news shocks account for most of the price pass-through, whereas flow-supply shocks drive the immediate output losses.

Figure F.1: *Energy-price responses to Euro Area flow-supply and supply-security news shocks*

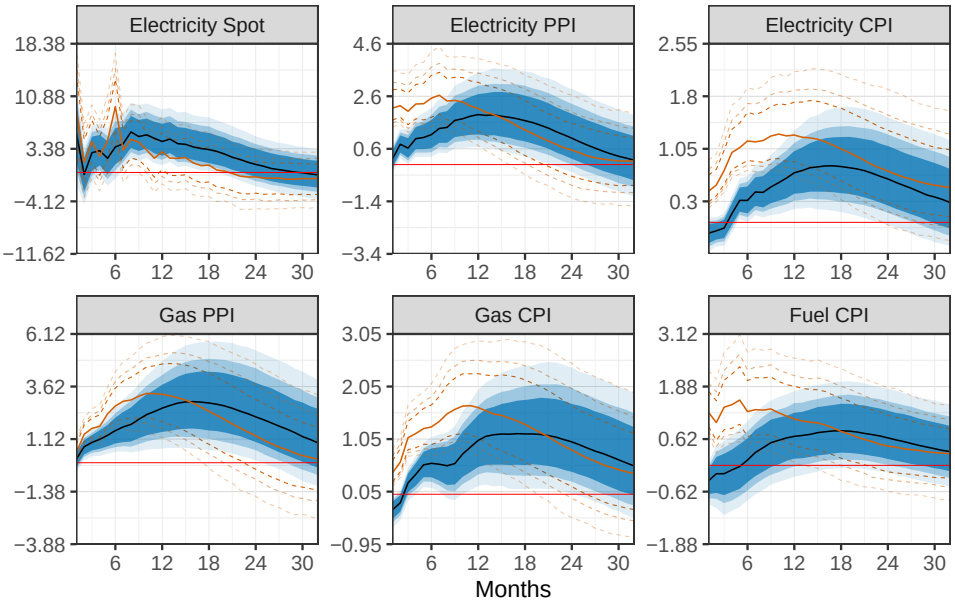
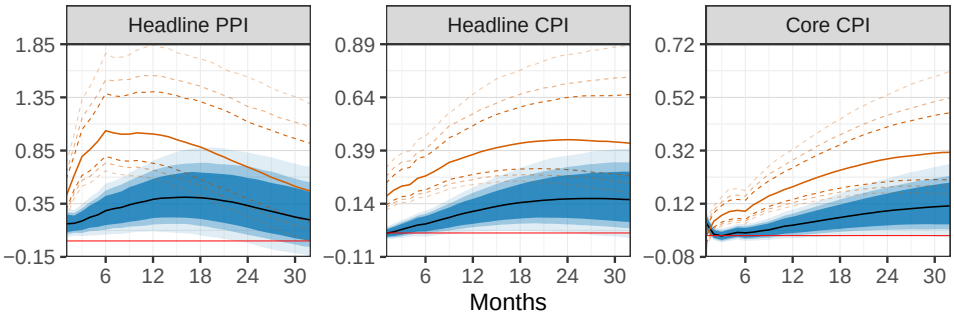
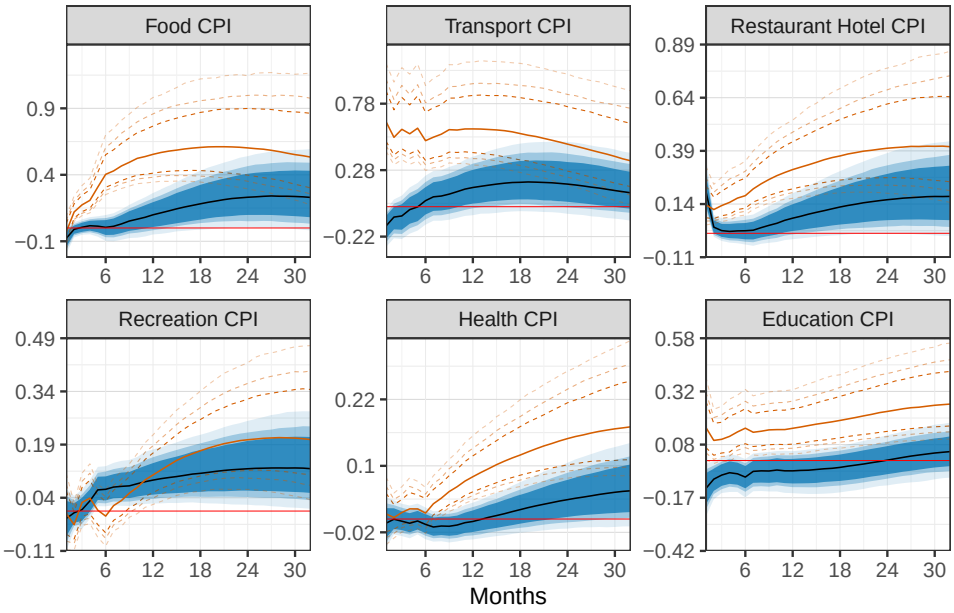


Figure F.1 shows that both shocks pass through to downstream energy prices, but unevenly along the pricing chain. Electricity spot prices respond immediately and most strongly, producer prices for gas and electricity build up over the first year, and consumer prices for gas, electricity, and fuels adjust more gradually, consistent with retail price stickiness. Across most consumer and producer price measures, supply-security news shocks generate larger near-term responses than flow-supply shocks, consistent with expected future scarcity being priced before physical flows adjust.

Figure F.2: *Price responses to Euro Area flow-supply and supply-security news shocks*



Aggregate consumer and producer prices

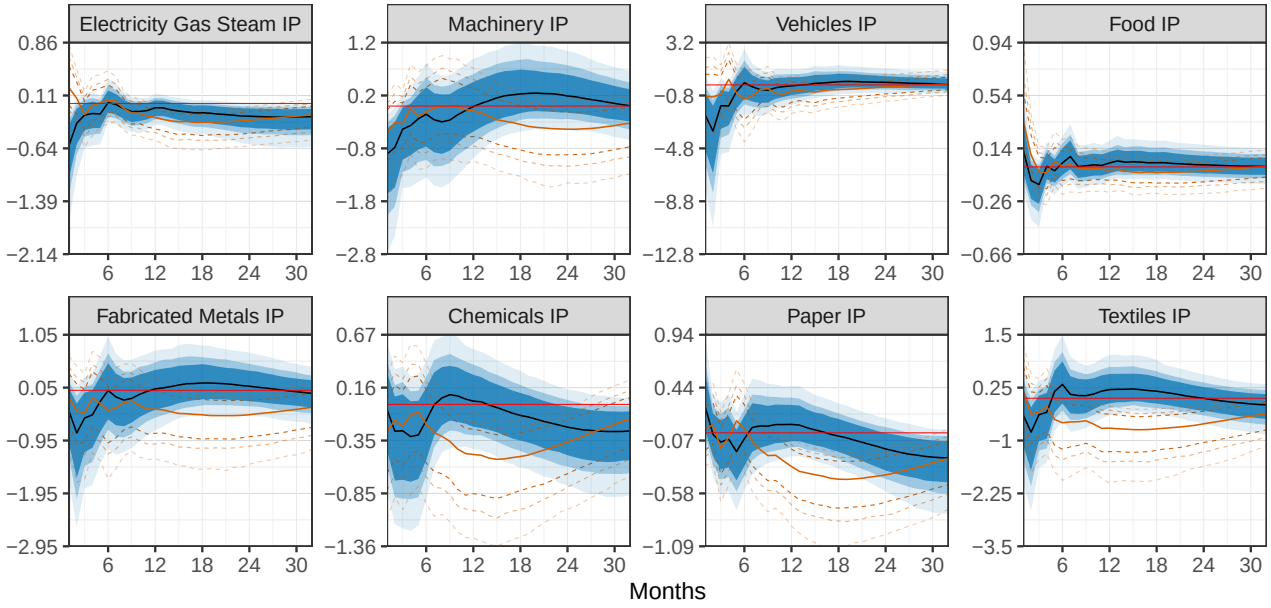


HICP consumer categories

Both panels of Figure F.2 show that the price effects obtained with the combined Euro Area supply instrument in Section V.D are driven mainly by supply-security news shocks. In the top panel, supply-security news shocks generate substantially larger responses of headline consumer prices, producer prices, and core consumer prices than flow-supply shocks; the difference is especially pronounced for headline and core consumer prices over medium horizons. The bottom panel shows a similar pattern across consumer categories. Flow-supply shocks generate comparatively modest price responses, while supply-security news shocks raise food and transport prices more strongly and generate larger increases in services over time, especially in restaurants and hotels, health, and education. This pattern is consistent with news about future supply security raising expected input costs, whereas part of the

price effect of contemporaneous flow disruptions unwinds as physical supply recovers.

Figure F.3: *Sectoral industrial-production responses to Euro Area flow-supply and supply-security news shocks*



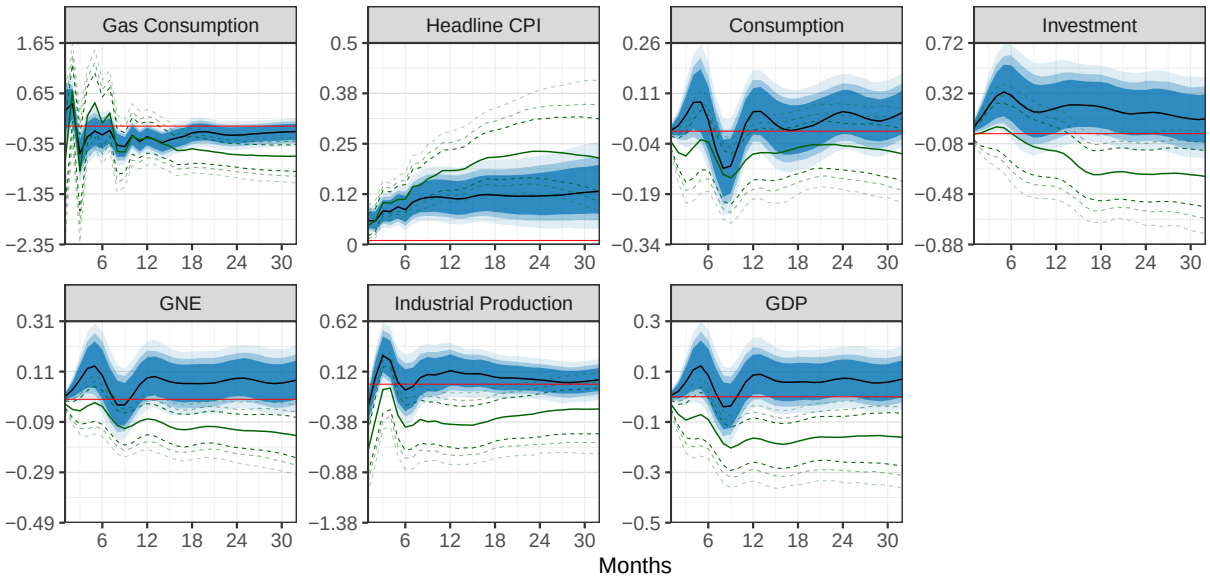
Finally, Figure F.3 considers sectoral output. Real effects are concentrated in energy supply and in energy-intensive manufacturing, including textiles, paper, chemicals, fabricated metals, and vehicles, while several other sectors respond only modestly. The two shocks differ primarily in the timing of the contraction. Flow-supply shocks generate more front-loaded declines in several exposed sectors, and these declines largely unwind within a few months. Supply-security news shocks instead generate contractions that build more gradually, particularly in chemicals and paper, consistent with the delayed price pass-through shown in the price responses above.

F.2 Real effects: Euro Area and Germany

Figure F.4 augments the Euro Area and Germany specifications underlying Figure XVI with real GDP. At the two-year horizon, German GDP declines by about 0.16%, more than twice the roughly 0.07% implied by weighting the industrial-production response by industry's share of value added, approximately one quarter.[6] Complementing this aggregate evidence, Figure F.5 compares the Euro Area sectoral responses presented in Figure XV with their German counterparts. In both figures, black solid lines with blue shaded credible

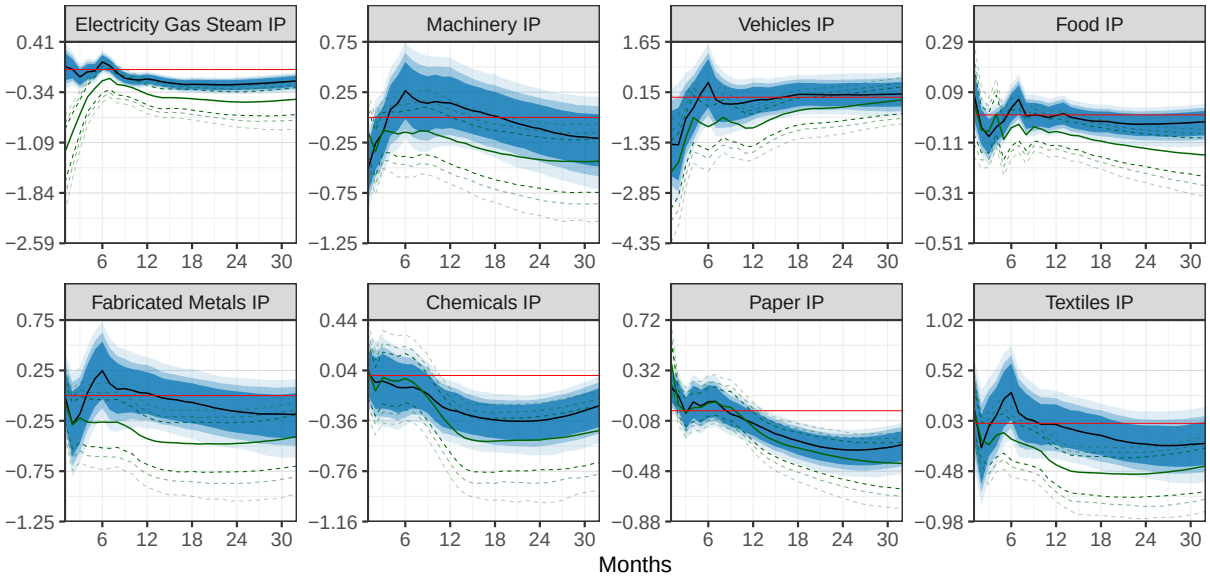
[6]Industry excluding construction (NACE Rev. 2 sections B–E) accounts on average for 24.5% of German gross value added at current prices over 2004–2023 (Eurostat 2026).

Figure F.4: *Euro Area and Germany: Real effects of a gas supply shock with GDP included*



Notes: GDP, consumption, investment, and GNE are measured at quarterly frequency and interpolated to monthly frequency using cubic splines.

Figure F.5: *Euro Area and Germany: Sectoral industrial-production responses to a gas supply shock*

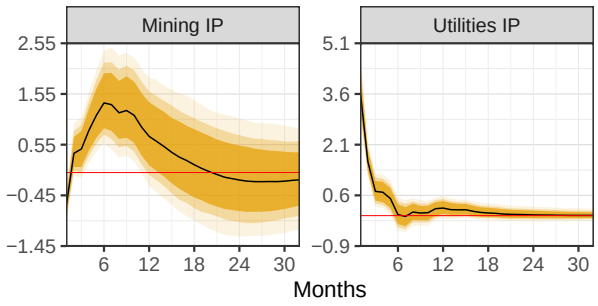


bands denote Euro Area responses; green solid lines with green dashed credible bands denote German responses. In both regions, output falls persistently in gas-intensive sectors, including chemicals, paper, and electricity and gas. Germany also experiences persistent losses in machinery, food, fabricated metals, and textiles, with generally more muted responses in the Euro Area. This broader contraction accompanies weaker German consumption and investment. Taken together, these output and expenditure responses are consistent with aggregate-demand amplification of the gas supply shock beyond the industries most directly exposed to higher gas costs.

F.3 U.S. industrial production after a gas demand shock

To examine the energy-related activity underlying the temporary increase in U.S. industrial production after a gas demand shock, Figure F.6 reports estimates from the baseline specification with total industrial production replaced jointly by mining and utilities output. Utilities output rises sharply on impact and the increase dissipates quickly, consistent with higher heating demand. Mining expands more gradually, consistent with the rise in domestic gas production in Figure VIII.

Figure F.6: *United States: Responses of mining and utilities output to a gas demand shock*

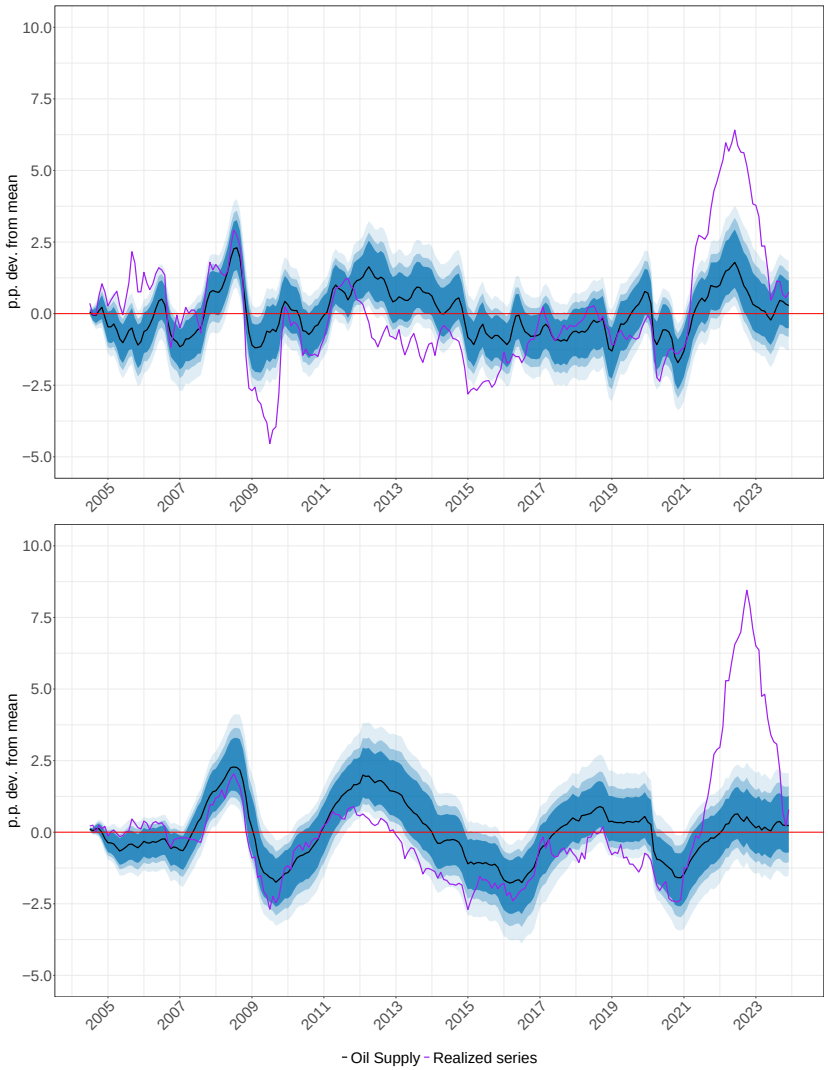


Notes: Impulse responses from the U.S. baseline specification with mining and utilities output jointly replacing total industrial production. Mining comprises oil and gas extraction (NAICS 211), mining except oil and gas (NAICS 212), and support activities for mining (NAICS 213); utilities comprise electric power generation, transmission, and distribution (NAICS 2211) and natural gas distribution (NAICS 2212) (Board of Governors of the Federal Reserve System 2026).

G Contribution of oil shocks to the historical decomposition of inflation

Figure G.1 reports the contribution of oil supply news shocks, identified as in Känzig (2021), to year-on-year headline inflation. These shocks contributed to the post-pandemic inflation surge in both regions, but played a more prominent role in the United States, whereas gas supply shocks contributed more in the Euro Area.

Figure G.1: *Historical decomposition of headline inflation: contribution of oil supply news shocks*



Notes: Contribution of oil supply news shocks (black line, with blue credible bands) and realized headline inflation (purple), in percentage points as deviations from their sample means. Top panel: United States; bottom panel: Euro Area.

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